



Additional Problems for Self Practice (APSP)

This Section is not meant for classroom discussion. It is being given to promote self-study and self testing amongst the Resonance students.

PART - I : PRACTICE TEST-1 (IIT-JEE (MAIN Pattern))

Max. Time : 1 Hr.

Max. Marks : 120

Important Instructions

- The test is of **1 hour** duration.
- The Test Booklet consists of **30** questions. The maximum marks are **120**.
- Each question is allotted **4 (four)** marks for correct response.
- Candidates will be awarded marks as stated above in Instructions No. 3 for correct response of each question.
 $\frac{1}{4}$ (**one fourth**) marks will be deducted for indicating incorrect response of each question. No deduction from the total score will be made if no response is indicated for an item in the answer sheet.
- There is only one correct response for each question. Filling up more than one response in any question will be treated as wrong response and marks for wrong response will be deducted accordingly as per instructions 4 above.

- Among them intensive property is :
(1) Mass (2) Volume (3) Surface tension (4) Enthalpy
- Which of the following is true for an adiabatic process :
(1) $\Delta H = 0$ (2) $\Delta W = 0$ (3) $\Delta Q = 0$ (4) $\Delta V = 0$
- Which of the following is not a state function :
(1) S (2) G (3) H (4) Q
- The relation between ΔU and ΔH at constant pressure is :
(1) $\Delta H = \Delta U - P\Delta V$ (2) $\Delta H = \Delta U + P\Delta V$ (3) $\Delta U = \Delta V + \Delta H$ (4) $\Delta U = \Delta H + P\Delta V$
- The work done in ergs for the reversible expansion of one mole of an ideal gas from a volume of 10 litres to 20 litres at 25°C is :
(1) $-2.303 \times 298 \times 0.082 \log 2$ (2) $-298 \times 10^7 \times 8.31 \times 2.303 \log 2$
(3) $-2.303 \times 298 \times 0.082 \log 0.5$ (4) $-8.31 \times 10^7 \times 298 - 2.303 \log 0.5$
- A coffee cup calorimeter initially contains 125 g of water, at a temperature of 24.2°C . 8 g of ammonium nitrate (NH_4NO_3), also at 24.2°C , is added to the water, and the final temperature is 18.2°C . What is the heat of solution of ammonium nitrate in kJ/mol ? The specific heat capacity of the solution is $4.2 \text{ J}^\circ\text{C} \text{ g}$.
(1) 33.51 kJ/mol (2) 39.5 kJ/mol (3) 32.2 kJ/mol (4) 37.3 kJ/mol
- The molar heat capacities at constant pressure (assumed constant with respect to temperature) at A, B and C are in ratio of 3 : 1.5 : 2.0 The enthalpy change for the exothermic reaction $\text{A} + 2\text{B} \longrightarrow 3\text{C}$ at 300 K and 310 K is ΔH_{300} and ΔH_{310} respectively then :
(1) $\Delta H_{300} > \Delta H_{310}$ (2) $\Delta H_{300} < \Delta H_{310}$ (3) $\Delta H_{300} = \Delta H_{310}$
(4) if $T_2 > T_1$ then $\Delta H_{310} > \Delta H_{300}$ and if $T_2 < T_1$ then $\Delta H_{310} < \Delta H_{300}$
- Heat of combustion of ethanol at constant pressure and at temperature T K (= 298 K) is found to be $-q \text{ J mol}^{-1}$. Hence, heat of combustion (in J mol^{-1}) of ethanol at the same temperature at constant volume will be :
(1) $RT - q$ (2) $-(q + RT)$ (3) $q - RT$ (4) $q + RT$
- When 1.0 g of oxalic acid ($\text{H}_2\text{C}_2\text{O}_4$) is burned in a bomb calorimeter whose heat capacity is 8.75 kJ/K , the temperature increases by 0.312 K . The enthalpy of combustion of oxalic acid at 27°C is :
(1) -245.7 kJ/mol (2) -244.452 kJ/mol (3) -241.947 kJ/mol (4) None of these



10. Benzene burns according to the following equation at 300 K ($R = 25/3 \text{ J mole}^{-1}\text{K}^{-1}$)
 $2\text{C}_6\text{H}_6(\ell) + 15 \text{O}_2(\text{g}) \longrightarrow 12 \text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\ell) \quad \Delta H^\circ = -6547.5 \text{ KJ}$
 What is the ΔE° for the combustion of 1.5 mol of benzene
 (1) -3271 kJ (2) -9813 kJ (3) -4905 kJ (4) -9810 kJ
11. Ethyl chloride ($\text{C}_2\text{H}_5\text{Cl}$), is prepared by reaction of ethylene with hydrogen chloride :
 $\text{C}_2\text{H}_4(\text{g}) + \text{HCl}(\text{g}) \longrightarrow \text{C}_2\text{H}_5\text{Cl}(\text{g}) \quad \Delta H = -72.3 \text{ kJ/mol.}$
 What is the value of ΔE (in kJ), if 98 g of ethylene and 109.5 g of HCl are allowed to react at 300 K.
 (1) -64.81 (2) -190.71 (3) -209.41 (4) -224.38
12. Which statement regarding entropy is correct ?
 (1) A completely ordered deck of cards has more entropy than a shuffled deck in which cards are arranged randomly.
 (2) A perfect ordered crystal of solid nitrous oxide has more entropy than a disordered crystal in which the molecules are oriented randomly.
 (3) 1 mole N_2 gas at STP has more entropy than 1 mole N_2 gas at 273 K in a volume of 11.2 litre.
 (4) 1 mole N_2 gas at STP has more entropy than 1 mole N_2 gas at 273 K and 0.25 atm.
13. One mole of solid Zn is placed in excess of dilute H_2SO_4 at 27°C in a cylinder fitted with a piston. Find the work done for the process if the area of piston is 500 cm^2 and it moves out by 50 cm against a pressure of 1 atm during the reaction.
 $\text{Zn}(\text{s}) + 2\text{H}^+(\text{aq}) \rightleftharpoons \text{Zn}^{2+}(\text{aq}) + \text{H}_2(\text{g})$
 (1) -1.53 KJ (2) -2.53 KJ (3) Zero (4) 2.53 KJ
14. The enthalpy change for the reaction of 50 mL of ethylene with 50.0 mL of H_2 at 1.5 atm pressure is $\Delta H = -0.31 \text{ KJ}$. What is the ΔE ?
 (1) -0.3024 (2) -0.6048 (3) -0.12 (4) None
15. When 0.1 mole of ice melt at 0°C and at constant pressure of 1 atm 144 calories of heat are absorbed by the system. Calculate ΔH for the reaction.
 (1) $\Delta H = 720 \text{ J}\cdot\text{mol}^{-1}$ (2) $\Delta H = 1440 \text{ Cal}\cdot\text{mol}^{-1}$ (3) $\Delta H = 1.4 \text{ kCal}\cdot\text{mol}^{-1}$ (4) $\Delta H = 0$
16. One mole of ideal monoatomic gas is carried through the reversible cyclic process as shown in figure. Calculate net heat absorbed by the gas in the path BC.

 (1) $\frac{1}{2} P^0 V^0$ (2) $\frac{7}{2} P^0 V^0$
 (3) $2 P^0 V^0$ (4) $\frac{5}{2} P^0 V^0$
17. 130 g of Zn is dissolved in dilute sulphuric acid in an open beaker at 27°C . Find the work done in the process assuming isothermal operation.
 (1) -1200 cal (2) -1800 cal (3) $+1800 \text{ cal}$ (4) $+1200 \text{ cal}$
18. The enthalpy of combustion of propane (C_3H_8) gas in terms of given data is :
 Bond energy (kJ/mol)
 $\begin{array}{ccccc} \epsilon_{\text{C-H}} & \epsilon_{\text{O=O}} & \epsilon_{\text{C=O}} & \epsilon_{\text{O-H}} & \epsilon_{\text{C-C}} \\ +X_1 & +X_2 & +X_3 & +X_4 & +X_5 \end{array}$
 Resonance energy of CO_2 is $-z \text{ kJ/mol}$ and $\Delta H_{\text{vaporization}} [\text{H}_2\text{O}(\ell)]$ is $y \text{ kJ/mol}$.
 (1) $8x_1 + 2x_5 + 5x_2 - 6x_3 - 8x_4 - 4y - 3z$ (2) $6x_1 + x_5 + 5x_2 - 3x_3 - 4x_4 - 4y - 3z$
 (3) $8x_1 + 2x_5 + 5x_2 - 6x_3 - 8x_4 - y - z$ (4) $8x_1 + x_5 + 5x_2 - 6x_3 - 8x_4 - 4y + 3z$
19. If x_1 , x_2 and x_3 are enthalpies of H-H, O=O and O-H bonds respectively, and x_4 is the enthalpy of vaporisation of water, estimate the standard enthalpy of combustion of hydrogen.
 (1) $x_1 + \frac{x_2}{2} - 2x_3 + x_4$ (2) $x_1 + \frac{x_2}{2} - 2x_3 - x_4$ (3) $x_1 + \frac{x_2}{2} - x_3 + x_4$ (4) $2x_3 - x_1 - \frac{x_2}{2} - x_4$



20. $\text{NH}_3(\text{g}) + 3\text{Cl}_2(\text{g}) \longrightarrow \text{NCl}_3(\text{g}) + 3\text{HCl}(\text{g})$; ΔH_1
 $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \longrightarrow 2\text{NH}_3(\text{g})$; ΔH_2
 $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \longrightarrow 2\text{HCl}(\text{g})$; ΔH_3
 The heat of formation of $\text{NCl}_3(\text{g})$ in the terms of ΔH_1 , ΔH_2 and ΔH_3 is
 (1) $\Delta H_f = -\Delta H_1 + \frac{\Delta H_2}{2} - \frac{3}{2}\Delta H_3$ (2) $\Delta H_f = \Delta H_1 + \frac{\Delta H_2}{2} - \frac{3}{2}\Delta H_3$
 (3) $\Delta H_f = \Delta H_1 - \frac{\Delta H_2}{2} - \frac{3}{2}\Delta H_3$ (4) None
21. ΔH°_f of water is $-285.5 \text{ kJ mol}^{-1}$. if enthalpy of neutralisation of monoacidic strong base with strong acid is $-57.3 \text{ kJ mol}^{-1}$, ΔH°_f of OH^- ion will be
 (1) $-228.5 \text{ kJ mol}^{-1}$ (2) $228.5 \text{ kJ mol}^{-1}$ (3) $114.5 \text{ kJ mol}^{-1}$ (4) $-114.5 \text{ kJ mol}^{-1}$
22. Ethanol can undergoes decomposition to form two sets of products.
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- If the molar ratio of C_2H_4 to CH_3CHO is 8 : 1 in a set of product gases, then the energy involved in the decomposition of 1 mole of ethanol is
 (1) 65.98 kJ (2) 48.137 kJ (3) 48.46 kJ (4) 57.22 kJ
23. Find the enthalpy of S-S bond from the following data.
 (i) $\text{C}_2\text{H}_5\text{S}-\text{C}_2\text{H}_5(\text{g})$ $\Delta H^\circ_f = -147.2 \text{ kJ/mol}$
 (ii) $\text{C}_2\text{H}_5\text{S}-\text{S}-\text{C}_2\text{H}_5(\text{g})$ $\Delta H^\circ_f = -201.9 \text{ kJ/mol}$
 (iii) $\text{S}(\text{g})$ $\Delta H^\circ_f = 222.8 \text{ kJ/mol}$
 (1) -168.1 kJ/mol (2) $+168.1 \text{ kJ/mol}$ (3) -277.5 kJ/mol (4) $+277.5 \text{ kJ/mol}$
24. Given the following equations and ΔH° values, determine the enthalpy of reaction at 298 K for the reaction :
- $\text{C}_2\text{H}_4(\text{g}) + 6\text{F}_2(\text{g}) \longrightarrow 2\text{CF}_4(\text{g}) + 4\text{HF}(\text{g})$
 $\text{H}_2(\text{g}) + \text{F}_2(\text{g}) \longrightarrow 2\text{HF}(\text{g})$; $\Delta H^\circ_1 = -537 \text{ kJ}$
 $\text{C}(\text{s}) + 2\text{F}_2(\text{g}) \longrightarrow \text{CF}_4(\text{g})$; $\Delta H^\circ_2 = -680 \text{ kJ}$
 $2\text{C}(\text{s}) + 2\text{H}_2(\text{g}) \longrightarrow \text{C}_2\text{H}_4(\text{g})$; $\Delta H^\circ_3 = 52 \text{ kJ}$
- (1) -1165 (2) -2486 (3) $+1165$ (4) $+2486$
25. Animals operate under conditions of constant pressure and most of the process that maintain life are isothermal (in a broad sense). How much energy is available for sustaining this type of muscular and nervous activity from the combustion of 1 mol of glucose molecules under standard conditions at 37°C (blood temperature) ? The entropy change is $+182.4 \text{ JK}^{-1}$ for the reaction stated above.
 $\Delta H_{\text{combustion}}[\text{glucose}] = -2808 \text{ kJ}$
 (1) -2754.4 kJ (2) -2864.5 kJ (3) -56.5 kJ (4) -2808 kJ
26. From the given table answer the following question :
- | | CO (g) | CO ₂ (g) | H ₂ O(g) | H ₂ (g) |
|---|--------|---------------------|---------------------|--------------------|
| $\Delta H^\circ_{298}(\text{Kcal/ mole})$ | -26.42 | -94.05 | -57.8 | 0 |
| $\Delta G^\circ_{298}(\text{Kcal/ mole})$ | -32.79 | -94.24 | -54.64 | 0 |
| $S^\circ_{298}(\text{Cal/ K mol})$ | 47.3 | 51.1 | ? | 31.2 |
- Reaction : $\text{H}_2\text{O}(\text{g}) + \text{CO}(\text{g}) \rightleftharpoons \text{H}_2(\text{g}) + \text{CO}_2(\text{g})$
 Calculate $S^\circ_{298}[\text{H}_2\text{O}(\text{g})]$
 (1) $-119.47 \text{ Cal/ K mole}$ (2) $+119.47 \text{ Cal/ K mole}$
 (3) $-45.13 \text{ Cal/ K mole}$ (4) $+45.13 \text{ Cal/ K mole}$
27. Calculate the free energy change at 298 K for the reaction; $\text{Br}_2(\ell) + \text{Cl}_2(\text{g}) \longrightarrow 2\text{BrCl}(\text{g})$. For the reaction $\Delta H^\circ = 29.3 \text{ kJ}$ & the entropies of $\text{Br}_2(\ell)$, $\text{Cl}_2(\text{g})$ & $\text{BrCl}(\text{g})$ at the 298K are 152.3, 223.0, 239.7 J $\text{mol}^{-1}\text{K}^{-1}$ respectively.
 (1) -1721.8 J (2) -60321.8 J (3) $+60321.8 \text{ J}$ (4) $+1721.8 \text{ J}$



28. One gram sample of oxygen undergoes free expansion in insulated container from 0.75 L to 3.0 L at 298 K. Calculate ΔS , q , w , ΔH and ΔE .
 (1) $\Delta S = 0.36 \text{ JK}^{-1}$, $\Delta E = 0$ (2) $W = 227.97 \text{ J}$, $\Delta E = 0$
 (3) $q = -227.97 \text{ J}$, $\Delta E = 0$ (4) $W = 227.97 \text{ J}$, $\Delta H = 0$
29. Given that : $\Delta G_f^\circ (\text{CuO}) = -30.4 \text{ kcal/mole}$
 $\Delta G_f^\circ (\text{Cu}_2\text{O}) = -34.98 \text{ kcal/mole}$ $T = 298 \text{ K}$
 Now on the basis of above data which of the following predictions will be most appropriate under the standard conditions and reversible reaction.
 (1) Finely divided form of CuO kept in excess O_2 would be completely converted to Cu_2O
 (2) Finely divided form of Cu_2O kept in excess O_2 would be completely converted to CuO
 (3) Finely divided form of CuO kept in excess O_2 would be converted to a mixture of CuO and Cu_2O (having more of CuO)
 (4) Finely divided form of CuO kept in excess O_2 would be converted to a mixture of CuO and Cu_2O (having more of Cu_2O)
30. Calculate $\Delta_r G^\circ$ for $(\text{NH}_4\text{Cl}, \text{s})$ at 310 K.
 Given : $\Delta_r H^\circ (\text{NH}_4\text{Cl}, \text{s}) = -314.5 \text{ kJ/mol}$; $\Delta_r C_p = 0$
 $S^\circ_{\text{N}_2(\text{g})} = 192 \text{ JK}^{-1} \text{ mol}^{-1}$; $S^\circ_{\text{H}_2(\text{g})} = 130.5 \text{ JK}^{-1} \text{ mol}^{-1}$;
 $S^\circ_{\text{Cl}_2(\text{g})} = 233 \text{ JK}^{-1} \text{ mol}^{-1}$; $S^\circ_{\text{NH}_4\text{Cl}(\text{s})} = 99.5 \text{ JK}^{-1} \text{ mol}^{-1}$
 All given data are at 300 K.
 (1) -198.56 kJ/mol (2) -426.7 kJ/mol (3) -202.3 kJ/mol (4) None of these

Practice Test-1 (IIT-JEE (Main Pattern))

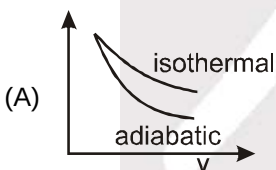
OBJECTIVE RESPONSE SHEET (ORS)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.										
Que.	11	12	13	14	15	16	17	18	19	20
Ans.										
Que.	21	22	23	24	25	26	27	28	29	30
Ans.										

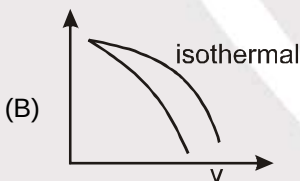
PART - II : NATIONAL STANDARD EXAMINATION IN CHEMISTRY (NSEC) STAGE-I

1. The enthalpy of hydrogenation for 1-pentene is +126 KJ/mol. The enthalpy of hydrogenation for 1 : 3-pentadiene is + 230 KJ/mol. Hence estimate the resonance (delocalization) energy of 1 : 3-pentadiene? [NSEC-2000]
 (A) 22 KJ (B) 104 KJ (C) 252 KJ
 (D) Cannot be calculated from this information.
2. The entropy change during an isothermal expansion of an ideal gas from V_1 to V_2 at temperature T is given by [NSEC-2000]
 (A) $\Delta S = 0$ (B) $\Delta S = 2.303 R \log_{10} V_2 / V_1$
 (C) $\Delta S = 2.303 RT \log_{10} V_2 / V_1$ (D) $\Delta S = 2.303 R \log_{10} V_1 / V_2$
3. An isothermal process is associated with [NSEC-2000]
 (A) constant entropy (B) constant temperature
 (C) constant enthalpy (D) large change in heat content
4. The heat change for the reaction, $\text{C}_{(\text{s})} + 2\text{S}_{(\text{s})} \rightarrow \text{CS}_{2(\text{l})}$ is known as [NSEC-2000]
 (A) heat of transition (B) heat of fusion
 (C) heat of vapourisation (D) heat of formation

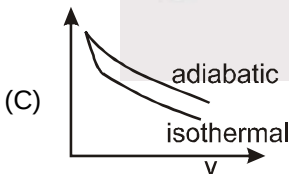


5. When a solid melts, there is : [NSEC-2001]
 (A) decreases in enthalpy (B) increase in enthalpy
 (C) no increase in enthalpy (D) decrease in entropy
6. The entropy change for the vapourisation of water at 100°C (molar latent heat of vapourisation of water = 40.8 kJ mol⁻¹) is : [NSEC-2001]
 (A) 109.4 JK⁻¹mol⁻¹ (B) 0.1094 JK⁻¹mol⁻¹ (C) 0.408 JK⁻¹mol⁻¹ (D) 40.8 JK⁻¹mol⁻¹
7. The free energy change for an irreversible process is : [NSEC-2001]
 (A) zero (B) negative (C) positive (D) very large positive
8. An isobaric process is associated with : [NSEC-2001]
 (A) constant temperature (B) constant volume
 (C) constant pressure (D) small change in enthalpy
9. Lattice energy of ionic compound is calculated by using : [NSEC-2002]
 (A) Hess's law (B) Kirchhoff's equation
 (C) Born-Haber cycle (D) Carnot cycle.
10. According to second law of thermodynamics [NSEC-2002]
 (A) total energy in the Universe remains constant
 (B) total enthalpy in the Universe remains constant
 (C) total entropy of the Universe always increases
 (D) total entropy of the Universe always decreases.
11. For a spontaneous reaction, ΔG and ΔS should be [NSEC-2002]
 (A) $\Delta G = +ve$ and $\Delta S = -ve$ (B) $\Delta G = -ve$ and $\Delta S = +ve$
 (C) $\Delta G = 0$ and $\Delta S = +ve$ (D) $\Delta G = +ve$ and $\Delta S = 0$
12. Which of the following expression is correct ? [NSEC-2002]
 (A) $E = H + PV$ (B) $H = E + PV$ (C) $H = E - PV$ (D) $P = E + HV$
13. The entropy change for the fusion of ice at 0°C (molar latent heat of fusion of ice = 6.025 kJ mol⁻¹) is : [NSEC-2002]
 (A) 22.07 JK⁻¹ mol⁻¹ (B) 1.226 JK⁻¹ mol⁻¹ (C) 2.207 JK⁻¹ mol⁻¹ (D) 25.0 JK⁻¹ mol⁻¹
14. The correct figure representing isothermal and adiabatic expansions of an ideal gas from a particular initial state is [NSEC-2003]
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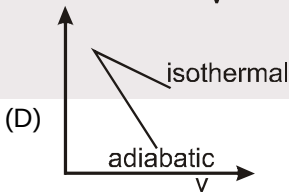
(A)



(B)



(C)

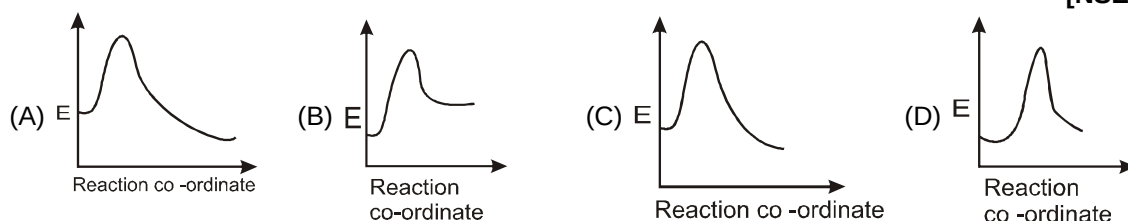


(D)
15. An ideal gas, on adiabatic expansion, gets cooled because [NSEC-2003]
 (A) the average distance between the gas molecules increases and hence the intermolecular interaction decreases
 (B) the internal energy of the gas is used in doing the work of expansion
 (C) the temperature of the gas is lower than the inversion temperature corresponding to Joule-Thomson effect
 (D) the pressure decreases and temperature is proportional to the pressure.



16. Which energy diagram corresponds to exothermic reaction and highest value of the rate constant ?

[NSEC-2003]



17. $\text{CCl}_4(\text{g}) + 2\text{H}_2\text{O}(\text{g}) \longrightarrow \text{CO}_2(\text{g}) + 4\text{HCl}(\text{g})$. The standard enthalpies of formation at 298 K for $\text{CCl}_4(\text{g})$, $\text{H}_2\text{O}(\text{g})$, $\text{CO}_2(\text{g})$ and $\text{HCl}(\text{g})$ are -106.7 , -241.8 , -393.7 and -92.5 kJ mol^{-1} respectively. The value of ΔH°_{298} for the above reaction is

[NSEC-2004]

- (A) -137.7 kJ (B) 173.4 kJ (C) -173.4 kJ (D) 137.7 kJ.

18. The temperature of an ideal gas increases in an

[NSEC-2004]

- (A) adiabatic expansion (B) isothermal expansion
(C) adiabatic compression (D) isothermal compression.

19. The value of Δn for the combustion reaction of one mole of benzene in gaseous state when both the reactants and products are at 298 K is

[NSEC-2004]

- (A) 1 (B) 0 (C) $1/2$ (D) $-1/2$.

20. For two moles of an ideal gas

[NSEC-2004]

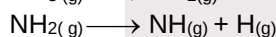
- (A) $C_V - C_P = R$ (B) $C_P - C_V = 2R$ (C) $C_P - C_V = R$ (D) $C_V - C_P = 2R$.

21. Heat of combustion of ethanol at constant pressure and at 298 K is found to be $-Q$ kJ mol^{-1} . Hence, heat of combustion of ethanol at the same temperature at constant volume will be

[NSEC-2004]

- (A) $(-Q + RT)$ kJ mol^{-1} (B) $(-Q - RT)$ kJ mol^{-1} (C) $(+Q - RT)$ kJ mol^{-1} (D) $(Q + RT)$ kJ mol^{-1} .

22. $\text{NH}_3(\text{g}) \longrightarrow \text{NH}_2(\text{g}) + \text{H}(\text{g})$



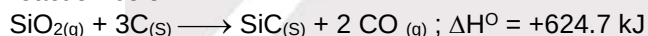
The enthalpy changes for the above reactions

[NSEC-2004]

- (A) are the same (B) are in increasing order
(C) are in decreasing order (D) cannot be correlated.

23. How much heat is absorbed in the complete reaction of 3.00 grams of SiO_2 with excess carbon in the reaction below ?

[NSEC-2005]

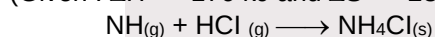


- (A) 1.33×10^4 kJ (B) 1.13×10^5 kJ (C) 5.06 kJ (D) 31.2 kJ

24. Estimate the temperature at which $\Delta G = 0$ for the following reaction:

[NSEC-2005]

(Given : $\Delta H = -176$ kJ and $\Delta S = -284.5$ J/K). Caution : Beware of units.



- (A) 582 K (B) 467 K (C) 634 K (D) 619 K.

25. A particular chemical reaction is characterised by $\Delta H_{\text{rxn}} = +250$ kJ/mol. Which of the following statements is/are true concerning this reaction ?

[NSEC-2005]

I. Heat is liberated to the surroundings.

II. The reaction is endothermic.

III. The total heat content of the products is lower than that of the reactants.

- (A) all are true (B) only II and III are true
(C) only I and II are true (D) only II is true.

26. Of the following composition for the mixture of $\text{H}_2(\text{g})$ and $\text{N}_2(\text{g})$ which gives rise to maximum entropy of mixing?

[NSEC-2006]

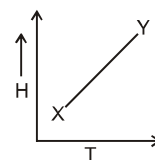
- (A) 2 mols of H_2 and 1 mol of N_2 (B) 1 mol of H_2 and 2 mols of N_2
(C) 1.5 mols of H_2 and 1.5 mols of N_2 (D) 0.5 mol of H_2 and 2.5 mols of N_2 .



27. The enthalpy change, ΔH due to reversible adiabatic expansion of an ideal gas from P_1, V_1, T_1 , to P_2, V_2, T_2 is [NSEC-2006]
 (A) $C_V \ln \frac{T_2}{T_1}$ (B) $C_P \ln \frac{T_2}{T_1}$ (C) $R \ln \frac{V_2}{V_1}$ (D) $R \ln \frac{P_2}{P_1}$
28. Which of the following processes is/are accompanied by an increase in the entropy of the system?
 (i) Water freezes
 (ii) Iodine sublimates at room temperature
 (iii) A building is constructed from bricks and mortar
 (iv) A partition is removed to allow two gases to mix [NSEC-2006]
 (A) (i) and (iv) (B) (i), (ii) and (iv) (C) (ii) and (iv) (D) only (iv).
29. A reference book lists the heat of formation of $\text{NH}_3(\text{g})$ to be -46 kJ mol^{-1} . The ΔH (in kJ mol^{-1}) of the reaction: $2\text{NH}_3(\text{g}) \longrightarrow \text{N}_2(\text{g}) + 3\text{H}_2(\text{g})$ is [NSEC-2006]
 (A) 46 (B) -46 (C) 92 (D) -92
30. Both the lattice energy (ΔU_o) and hydration enthalpy (ΔH_h) of a binary salt are negative quantities. However, if ΔU_o is more negative than ΔH_h then [NSEC-2006]
 (A) the salt will not dissolve in water (B) salt will dissolve in water
 (C) dissolution of salt in water is exothermic (D) dissolution of salt in water is endothermic
31. Preferred conditions for spontaneity in nature are [NSEC-2006]
 (A) low enthalpy and low entropy (B) high enthalpy and high entropy
 (C) high enthalpy and low entropy (D) low enthalpy and high entropy.
32. The change in free energy accompanied by the isothermal reversible expansion of 1 mol of an ideal gas when it doubles its volume is ΔG_1 . The change in free energy accompanied by sudden isothermal irreversible doubling of volume of 1 mol of the same gas is ΔG_2 . Ratio of ΔG_1 to ΔG_2 is : [NSEC-2007]
 (A) 1 (B) 1/2 (C) -1 (D) -1/2
33. The degree of freedom for $\text{NH}_4\text{Cl}(\text{s}) \rightleftharpoons \text{NH}_4\text{Cl}(\text{g}) \rightleftharpoons \text{NH}_3(\text{g}) + \text{HCl}(\text{g})$ will be [NSEC-2008]
 (A) 4 (B) 3 (C) 2 (D) 1
34. A reversible adiabatic expansion is an [NSEC-2008]
 (A) isothermal process (B) isobaric process (C) isentropic process (D) isochoric process
35. If the bond energies are as follows [NSEC-2008]
 (i) C-H = 413.8 kJ (ii) Cl-Cl = 238.0 kJ (iii) C-Cl = 327.2 kJ (iv) H-Cl = 429.8 kJ
 the enthalpy of the reaction: $\text{CH}_4 + 2\text{Cl}_2 \longrightarrow \text{CH}_2\text{Cl}_2 + 2\text{HCl}$ will be
 (A) +202.6 kJ (B) -202.6 kJ (C) +220.1 kJ (D) +870 kJ
36. For the isobaric gaseous reaction, the enthalpy change given by - [NSEC-2009]
 (A) $\Delta H = \Delta E + \Delta nRT$ (B) $\Delta H = \Delta E + P\Delta V$ (C) $\Delta H = \Delta E + \Delta nP\Delta V$ (D) $\Delta H = \Delta E + V\Delta P$
37. A reaction will never be spontaneous at any temperature and pressure if [NSEC-2010]
 (A) $\Delta S = +ve, \Delta H = +ve$ (B) $\Delta S = +ve, \Delta H = -ve$ (C) $\Delta S = -ve, \Delta H = +ve$ (D) $\Delta S = -ve, \Delta H = -ve$
38. The internal energy change involved when a system goes from state A to state B is 40 kJ. If the system goes from A to B by a reversible path and returns to A by an irreversible path, the net change in internal energy is [NSEC-2010]
 (A) 40 kJ (B) > 40 kJ (C) < 40 kJ (D) zero
39. The enthalpy of formation of carbon dioxide and water are -395 kJ and -285 kJ , respectively, and the enthalpy of combustion of acetic acid is -869 kJ . Hence the enthalpy of formation of acetic acid is [NSEC-2010]
 (A) 235 kJ (B) -235 kJ (C) 420 kJ (D) 491 kJ
40. For the reaction $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \longrightarrow 2\text{NH}_3(\text{g})$
 Which of the following is correct ? [NSEC-2010]
 (A) $\Delta H = \Delta E$ (B) $\Delta H > \Delta E$ (C) $\Delta H < \Delta E$ (D) $\Delta H = 2\Delta E$



41. The enthalpies of decomposition of methane (CH_4)(g) and ethane (C_2H_6)(g) are 400 and 670 kJ mol^{-1} , respectively. The $\Delta H_{\text{C-C}}$ in kJ mol^{-1} is [NSEC-2011]
(A) 270 (B) 70 (C) 200 (D) 240
42. The degrees of freedom for the system $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$ are [NSEC-2011]
(A) 1 (B) 2 (C) 3 (D) 4
43. Which of the following has a positive entropy change ? [NSEC-2012]
(A) $\text{H}_2\text{O}(\text{g}) \longrightarrow \text{H}_2\text{O}(\text{l})$ (B) $\text{BF}_3(\text{g}) + \text{NH}_3(\text{g}) \longrightarrow \text{F}_3\text{B} \cdot \text{NH}_3(\text{s})$
(C) $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{SO}_3(\text{g})$ (D) $2\text{NH}_4\text{NO}_3(\text{s}) \longrightarrow 2\text{N}_2(\text{g}) + 4\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$
44. The enthalpy changes for the following reactions are : [NSEC-2012]
 $\text{C}_{\text{diamond}} + \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g}) \quad \Delta H = -395.3 \text{ kJ mol}^{-1}$
 $\text{C}_{\text{graphite}} + \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g}) \quad \Delta H = -393.4 \text{ kJ mol}^{-1}$
The enthalpy change for the transition $\text{C}_{\text{diamond}} \longrightarrow \text{C}_{\text{graphite}}$ will be :
(A) -3.8 kJ mol^{-1} (B) $+3.8 \text{ kJ mol}^{-1}$ (C) -1.9 kJ mol^{-1} (D) $+1.9 \text{ kJ mol}^{-1}$
45. For a chemical reaction ΔH is negative and ΔS is positive. This reaction is [NSEC-2013]
(A) spontaneous at all temperatures (B) nonspontaneous at all temperatures
(C) spontaneous only at high temperatures (D) spontaneous only at low temperatures
46. 6.24 g of ethanol are vaporized by supplying 5.89 kJ of heat energy. What is the enthalpy of vapourisation of ethanol ? [NSEC-2013]
(A) 43.42 kJ (B) 47.0 kJ (C) 21.75 kJ (D) 435.0 kJ
47. Two samples A and B of an ideal gas, initially at same temperature and pressure, are compressed from volume V to $V/2$, isothermally for A and adiabatically for B. The final pressure of A will be [NSEC-2014]
(A) greater than that of B (B) less than that of B
(C) twice that of B (D) equal to that at B
48. The **heat of formation** of ethanol, from the following data is [NSEC-2014]
 $\text{C}_2\text{H}_5\text{OH}(\text{l}) + 3\text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) + 3\text{H}_2\text{O}(\text{l}) \quad -1368 \text{ kJ}$
 $\Delta H_{\text{f}}\text{CO}_2(\text{g}) = -393.5 \text{ kJ/mol}; \quad \Delta H_{\text{f}}\text{H}_2\text{O}(\text{l}) = -286 \text{ kJ/mol}$
(A) -277 kJ/mol (B) -1260.5 kJ/mol (C) -688.5 kJ/mol (D) -3013 kJ/mol
49. N_2 gas stored in a cylinder, fixed with a movable piston, undergoes adiabatic expansion. The statement that is true for the given situation is [NSEC-2014]
(A) $q = W$ (B) $\Delta U = W$ (C) $\Delta U = 0$ (D) $\Delta U = q$
50. The enthalpy of vaporization of benzene is $+35.3 \text{ kJ/mol}$ at its boiling point of 80°C . The entropy change in the transition of vapour to liquid at its boiling point is [NSEC-2014]
(A) -100 (B) $+100$ (C) $+342$ (D) -342
51. The process in which an ideal gas undergoes change from X to Y as shown in the following diagram is [NSEC-2015]
(A) isothermal compression (B) adiabatic compression
(C) isothermal expansion (D) adiabatic expansion
52. Glucose when dissolved in water leads to cooling of the solution. Suppose you take 250 mL water at room temperature in an open container (such as a bowl) made of thermally insulated material and dissolve a spoonful of glucose in it. If you are able to accurately measure the heat absorbed by this solution in reaching back to room temperature (assuming negligible changes in the composition and the amount of solution during this process), you will be measuring [NSEC-2015]
(A) The enthalpy of dissolution of the glucose in water
(B) The Gibbs free energy of dissolution of the glucose in Water
(C) The work done by the atmosphere on the system during the dissolution process
(D) The heat capacity of the solution





53. An ideal gas taken in an insulated chamber is released into interstellar space. The statement that is nearly true for this process is [NSEC-2015]
 (A) $Q = 0, W \neq 0$ (B) $W = 0, Q \neq 0$ (C) $\Delta U = 0, Q \neq 0$ (D) $Q = W = \Delta U = 0$
54. Standard molar enthalpy of formation of $\text{CO}_2(\text{g})$ is equal to [NSEC-2016]
 (A) Zero
 (B) The standard molar enthalpy of combustion of carbon (graphite)
 (C) The standard molar enthalpy of combustion of $\text{C}(\text{g})$
 (D) The standard molar enthalpy of combustion of $\text{CO}(\text{g})$
55. For the conversion $\text{CCl}_4(\ell) \rightarrow \text{CCl}_4(\text{g})$ at 1 bar and 350 K, the correct set of thermodynamic parameters is (Boiling point of CCl_4 is 77°C) [NSEC-2016]
 (A) $\Delta G = 0, \Delta S = +ve$ (B) $\Delta G = 0, \Delta S = -ve$ (C) $\Delta G = -ve, \Delta S = 0$ (D) $\Delta G = -ve, \Delta S = +ve$
56. Which reaction is spontaneous at all temperatures at standard pressure and concentration ? [NSEC-2016]
 (A) exothermic reaction with a decrease in entropy
 (B) exothermic reaction with an increase in entropy
 (C) endothermic reaction with a decrease in entropy
 (D) endothermic reaction with an increase in entropy
57. Given $\Delta_r H^\circ = -54.08 \text{ kJ mol}^{-1}$ and $\Delta_r S^\circ = 10.0 \text{ J mol}^{-1}$ at 25°C , the value of $\log_{10} K$ for the reaction $\text{A} \rightleftharpoons \text{B}$ is [NSEC-2016]
 (A) 3.4 (B) 10 (C) 0.53 (D) 113
58. Three samples of 100 g of water (samples I, II and III), initially kept at 1 atm pressure and 298 K were given the following treatments.
 Sample I was heated to 320 K and cooled to 298 K
 Sample II was heated to 300 K, cooled to 273 K and heated to 298 K
 Sample III was heated to 373 K and cooled to 298 K
 At the end of these processes, the internal energy of [NSEC-2017]
 (A) III is the highest (B) II is the highest
 (C) I and III are the same, II is lower than that of I and III (D) I, II and III are the same
59. For the following spontaneous process $\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{O}(\text{s})$ at 268 K, which of the following is true?
 (A) $\Delta S_{\text{sys}} < 0$ (B) $\Delta S_{\text{sys}} > 0$ (C) $\Delta S_{\text{surr}} < 0$ (D) $\Delta S_{\text{sys}} = -\Delta S_{\text{surr}}$
60. At constant volume, 6.0 mol of H_2 gas at 0°C and 100 kPa was heated to 250 kPa. The molar heat of H_2 at constant pressure (C_p) = 28.9 J mol^{-1} . (Assume that the heat capacity values do not change with temperature). The final temperature of the H_2 gas and the change in entropy of the process are [NSEC-2017]
 (A) 273°C and $113 \text{ kJ mol}^{-1} \text{ K}^{-1}$ (B) 410°C and $158.8 \text{ J mol}^{-1} \text{ K}^{-1}$
 (C) 682.5°C and $113 \text{ J mol}^{-1} \text{ K}^{-1}$ (D) 682.5 K and $113 \text{ J mol}^{-1} \text{ K}^{-1}$
61. Hydrazine used in rocket fuels can be obtained by the reaction of ammonia and hydrogen peroxide according to the following equations
 $2\text{NH}_3(\text{g}) + \text{H}_2\text{O}_2(\text{l}) \longrightarrow \text{N}_2\text{H}_4(\text{l}) + 2\text{H}_2\text{O}(\text{l})$ ($\Delta H^\circ_{\text{reaction}} = -241 \text{ kJ/mol}$)
 If ΔH° (formation) of NH_3 , H_2O_2 and H_2O are -46.1 , -187.8 and -285.8 kJ/mol respectively, ΔH° for the decomposition of hydrazine into N_2 and H_2 is [NSEC-2017]
 (A) 50.6 kJ/mol (B) 241 kJ/mol (C) -50.6 kJ/mol (D) 120.5 kJ/mol
62. At 298 K, change in internal energy for the complete combustion of fullerene, $\text{C}_{60}(\text{s})$, an allotrope of carbon, and the enthalpy of formation of $\text{CO}_2(\text{g})$ are $-25970 \text{ kJ mol}^{-1}$ and -393 kJ mol^{-1} respectively. The enthalpy of formation of $\text{C}_{60}(\text{s})$ at 298 K is [NSEC-2018]
 (A) -2390 kJ (B) $4.95 \times 10^4 \text{ kJ}$ (C) $2.60 \times 10^4 \text{ kJ}$ (D) 2390 kJ
63. The lattice enthalpy and enthalpy of solution in water for solid NaCl are 753 kJ mol^{-1} and 5 kJ mol^{-1} respectively (Fig. above). If the solution enthalpies of Na^+ and Cl^- are in the ratio 6 : 5, the enthalpy of hydration of Na^+ ion is [NSEC-2018]
 (A) 408 kJ mol^{-1} (B) -412 kJ mol^{-1} (C) -408 kJ mol^{-1} (D) -412 kJ mol^{-1}



64. For the reaction $4\text{NO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{N}_2\text{O}_5(\text{g})$, $\Delta H_{\text{reaction}} = -112 \text{ kJ}$. If the N_2O_5 is assumed to be formed in the reaction as a solid, $\Delta H_{\text{reaction}}$ will be ($\Delta H_{\text{sublimation}}$ of N_2O_5 is 54 kJ mol^{-1}) [NSEC-2018]
 (A) -220 kJ (B) -4 kJ (C) -166 kJ (D) -332 kJ
65. 2.0 L of N_2 gas kept at 25°C and 5 atm pressure were expanded isothermally against a constant pressure of 1 atm until the pressure of the gas reaches 1 atm . Assuming ideal behavior, reversible work of expansion in this process (in J) is close to [NSEC-2018]
 (A) 810 J (B) -194 kJ (C) -810 kJ (D) 3390 kJ
66. The daily energy requirement of a teenager is 7800 kJ . As calculated from the data given in the table below, the amount of glucose he has to consume (g) per day assuming that the entire energy he requires comes from the combustion of glucose is [NSEC-2018]

Molecule	$\Delta H_f (\text{kJ mol}^{-1})$
$\text{C}_6\text{H}_{12}\text{O}_6$	-1273
$\text{CO}_2(\text{g})$	-394
H_2O	-286

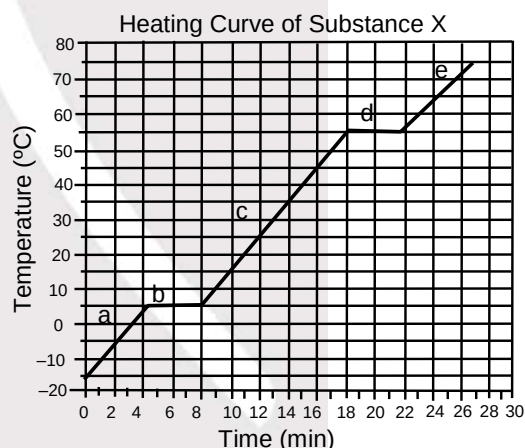
- (A) 262 (B) 500 (C) 131 (D) 250
67. The standard molar entropies of $\text{H}_2(\text{g})$, $\text{N}_2(\text{g})$ and $\text{NH}_3(\text{g})$ are 130 , 190 and $193 \text{ J mol}^{-1}\text{K}^{-1}$ respectively. For the reaction $\frac{1}{2}\text{N}_2(\text{g}) + \frac{3}{2}\text{H}_2(\text{g}) \rightleftharpoons \text{NH}_3(\text{g})$ ($\Delta H_{\text{reaction}} = -45 \text{ kJ}$) to be in equilibrium, the temperature must be equal to [NSEC-2018]
 (A) 464 K (B) 928 K (C) 737 K (D) 354 K

68. A substance X was heated at constant pressure and the temperature observed at various times of heating was plotted as given below [NSEC-2018]

Which of the following is/are correct ?

- I. Melting point of X is -5°C
 II. Solid and liquid forms of X coexist in the region b
 III. Boiling point of X is 55°C
 IV. Solid and liquid forms of X coexist in the region d

- (A) I and IV
 (B) II and III
 (C) III only
 (D) I, II and III



PART - III : HIGH LEVEL PROBLEMS (HLP)

SUBJECTIVE QUESTIONS

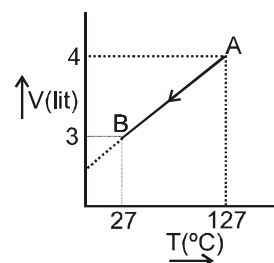
- 6 moles of an ideal gas expand isothermally and reversibly from a volume of 1 litre to a volume of 10 litres at 27°C . What is the maximum work done.
- Calculate work done by 1 mole of Ideal gas expand isothermally and irreversibly from pressure of 5 atm to 2 atm against a constant external pressure of 1 atm at 300 K temperature.



3. One mole of Ideal gas ($C_{P,m} = 15 \text{ JK}^{-1} \text{ mole}^{-1}$) follow the process as shown in figure.

Predict the following :

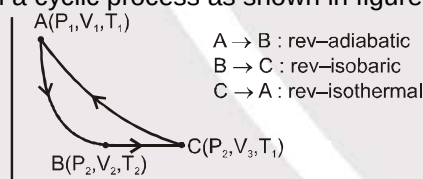
- (i) Nature of process
(ii) Calculate work done, Heat of process.



4. 1 mole of CO_2 gas at 300 K is expanded under reversible adiabatic condition such that its volume becomes 27 times.
(a) What is the final temperature ? (b) What is work done ?
Given $\gamma = 1.33$ and $C_v = 25.08 \text{ Jmol}^{-1}\text{K}^{-1}$ for CO_2 .
5. (a) A certain mass of a gas initially at (1 L, 5 atm, 300 K) is expanded reversibly and isothermally to a final volume of 5L, calculate work done by the gas and heat supplied in this process to the gas.
(b) Now, if the gas is restored to initial position by compressing it using an external constant pressure of 5 atm. Find work done on the gas in this process and heat rejected by gas
(c) In the above two processes, what is the net heat gained by surroundings?
[Note : From above question see that surroundings has done extra work on the system but system has returned that work in the form of heat to surroundings and work is considered on organized form of energy while heat as an unorganised form hence in the above process, there must be net increment in randomness of universe which will be called **Entropy**, soon.]
6. The molar heat capacity, C_v of helium gas is $3/2 R$ and is independent of temperature. For hydrogen gas, C_v approaches $3/2 R$ at very low temperatures, equal $5/2 R$ at moderate temperatures and is higher than $5/2 R$ at high temperatures. Give a reason for the temperature dependence of C_v in case of hydrogen, in not more than two or three sentences. [JEE 2003, 2/60]
7. There is 1 mol liquid (molar volume 100 ml) in an adiabatic container initial, pressure being 1 bar Now the pressure is steeply increased to 100 bar, and the volume decreased by 1 ml under constant pressure of 100 bar. Calculate ΔH and ΔE . [Given 1 bar = 10^5 N/m^2] [JEE 2004, 2/60]
8. One mole of monoatomic gas was taken through a cyclic process as shown in figure.

Calculate .

$$\sum_{ABCA} \frac{dq_{\text{rev}}}{T}$$



9. One mole of ideal monoatomic gas was taken through isochoric heating from 100 K to 1000 K. Calculate ΔS_{system} , ΔS_{surr} and ΔS_{total} in
(i) when the process carried out reversibly (ii) when the process carried out irreversibly (one step)
10. Comment on the following statements :
(a) An exothermic reaction is always thermodynamically spontaneous.
(b) Reaction with $\Delta G^\circ < 0$ always have an equilibrium constant greater than 1.
11. Following reaction occurs at 25°C :
 $2\text{NO}(\text{g}, 1 \times 10^{-5} \text{ atm}) + \text{Cl}_2(\text{g}, 1 \times 10^{-2} \text{ atm}) \rightleftharpoons 2\text{NOCl}(\text{g}, 1 \times 10^{-2} \text{ atm})$
Calculate ΔG° [$R = 8 \text{ J/mole K}$]
12. For the reaction : $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$
(i) In a mixture of 5 mol NO_2 and 5 mol N_2O_4 and pressure of 20 bar. Calculate the value of ΔG for the reaction. Given $\Delta G_f^\circ(\text{NO}_2) = 50 \text{ kJ/mol}$, $\Delta G_f^\circ(\text{N}_2\text{O}_4) = 100 \text{ kJ/mol}$ and $T = 298 \text{ K}$.
(ii) Predict the direction in which the reaction will shift, in order to attain equilibrium
[Given at $T = 298 \text{ K}$, $2.303 RT = 5.7 \text{ kJ/mol}$.] [JEE 2004, 2/60]



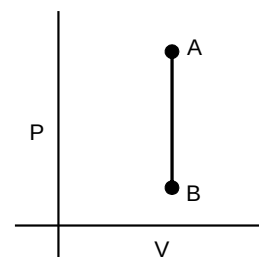
13. The difference between heats of reaction at constant pressure and at constant volume for the reaction : $2\text{C}_6\text{H}_6(\ell) + 15\text{O}_2(\text{g}) \longrightarrow 12\text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\ell)$ at 25°C in kJ is :
14. Classify the following processes as exothermic or endothermic :
 (A) Burning of a match stick (B) Melting of ice
 (C) Molten metal solidifies (D) Reaction between Na and H_2O
 (E) Rubbing alcohol evaporates.
15. Why is the enthalpy of sublimation equal to the sum of enthalpy of fusion and enthalpy of vaporisation ?
16. For a chemical reaction, ΔC_P is negative ($\Delta C_P < 0$).
 The heat required to increase temperature of reactants of this reaction by a certain amount = q_1 and heat required to increase temperature of products of the same reaction by same amount = q_2 , Relate q_1 and q_2
17. The specific heats of iodine vapour and solid are $0.031 \text{ cal g}^{-1} ^\circ\text{C}^{-1}$ and $0.055 \text{ cal g}^{-1} ^\circ\text{C}^{-1}$ respectively. If heat of sublimation of iodine is 24 cal/g at 200°C , what is its value at 250°C ?
18. Find out the heat evolved in combustion if 112 litres (at STP) of water gas (mixture of equal volume of $\text{H}_2(\text{g})$ and $\text{CO}(\text{g})$).
 $\text{H}_2(\text{g}) + 1/2 \text{O}_2(\text{g}) \longrightarrow \text{H}_2\text{O}(\text{g}) \quad \Delta H = -241.8 \text{ kJ}$
 $\text{CO}(\text{g}) + 1/2 \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g}) \quad \Delta H = -283 \text{ kJ}$
19. Substance $\text{A}_2\text{B}(\text{g})$ can undergoes decomposition to form two set of products :
 $\text{A}_2\text{B}(\text{g}) \longrightarrow \text{A}_2(\text{g}) + \text{B}(\text{g}) ; \Delta H^\circ = 40 \text{ kJ/mole}$
 $\text{A}_2\text{B}(\text{g}) \longrightarrow \text{A}(\text{g}) + \text{AB}(\text{g}) ; \Delta H^\circ = 50 \text{ kJ/mole}$
 If the molar ratio of $\text{A}_2(\text{g})$ to $\text{A}(\text{g})$ is 5 : 3 in a set of product gases, calculate the energy involved in the decomposition of 1 mole of A_2B ,
20. The ΔH°_f (KF, s) is -563 kJ mol^{-1} . The ionization enthalpy of $\text{K}(\text{g})$ is 419 kJ mol^{-1} . and the enthalpy of sublimation of potassium is 88 kJ mol^{-1} . The electron affinity of $\text{F}(\text{g})$ is 322 kJ mol^{-1} and $\text{F}-\text{F}$ bond enthalpy is 158 kJ mol^{-1} . Calculate the lattice enthalpy of $\text{KF}(\text{s})$.
 The given data are as follows :
 (i) $\text{K}(\text{s}) + 1/2 \text{F}_2(\text{g}) \rightarrow \text{KF}(\text{s}) \quad \Delta H^\circ_f = -563 \text{ kJ mol}^{-1}$
 (ii) $\text{K}(\text{g}) \rightarrow \text{K}^+(\text{g}) + \text{e}^- \quad \Delta H^\circ_{\text{ioniz}} = 419 \text{ kJ mol}^{-1}$
 (iii) $\text{K}(\text{s}) \rightarrow \text{K}(\text{g}) \quad \Delta H^\circ_{\text{sub}} = 88 \text{ kJ mol}^{-1}$
 (iv) $\text{F}(\text{g}) + \text{e}^- \rightarrow \text{F}^-(\text{g}) \quad \Delta H^\circ_{\text{eg}} = -322 \text{ kJ mol}^{-1}$
 (v) $\text{F}_2(\text{g}) \rightarrow 2\text{F}(\text{g}) \quad \Delta H^\circ_{\text{diss}} = 158 \text{ kJ mol}^{-1}$
 (vi) $\text{K}^+(\text{g}) + \text{F}^-(\text{g}) \rightarrow \text{KF}(\text{s}) \quad \Delta H^\circ_{\text{L}} = ?$

ONLY ONE OPTION CORRECT TYPE

21. In thermodynamics, a process is called reversible when -
 (A) surroundings and system change into each other
 (B) there is no boundary between system and surroundings
 (C) the surroundings are always in equilibrium with the system
 (D) the system changes into the surroundings spontaneously
22. Two moles of an ideal gas expand spontaneously in vacuum. The work done is :
 (A) 2 Joule (B) 4 Joule (C) Zero (D) Infinite
23. Temperature of 1 mole of an ideal gas is increased by 2°C at constant pressure, then magnitude of work done is :
 (A) R (B) $2R$ (C) $R/2$ (D) Data insufficient
24. Mechanical work is specially important in system that contain
 (A) Solid-liquid (B) Liquid-Liquid (C) Solid-Solid (D) Gases



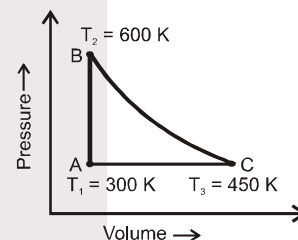
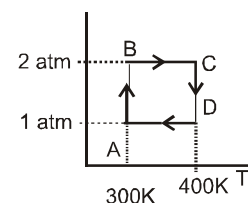
25. Which of the following statements is/are correct regarding the following process :



- (A) $dW = -P_{\text{ext}}.dv = 0$; Only if the system is undergoing only P-V work.
 (B) $dW = -P_{\text{ext}}.dv = 0$; Irrespective of any type of work done by the system.
 (C) $dU \neq dq_v$
 (D) $dW \neq 0$
26. A piece of zinc at a temperature of 20°C weighing 65.38 g is dropped into 180 g of boiling water ($T = 100^\circ\text{C}$). The specific heat of zinc is $0.4 \text{ J g}^{-1} \text{ C}^{-1}$ and that of water is $4.2 \text{ J g}^{-1} \text{ C}^{-1}$. What is the final common temperature reached by both the zinc and water ?
 (A) 97.3°C (B) 33.4°C (C) 80.1°C (D) 60.0°C
27. Which has maximum internal energy at 290 K ?
 (A) Neon gas (B) Nitrogen gas
 (C) Ozone gas (D) All have equal value
28. A sample of liquid in a thermally insulated container (a calorimeter) is stirred for 2 hr. by a mechanical linkage to a motor in the surrounding, for this process :
 (A) $w < 0$; $q = 0$; $\Delta U = 0$ (B) $w > 0$; $q > 0$; $\Delta U > 0$
 (C) $w < 0$; $q > 0$; $\Delta U = 0$ (D) $w > 0$; $q = 0$; $\Delta U > 0$
29. A balloon of 1 L volume is fill with O_2 gas, kept in a spaceship at STP. Now this spaceship is send in space & the balloon is released out of the spaceship. Then balloon expands upto bursting then calculate work done by balloon.
 (A) 0 (B) 1 (C) 2 (D) 3
30. 1 mole of an ideal diatomic gas undergoes a reversible polytropic process ($PV^2 = \text{constant}$). The gas expand from initial volume of 1 litre and temp 300 K to final volume 3 lit. Calculate change in internal energy (approx.)
 (A) -4.0 kJ (B) -4.2 kJ (C) -4.4 kJ (D) -4.6 kJ
31. A system containing real gas changes its state from state-1 to state-2.
 State-1 (2 atm, 3L, 300 K)
 State-2 (5 atm, 4L, 500 K)
 If change in internal energy = 30 L atm then calculate change in enthalpy.
 (A) 44 L atm (B) 35 L atm (C) 40 L atm (D) None of these
32. A container of volume 2L is separated into equal compartments.
 In one compartment one mole of an ideal monoatomic gas is filled at 1 bar pressure and the other compartment is completely evacuated. A pinhole is made in the separator so gas expands to occupy full 2L and heat is supplied to gas so that finally pressure of gas equals 1 bar. Then :
 (A) $\Delta U = \Delta H = 150 \text{ J}$ (B) $\Delta H = 250 \text{ J}$ (C) $\Delta U = 100 \text{ J}$ (D) $\Delta U = \Delta H = 0$
- | | |
|--------|-----|
| vacuum | 1 L |
| 1 bar | 1 L |
33. Which one of the following statement is false :
 (A) work is a state function
 (B) temperature is a state function
 (C) change in the state is completely defined when the initial and final states are specified
 (D) work appears at the boundary of the system.
34. Identify the state function among the following
 (A) q (B) $q - w$ (C) q/w (D) $q + w$



35. Two moles of Helium gas undergo a reversible cyclic process as shown in figure. Assuming gas to be ideal, what is the net work involved in the cyclic process ?
- (A) $-100 R \ln 4$ (B) $+100 R \ln 4$
(C) $+200 R \ln 4$ (D) $-200 R \ln 4$
36. Ice–Water mass ratio is maintained as 1 : 1 in a given system containing water in equilibrium with ice at constant pressure. If $C_P(\text{ice}) = C_P(\text{water}) = 4.18 \text{ J mol}^{-1} \text{ K}^{-1}$ molar heat capacity of such a system is
- (A) Zero (B) Infinity (C) $4.182 \text{ JK}^{-1} \text{ mol}^{-1}$ (D) $75.48 \text{ JK}^{-1} \text{ mol}^{-1}$
37. Two mole of an ideal gas is heated at constant pressure of one atmosphere from 27°C to 127°C . If $C_{v,m} = 20 + 10^{-2} T \text{ JK}^{-1} \text{ mol}^{-1}$, then q and ΔU for the process are respectively :
- (A) 6362.8 J , 4700 J (B) 3037.2 J , 4700 J (C) 7062.8 , 5400 J (D) 3181.4 J , 2350 J
38. The increase in internal energy of 1 kg of water at 100°C when it is converted into steam at the same temperature and at 1 atm (100 k Pa) will be [The density of water and steam are 1000 kg/m^3 & 0.6 kg/m^3 respectively. The latent heat of vapourisation of water is $2.25 \times 10^6 \text{ J/kg}$.]
- (A) $2.08 \times 10^6 \text{ J}$ (B) $4 \times 10^7 \text{ J}$ (C) $3.27 \times 10^8 \text{ J}$ (D) $5 \times 10^9 \text{ J}$
39. Consider a classroom that is roughly $5 \text{ m} \times 10 \text{ m} \times 3 \text{ m}$. Initially $t = 27^\circ\text{C}$ and $P = 1 \text{ atm}$. There are 50 people in an insulated class losing energy to the room at the average rate of 150 watt per person. How long can they remain in class if the body temperature is 42°C and person feels uncomfortable above this temperature. Molar heat capacity of air = $(7/2) R$.
- (A) 4.34 minutes (B) 5.91 minutes (C) 6.86 minutes (D) 7.79 minutes
40. A heat engine carries one mole of an ideal mono-atomic gas around the cycle as shown in the figure, the amount of heat added in the process AB and heat removed in the process CA are :
- (A) $q_{AB} = 450 R$ and $q_{CA} = -450 R$
(B) $q_{AB} = 450 R$ and $q_{CA} = -225 R$
(C) $q_{AB} = 450 R$ and $q_{CA} = -375 R$
(D) $q_{AB} = 375 R$ and $q_{CA} = -450 R$
41. There are two samples of same gas initially under similar initial state. Gases of both the samples are expanded. Ist sample using reversible isothermal process and IInd sample using reversible adiabatic process till final pressures of both the samples becomes half of initial pressure. Then :
- (A) Final volume of Ist sample < final volume of IInd sample
(B) Final volume of IInd sample < final volume of Ist sample
(C) Final volumes will be equal
(D) Information is insufficient
42. In the above problem :
- (A) work done by gas in Ist sample > work done by gas in IInd sample
(B) work done by gas in IInd sample > work done by gas in Ist sample
(C) work done by gas in Ist sample = work done by gas in IInd sample
(D) none of these
43. A new fluoro-carbon of molar mass 102 g mol^{-1} was placed in an electrically heated vessel. When the pressure was 650 torr , the liquid boiled at 77°C . After the boiling point had been reached, it was found that a current of 0.25 A from a 12.0 volt supply passed for 600 sec vaporises 1.8 g of the sample. The molar enthalpy & internal energy of vaporisation of new fluoro-carbon will be :
- (A) $\Delta H = 102 \text{ kJ/mol}$, $\Delta E = 99.1 \text{ kJ/mol}$ (B) $\Delta H = 95 \text{ kJ/mol}$, $\Delta E = 100.3 \text{ kJ/mol}$
(C) $\Delta H = 107 \text{ kJ/mol}$, $\Delta E = 105.1 \text{ kJ/mol}$ (D) $\Delta H = 92.7 \text{ kJ/mol}$, $\Delta E = 97.4 \text{ kJ/mol}$
44. Internal pressure of a perfect gas (ideal gas) is :
- (A) Zero (B) infinite
(C) Calculated by $PV = nRT$ (D) Can not say directly





45. In a laboratory, liquid in a thermally insulated container is stirred for one hr, by a mechanical linkage to a in surrounding, for this process :
 (A) $W < 0$; $q = 0$ (B) $W < 0$; $q > 0$ (C) $W < 0$; $q > 0$ (D) $W > 0$; $q = 0$
46. If W_1 ; W_2 & W_3 are magnitude of work done of an ideal gas in isothermal, adiabatic & isobaric reversible expansion process from same initial stage to same final volume. Then correct order will be :
 (A) $W_3 > W_1 > W_2$ (B) $W_3 > W_2 > W_1$ (C) $W_2 > W_1 > W_3$ (D) $W_1 > W_2 > W_3$
47. A certain mass of gas is expanded from (1L, 10 atm) to (4L, 5 atm) against a constant external pressure of 1 atm. If initial temperature of gas is 300 K and the heat capacity of process is $50 \text{ J/}^\circ\text{C}$. Then the enthalpy change during the process is (1L atm $\approx$ 100 J)
 (A) $\Delta H = 15 \text{ kJ}$ (B) $\Delta H = 15.7 \text{ kJ}$ (C) $\Delta H = 14.4 \text{ kJ}$ (D) $\Delta H = 14.7 \text{ kJ}$
48. 2 moles of ideal gas is expanded isothermally & reversibly from 1 litre to 10 litre. Find the enthalpy change in kJ mol^{-1} . [JEE 2004, 3/84]
 (A) 0 (B) 11.7 (C) - 11.7 (D) 25
49. One mole of an ideal monoatomic gas at temperature T and volume 1L expands to 2L against a constant external pressure of one atm under adiabatic conditions, then final temperature of gas will be : [JEE 2005, 3/84]
 (A) $T + \frac{2}{3 \times 0.0821}$ (B) $T - \frac{2}{3 \times 0.0821}$ (C) $\frac{T}{2^{5/3-1}}$ (D) $\frac{T}{2^{5/3+1}}$
50. The ratio of P to V at any instant is constant and is equal to 1, for a monoatomic ideal gas under going a process. What is the molar heat capacity of the gas. [JEE 2006, 3/184]
 (A) $\frac{3R}{2}$ (B) $\frac{4R}{2}$ (C) $\frac{5R}{2}$ (D) 0
51. An ideal gas expands in volume from $1 \times 10^{-3} \text{ m}^3$ to $1 \times 10^{-2} \text{ m}^3$ at 300 K against a constant pressure of $1 \times 10^5 \text{ Nm}^{-2}$. The work is : [AIEEE 2004]
 (A) - 900 J (B) - 900 kJ (C) 270 kJ (D) + 900 kJ
52. Consider the reaction, $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \longrightarrow 2\text{NH}_3(\text{g})$; carried out at constant temperature and pressure. If ΔH and ΔU are enthalpy change and internal energy change respectively, which of the following expressions is true ? [AIEEE 2005]
 (A) $\Delta H = 0$ (B) $\Delta H = \Delta U$ (C) $\Delta H < \Delta U$ (D) $\Delta H > \Delta U$
53. An ideal gas is allowed to expand both reversibly and irreversibly in an isolated system. If T_i is the initial temperature and T_f is the final temperature, which of the following statements is correct ? [AIEEE 2006]
 (A) $T_f > T_i$ for reversible process but $T_f = T_i$ for irreversible process (B) $(T_f)_{\text{rev}} = (T_f)_{\text{irrev}}$
 (C) $T_f = T_i$ for both reversible and irreversible processes (D) $(T_f)_{\text{irrev}} > (T_f)_{\text{rev}}$
54. When one mole of an ideal gas is compressed to half of its initial volume and simultaneously heated to twice its initial temperature, the change in entropy of gas (ΔS) is :
 (A) $C_{p,m} \ln 2$ (B) $C_{v,m} \ln 2$ (C) $R \ln 2$ (D) $(C_{v,m} - R) \ln 2$
55. Two mole of an ideal gas is expanded irreversibly and isothermally at 37°C until its volume is doubled and 3.41 kJ heat is absorbed from surrounding. ΔS_{total} (system + surrounding) is :
 (A) -0.52 J/K (B) 0.52 J/K (C) 22.52 J/K (D) 0
56. 1 mole of an ideal gas at 25°C is subjected to expand reversibly and adiabatically to ten times of its initial volume. Calculate the change in entropy during expansion (in $\text{J K}^{-1} \text{ mol}^{-1}$)
 (A) 19.15 (B) - 19.15 (C) 4.7 (D) zero
57. What is the change in entropy when 2.5 mole of water is heated from 27°C to 87°C ? Assume that the heat capacity is constant. ($C_{p,m}(\text{H}_2\text{O}) = 4.2 \text{ J/g-K}$ $\ln(1.2) = 0.18$)
 (A) 16.6 J/K (B) 9 J/K (C) 34.02 J/K (D) 1.89 J/K



58. $\text{C}_2\text{H}_6(\text{g}) + 3.5 \text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) + 3\text{H}_2\text{O}(\text{g})$
 $\Delta S_{\text{vap}}(\text{H}_2\text{O}, \ell) = x_1 \text{ cal K}^{-1}$ (boiling point = T_1)
 $\Delta H_f(\text{H}_2\text{O}, \ell) = x_2$
 $\Delta H_f(\text{CO}_2) = x_3$
 $\Delta H_f(\text{C}_2\text{H}_6) = x_4$
 Hence, ΔH for the reaction is -
 (A) $2x_3 + 3x_2 - x_4$ (B) $2x_3 + 3x_2 - x_4 + 3x_1T_1$
 (C) $2x_3 + 3x_2 - x_4 - 3x_1T_1$ (D) $x_1T_1 + x_2 + x_3 - x_4$
59. Consider the ΔG°_f and ΔH°_f (kJ/mol) for the following oxides. Which oxide can be most easily decomposed to form the metal and oxygen gas?
 (A) ZnO ($\Delta G^\circ = -318.4$, $\Delta H^\circ = -348.3$) (B) Cu_2O ($\Delta G^\circ = -146.0$, $\Delta H^\circ = -168.8$)
 (C) HgO ($\Delta G^\circ = -58.5$, $\Delta H^\circ = -90.8$) (D) PbO ($\Delta G^\circ = -187.9$, $\Delta H^\circ = -217.3$)
60. If a refrigerator's door is opened, then we get
 (A) Room heated (B) Room cooled
 (C) More amount of heat is passed out (D) No effect on room
61. Which of the following statement (s) is/are correct?
 Statement (i) : The entropy of isolated system with P-V work only, is always maximized at equilibrium.
 Statement (ii) : It is possible for the entropy of close system to decrease substantially in an irreversible process.
 Statement (iii) : Total entropy can be created but not destroyed.
 Statement (iv) : ΔS_{system} is zero for reversible process in an isolated system.
 (A) Statement i, ii, iii, (B) Statement ii, iv (C) Statement i, ii, iv (D) All of these
62. During winters, moisture condenses in the form of dew and can be seen on plant leaves and grass. The entropy of the system in such cases decreases as liquids possess lesser disorder as compared to gases. With reference to the second law, which statement is correct, for the above process?
 (A) The randomness of the universe decreases
 (B) The randomness of the surroundings decreases
 (C) Increase in randomness of surroundings equals to the decrease in randomness of system
 (D) The increase in randomness of the surroundings is greater as compared to the decrease in randomness of the system.
63. When $\text{CH}_3\text{OH}(\ell)$ is mixed in water (assume endothermic) :
 (A) Degree of Randomness (entropy) decreases due to formation of Hydrogen bond.
 (B) Entropy of surrounding decreases
 (C) Mixing is spontaneous and reversible
 (D) All of the above are correct
64. Although graphite is thermodynamically most stable allotrope of Carbon, but it has more entropy than diamond because:
 (A) It has layer structure which slides on each other.
 (B) In diamond Carbon is bonded with covalent bond.
 (C) In graphite, only covalent bond is present within sheet and weak vanderwaal forces are present between sheets which allow easy sliding.
 (D) Two types of bond lengths are present in graphite.
65. By how much does the entropy of universe change if 3 mole of an ideal gas is going from a pressure of 2 bar to a pressure of 1 bar without any change in temperature. If the surrounding is at 1 bar and 300 K. (Expansion is against the constant external pressure of surrounding).
 (A) $+ 7.29 \text{ J-K}^{-1}$ (B) $+ 4.82 \text{ J-K}^{-1}$ (C) $- 5.29 \text{ J-K}^{-1}$ (D) $- 8.35 \text{ J-K}^{-1}$
66. For a perfectly crystalline solid $C_{p,m} = aT^3 + bT$, where a and b constant. If $C_{p,m}$ is 0.40 J/K mol at 10 K and 0.92 J/K mol at 20 K, then molar entropy at 20 K is :
 (A) 0.92 J/K mol (B) 8.66 J/K mol (C) 0.813 J/K mol (D) None of these



67. For the hypothetical reaction $A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$
If $\Delta_r G^\circ$ and $\Delta_r S^\circ$ are 20 kJ/mol and $-20 \text{ JK}^{-1} \text{ mol}^{-1}$ respectively at 200 K.
 $\Delta_r C_p$ is $20 \text{ JK}^{-1} \text{ mol}^{-1}$ then $\Delta_r H^\circ$ at 400 K is :
(A) 20 kJ/mol (B) 7.98 kJ/mol (C) 28 kJ/mol (D) None of these
68. Given $\Delta_r S^\circ = -266$ and the listed $[S^\circ_m \text{ values}]$.
Calculate S° for $\text{Fe}_3\text{O}_4(s)$:
 $4\text{Fe}_3\text{O}_4(s) [\dots\dots\dots] + \text{O}_2(g) [205] \longrightarrow 6\text{Fe}_2\text{O}_3(s) [87]$
(A) +111.1 (B) +122.4 (C) 145.75 (D) 248.25
69. The enthalpy change for a given reaction at 298 K is $-x \text{ J mol}^{-1}$ (x being positive). If the reaction occurs spontaneously at 298 K, the entropy change at that temperature
(A) can be negative but numerically larger than $x/298$
(B) can be negative but numerically smaller than $x/298$
(C) cannot be negative
(D) cannot be positive
70. Select the correct statement(s) :
 S_1 : AlCl_3 when dissolve in H_2O its entropy decreases therefore it is a non-spontaneous process.
 S_2 : When H_2 gas adsorbed at the surface of Pd, some amount of heat is released.
 S_3 : Entropy of D_2 gas is greater than H_2 gas.
(A) S_1, S_2 & S_3 (B) S_1 & S_3 (C) S_2 & S_3 (D) S_1 & S_2
71. One mole of an ideal diatomic gas ($C_v = 5 \text{ cal}$) was transformed from initial 25°C and 1 L to the state when temperature is 100°C and volume 10 L. The entropy change of the process can be expressed as ($R = 2 \text{ calories/mol/K}$)
(A) $3 \ln \frac{298}{373} + 2 \ln 10$ (B) $5 \ln \frac{373}{298} + 2 \ln 10$
(C) $7 \ln \frac{373}{298} + 2 \ln \frac{1}{10}$ (D) $5 \ln \frac{373}{298} + 2 \ln \frac{1}{10}$
72. Combustion of sucrose is used by aerobic organisms for providing energy for the life sustaining processes. If all the capturing of energy from the reaction is done through electrical process (non P-V work) then calculate maximum available energy which can be captured by combustion of 34.2 g of sucrose.
Given : $\Delta H_{\text{combustion}}(\text{sucrose}) = -6000 \text{ kJ mol}^{-1}$
 $\Delta S_{\text{combustion}} = 180 \text{ J/K-mol}$ and body temperature is 300 K
(A) 600 kJ (B) 594.6 kJ (C) 5.4 kJ (D) 605.4 kJ
73. Given the following data (at 298 K):
- | Substance | $\Delta H^\circ (\text{kJ/mol})$ | $S^\circ (\text{J/mol K})$ | $\Delta G^\circ (\text{kJ/mol})$ |
|-----------------------|----------------------------------|----------------------------|----------------------------------|
| $\text{FeO}(s)$ | -266.3 | 57.49 | -245.12 |
| C (Graphite) | 0 | 5.74 | 0 |
| $\text{Fe}(s)$ | 0 | 27.28 | 0 |
| $\text{CO}(g)$ | -110.5 | 197.6 | -137.15 |
- Determine at what temperature the following reaction is spontaneous ? (Assume ΔH_r° and S_m° to be independent of temperature)
 $\text{FeO}(s) + \text{C (Graphite)} \longrightarrow \text{Fe}(s) + \text{CO}(g)$
(A) 298 K (B) 668 K (C) 964 K
(D) ΔG° is +ve, hence the reaction will never be spontaneous
74. $\Delta H = 30 \text{ kJ mol}^{-1}$, $\Delta S = 75 \text{ J / K / mol}$. Find boiling temperature at 1 atm. [JEE 2004, 3/84]
(A) 400 K (B) 300 K (C) 150 K (D) 425 K
75. For a spontaneous reaction under standard conditions, the ΔG , equilibrium constant (K) and E°_{cell} will be respectively [AIEEE 2005]
(A) -ve, > 1 , +ve (B) +ve, > 1 , -ve (C) -ve, < 1 , -ve (D) -ve, > 1 , -ve



76. If ΔH is the change in enthalpy and ΔU the change in internal energy accompanying a gaseous reaction
 (A) ΔH is always greater than ΔU
 (B) $\Delta H < \Delta U$ only if the number of moles of the products is greater than the number of the reactants
 (C) ΔH is always less than ΔU
 (D) $\Delta H < \Delta U$ only if the number of moles of the products is less than the number of moles of the reactants
77. In the reaction $\text{CS}_2(\ell) + 3\text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g}) + 2\text{SO}_2(\text{g})$, $\Delta H = -265 \text{ kcal}$
 The enthalpies of formation of CO_2 and SO_2 are both negative and are in the ratio 4 : 3. The enthalpy of formation of CS_2 is +26 kcal/mol. Calculate the enthalpy of formation of SO_2 .
 (A) -90 kcal/mol (B) -52 kcal/mol (C) -78 kcal/mol (D) -71.7 kcal/mol
78. Heat of hydrogenation of ethene is x_1 and that of benzene is x_2 . Hence, resonance energy is :
 (A) $x_1 - x_2$ (B) $x_1 + x_2$ (C) $3x_1 - x_2$ (D) $x_1 - 3x_2$
79. Calculate the resonance energy of gaseous benzene from the given data
 $\Delta H_{\text{C-H}} = 416$; $\Delta H_{\text{C-C}} = 331$; $\Delta H_{\text{C=C}} = 591$; $\Delta H_{\text{sub}}(\text{C, graphite}) = 718$
 $\Delta H_{\text{diss}}(\text{H}_2, \text{g}) = 436$ and $\Delta H_f(\text{benzene, g}) = 83$ (all unit in kJ mole^{-1})
 (A) 354 (B) 271 (C) 437 (D) 83
80. Born-Haber cycle is used to determine :
 (A) Lattice energy (B) Electron affinity (C) Sublimation energy (D) All of these
81. Heat of neutralization of strong acid by a strong base is a constant value due to :
 (A) salt formed does not hydrolysis
 (B) only H^+ and OH^- ions react
 (C) The strong base and strong acid react completely
 (D) The strong base and strong acid react in aqueous solution
 Heat of neutralization of strong acid by a strong base is a constant value due to :
82. If $\text{CH}_3\text{COOH} + \text{OH}^- \longrightarrow \text{CH}_3\text{COO}^- + \text{H}_2\text{O} + q_1$
 $\text{H}^+ + \text{OH}^- \longrightarrow \text{H}_2\text{O} + q_2$
 then the enthalpy change for the reaction $\text{CH}_3\text{COOH} \longrightarrow \text{CH}_3\text{COO}^- + \text{H}^+$ is equal to:
 (A) $(q_1 + q_2)$ (B) $(q_1 - q_2)$ (C) $(q_2 - q_1)$ (D) $-(q_1 + q_2)$
83. Enthalpy of neutralization of H_3PO_3 acid is -106.68 kJ/mol using NaOH . If enthalpy of neutralization of HCl by NaOH is -55.84 kJ/mol . Calculate $\Delta H_{\text{ionization}}$ of H_3PO_3 into its ions :
 (A) 50.84 kJ/mol (B) 5 kJ/mol (C) 2.5 kJ/mol (D) None of these
84. The difference between ΔH and ΔE (on a molar basis) for the combustion of n-octane (ℓ) at 25°C would be :
 (A) -13.6 kJ (B) -1.14 kJ (C) -11.15 kJ (D) $+11.15 \text{ kJ}$
85. The heat of formation of HCl at 348 K from the following data, will be
 $0.5 \text{ H}_2(\text{g}) + 0.5 \text{ Cl}_2(\text{g}) \longrightarrow \text{HCl} \quad \Delta H^\circ_{298} = -22060 \text{ cal}$
 The mean heat capacities over this temperature range are,
 $\text{H}_2(\text{g})$, $C_p = 6.82 \text{ cal mol}^{-1} \text{ K}^{-1}$; $\text{Cl}_2(\text{g})$, $C_p = 7.71 \text{ cal mol}^{-1} \text{ K}^{-1}$; $\text{HCl}(\text{g})$, $C_p = 6.81 \text{ cal mol}^{-1} \text{ K}^{-1}$
 (A) -20095 cal (B) -32758 cal (C) -37725 cal (D) -22083 cal
86. From the following data of ΔH , of the following reactions,
 $\text{C}(\text{s}) + \frac{1}{2} \text{O}_2(\text{g}) \longrightarrow \text{CO}(\text{g}) \quad \Delta H = -110 \text{ kJ}$
 $\text{C}(\text{s}) + \text{H}_2\text{O}(\text{g}) \longrightarrow \text{CO}(\text{g}) + \text{H}_2(\text{g}) \quad \Delta H = 132 \text{ kJ}$
 What is the mole composition of the mixture of steam and oxygen on being passed over coke at 1273 K , to maintain constant temperature :
 (A) 0.5 : 1 (B) 1 : 0.6 (C) 0.8 : 1 (D) 1 : 1
87. When 12.0 g of carbon reacted with limited quantity of oxygen, 57.5 kcal of heat was produced, Calculate the number of moles of CO produced ($\Delta_f H(\text{CO}_2) = -94.05 \text{ kcal}$, $\Delta_f H(\text{CO}) = -21.41 \text{ kcal}$.
 (A) 0.5 mol (B) 0.46 mol (C) 0.64 mol (D) 0.74 mol



88. The standard enthalpy of formation of FeO & Fe₂O₃ is – 65 kcal mol⁻¹ and –197 kcal mol⁻¹ respectively. A mixture of two oxides contains FeO & Fe₂O₃ in the mole ratio 2 : 1. If by oxidation, it is changed into a 1 : 2 mole ratio mixture, how much of thermal energy will be released per mole of the initial mixture ?
(A) 13.4 kcal/mole (B) 14.6 kcal/mole (C) 15.7 kcal/mole (D) 16.8 kcal/mole
89. Reactions involving gold have been of particular interest to alchemists. Consider the following reactions,

$$\text{Au(OH)}_3 + 4 \text{HCl} \longrightarrow \text{HAuCl}_4 + 3 \text{H}_2\text{O}, \quad \Delta H = -28 \text{ kcal}$$

$$\text{Au(OH)}_3 + 4 \text{HBr} \longrightarrow \text{HAuBr}_4 + 3 \text{H}_2\text{O}, \quad \Delta H = -36.8 \text{ kcal}$$
 In an experiment, there was an absorption of 0.44 kcal when one mole of HAuBr₄ was mixed with 4 moles of HCl. Then, the fraction of HAuBr₄ converted into HAuCl₄ (percentage conversion)
(A) 5% (B) 6% (C) 7% (D) 8%
90. The average Xe–F bond energy is 34 kcal/mol, first I.E. of Xe is 279 kcal/mol, electron affinity of F is 85 kcal/mol & bond dissociation energy of F₂ is 38 kcal/mol. Then, the enthalpy change for the reaction

$$\text{XeF}_4 \longrightarrow \text{Xe}^+ + \text{F}^- + \text{F}_2 + \text{F}$$
 will be
(A) 367 kcal/mole (B) 425 kcal/mole (C) 292 kcal/mole (D) 392 kcal/mole
91. Equal volumes of molar hydrochloric acid and sulphuric acid are neutralized by dil. NaOH solution and x kcal and y kcal of heat are liberated respectively. Which of the following is true ?
(A) x = y (B) $x = \frac{1}{2}y$ (C) x = 2y (D) None of these
92. A solution is 500 ml of 2 M KOH is added to 500 ml of 2 M HCl and the mixture is well shaken. The rise in temperature T₁ is noted. The experiment is then repeated using 250 ml of each solution and rise in temperature T₂ is again noted. Assume all heat is taken by the solution
(A) T₁ = T₂ (B) T₁ is 2 times as large as T₂
(C) T₂ is twice of T₁ (D) T₁ is 4 times as large as T₂
93. 50.0 mL of 0.10 M HCl is mixed with 50.0 mL of 0.10 M NaOH. The solution's temperature rises by 3.0°C. Calculate the enthalpy of neutralization per mole of HCl. (Assuming density of sol. = 1 g/ml & specific heat of water is 4.2 J/g·K) is
(A) – 2.5 × 10² kJ/mole (B) – 1.3 × 10² kJ/mole (C) – 8.4 × 10¹ kJ/mole (D) – 6.3 × 10¹ kJ/mole
94. A piece of NH₄Cl is kept inside a balloon filled with some air, having thermally insulated walls. The temperature is kept just above sublimation temperature of NH₄Cl.
- $$\text{NH}_4\text{Cl (s)} \xrightarrow{\text{sublime}} \text{NH}_3 \text{ (g)} + \text{HCl (g)}.$$
- Which of the following is correct regarding this :
(A) q_{system} = positive
(B) ΔU = negative
(C) w = zero
(D) ΔU = positive
-

95. Equal volume of HCOOH and NaOH are mixed. If x is the heat of formation of water from strong acid & strong base, then heat evolved of neutralization is :
(A) more than x (B) equal to x (C) twice of x (D) less than x
96. Which of the following equation gives the values of heat of formation (ΔH_f⁰) [JEE 2003, 3/84]
 (A) C (diamond) + O₂(g) → CO₂(g) (B) $\frac{1}{2} \text{H}_2(\text{g}) + \frac{1}{2} \text{F}_2(\text{g}) \longrightarrow \text{HF}(\text{g})$
 (C) N₂(g) + 3H₂(g) → 2NH₃(g) (D) H₂(g) + F₂(g) → 2HF(g)
97. The enthalpies of combustion of carbon and carbon monoxide are – 393.5 and – 283 kJ mol⁻¹ respectively. The enthalpy of formation of carbon monoxide per mole is [AIEEE 2004]
(A) 110.5 kJ (B) 676.5 kJ (C) – 676.5 kJ (D) – 110.5 kJ



98. If the bond dissociation energies of XY, X₂ and Y₂ (all diatomic molecules) are in the ratio of 1 : 1 : 0.5 and ΔH_f for the formation of XY is - 200 KJ mol⁻¹. The bond dissociation energy of X₂ will be [AIEEE 2005]
 (A) 100 KJ mol⁻¹ (B) 200 KJ mol⁻¹ (C) 300 KJ mol⁻¹ (D) 800 KJ mol⁻¹
99. The enthalpy changes for the following processes are listed below: [AIEEE 2006]
 $\text{Cl}_2(\text{g}) \longrightarrow 2\text{Cl}(\text{g}); 242.3 \text{ kJ mol}^{-1}$; $\text{I}_2(\text{g}) \longrightarrow 2\text{I}(\text{g}); 151.0 \text{ kJ mol}^{-1}$
 $\text{ICl}(\text{g}) \longrightarrow \text{I}(\text{g}) + \text{Cl}(\text{g}); 211.3 \text{ kJ mol}^{-1}$; $\text{I}_2(\text{s}) \longrightarrow \text{I}_2(\text{g}); 62.76 \text{ kJ mol}^{-1}$
 Given that the standard state for iodine and chlorine are I₂(s) and Cl₂(g), the standard enthalpy of formation for ICl(g) is
 (A) -16.8 kJ mol⁻¹ (B) +16.8 kJ mol⁻¹ (C) +244.8 kJ mol⁻¹ (D) -14.6 kJ mol⁻¹

MATCH THE COLUMN

100. Match column-I to column-II standard entropy in J/K-molar at 298 K

	Column-I		Column-II
(A)	C (s, diamond)	(p)	5.7
(B)	C (s, graphite)	(q)	2.37
(C)	H ₂ (g)	(r)	117.6
(D)	H (g)	(s)	130.6

SINGLE AND DOUBLE VALUE INTEGER TYPE

101. A gas expands against a variable pressure given by $P = \frac{20}{V}$ bar-lt. During expansion from volume of 1 litre to 10 litre, the gas undergoes a change in internal energy of 400 J. Heat absorbed by the gas during expansion (in kJ) :
102. A sample of 4 mole He gas is originally confined in 20 L at 270 K and then undergoes adiabatic expansion against a constant pressure of 125 mbar until the volume becomes 3 times. Calculate $|\Delta T|$ during process: (Assume ideal behaviour)
103. The molar enthalpy of vaporization of benzene at its boiling point (353K) is 29.7 KJ/mole. For how long (minute) would a 11.4 Volt source need to supply a 0.5 A current in order to vaporise 7.8 g of the sample at its boiling point ? (Round off to nearest integer)
104. At 5×10^5 bar pressure, density of diamond and graphite are 3 g/cc and 2 g/cc respectively, at certain temperature T. Find the value of $\left(\frac{\Delta U - \Delta H}{20}\right)$ for the conversion of 1 mole of graphite to 1 mole of diamond (in KJ) at temperature T. (1L.atm = 100 J)
105. 1 mole of an ideal gas A ($C_{v,m} = 3R$) and 2 mole of an ideal gas B are $\left(C_{v,m} = \frac{3}{2}R\right)$ taken in a container and expanded reversibly and adiabatically from 1 litre to 4 litre starting from initial temperature of 320 K. Find the magnitude value of $\Delta U/10R$ for the process is :
106. 10 litres of a monoatomic ideal gas at 0°C and 10 atm pressure is suddenly released to 1 atm pressure and the gas expands adiabatically against this constant pressure. The final volume (L) of the gas.
107. A system is provided 50 joule of heat and the change in internal energy during the process is 60 J. Magnitude of work done on the system is :
108. A system works in a cyclic process. It absorbs 20 calories of heat and rejects 60 J of heat during the process. The magnitude of work done (J) is [1 calorie = 4.2 J] :
109. Two moles of an ideal gas is expanded isothermally and irreversibly at 27°C from volume V₁ to 7.5 V₁ and 4.17 kJ heat is absorbed from surroundings. Determine ΔS_{sys} ? (Approximate integer in Cal)



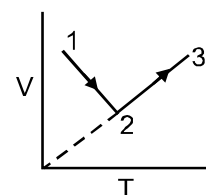
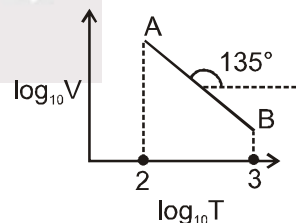
110. The vapour pressure of liquid Hg at 433 K is 5 mm Hg. Calculate the free energy change accompanying the expansion of one mole of Hg vapour in equilibrium with liquid at 433 K to a pressure of 750 mm Hg at the same temperature assuming the vapour behaves like an ideal monoatomic gas in kJ mole^{-1} . (Approximate integer and $e^5 = 150$)
111. Two mole of an ideal gas originally at a volume of 8 L. at 1000 K, is allowed to expand adiabatically until final volume is 24 L. For the gas $C_V = 1.5 R$. Calculate values of ΔS for the process when :
 (i) The expansion takes place reversibly.
 (ii) The change in volume involves a free expansion.
 Sum of ΔS of these two case is _____. (Used : $e^{1.09} = 3$ and $1.09 \times 8.314 = 9$)
112. One litre sample of a mixture of CH_4 and O_2 measured at 32°C and 760 torr, was allowed to react at constant pressure in a calorimeter. The complete combustion of CH_4 to CO_2 and water caused a temperature rise in calorimeter of 1 K. Calculate mole % of CH_4 in original mixture.
 [Given : Heat of combustion of CH_4 is -210.8 Kcal/mol . Total heat capacity of the calorimeter = 2108 cal/K]
113. 150 mL of 0.5 N HCl solution at 25°C was mixed with 150 mL of 0.5 N NaOH solution at same temperature. Calculate heat of neutralization of HCl with NaOH, if final temperature was recorded to be 29°C . ($\rho_{\text{H}_2\text{O}} = 1 \text{ g/mL}$) (Give your answer magnitude wise).
114. Calculate the resonance energy of isoprene (C_5H_8) from the data given.
 Given that $\Delta H_{\text{C}=\text{C}} = 615 \text{ kJ mole}^{-1}$; $\Delta H_{\text{C}-\text{C}} = 348 \text{ kJ mole}^{-1}$; $\Delta H_{\text{C}-\text{H}} = 413 \text{ kJ mole}^{-1}$; $\Delta H_{\text{H}-\text{H}} = 435 \text{ kJ mole}^{-1}$
 The standard heat of sublimation of graphite is 718 kJ mole^{-1} and heat of formation of $\text{C}_5\text{H}_8(\text{g})$ is 79 kJ mole^{-1}
 (Give your answer in kcal mole^{-1} ; approximate integer).
115. Calculate the amount of heat evolved during the complete combustion of 100 ml of liquid benzene from the following data. Predict your answer as $\frac{\Delta H}{100}$ (in KJ/mole).
 (i) 18 g of graphite on complete combustion evolve 585 KJ heat
 (ii) 15540 KJ heat is required to dissociate all the molecules of 1 litre water into H_2 and O_2 .
 (iii) The heat of formation of liquid benzene is 48 kJ/mol
 (iv) Density of C_6H_6 (ℓ) = 0.87 g/mL
116. The enthalpy of neutralisation of a weak acid in 1 M solution with a strong base is $-56.1 \text{ kJ mol}^{-1}$. If the enthalpy of ionization of the acid is 1.5 kJ mol^{-1} and enthalpy of neutralization of the strong acid with a strong base is $-57.3 \text{ kJ equiv}^{-1}$, what is the % ionization of the weak acid in molar solution (assume the acid to be monobasic)?
117. For reduction of ferric oxide by hydrogen, $\text{Fe}_2\text{O}_3(\text{s}) + 3 \text{H}_2(\text{g}) \longrightarrow 2 \text{Fe}(\text{s}) + 3 \text{H}_2\text{O}(\ell)$;
 $\Delta H^\circ_{300} = -26.72 \text{ kJ}$. The reaction was found to be too exothermic. To be convenient, it is desirable that ΔH° should be at the most -26 kJ . At what temperature difference it is possible ?
 $C_P[\text{Fe}_2\text{O}_3] = 105$, $C_P[\text{Fe}(\text{s})] = 25$, $C_P[\text{H}_2\text{O}(\ell)] = 75$, $C_P[\text{H}_2(\text{g})] = 30$ (all are in J/mol)
118. Calculate the magnitude of enthalpy change for 0.1 mole of NaBr in kcal, rounding it off to nearest whole number.
 $\text{Na}(\text{s}) + \frac{1}{2} \text{Br}_2(\text{g}) \longrightarrow \text{NaBr}(\text{s})$
 Given : $\Delta H_{\text{sub}}(\text{Na}) = 137 \text{ kJ mole}^{-1}$; $\Delta H_{\text{bond dissociation}}(\text{Br}_2(\text{g})) = 144 \text{ kJ mole}^{-1}$
 $\Delta H_{1\text{st ionisation}}(\text{Na}(\text{g})) = 496 \text{ kJ mole}^{-1}$; $\Delta H_{1\text{st electron gain enthalpy}}(\text{Br}(\text{g})) = -325 \text{ kJ mole}^{-1}$
 $\Delta H_{\text{Lattice energy}}(\text{NaBr}) = +742 \text{ kJ mole}^{-1}$
119. In a constant volume calorimeter, 3.5 g of a gas with molecular weight 28 was burnt in excess oxygen at 298.0 K . The temperature of the calorimeter was found to increase from 298.0 K to 298.45 K due to the combustion process. Given that the heat capacity of the calorimeter is 2.5 kJ K^{-1} , the numerical value for the enthalpy of combustion of the gas in kJ mol^{-1} is : [JEE 2009, 4/160]



120. One mole of an ideal gas is expanded isothermally at 300 K until its volume is tripled. Find the values of ΔS_{total} under the condition.
Expansion is carried out irreversibly where 900 J of heat is less absorbed than expansion is carried out reversibly. (Approximate integer value, Used $e^{1.09} = 3$ & $8.314 \times 1.09 = 9$)

ONE OR MORE THAN ONE OPTIONS CORRECT TYPE

121. Which of the following are incorrect :
(A) Work done by the surrounding in case of infinite stage expansion is more than single stage expansion
(B) Irreversible work is always greater than reversible work. (with sign)
(C) On an ideal gas in case of single stage expansion and compression system as well as surrounding are restored back to their original states
(D) If a gas in thermodynamic equilibrium is taken from state A to state B, by four successive single stage expansions, then we can plot maximum 4 points on the P-V indicator diagram.
122. Identify the following processes in which non P-V work is involved :
(A) Transfer of electron through a resistor.
(B) Stirring of liquid through mechanical agitator.
(C) Falling of an apple on Newton's head.
(D) Attraction of iron filings towards a magnet.
123. 0.5 mole each of two ideal gases A and B ($C_{v,m} = \frac{5}{2}R$) ($C_{v,m} = 3R$) are taken in a container and expanded reversibly and adiabatically, during this process temperature of gaseous mixture decreased from 350 K and 250 K. Then for the process :
(A) $\Delta U = -100 R$ (B) $\Delta U = -275 R$ (C) $\Delta H = -375 R$ (D) $\Delta H = -300 R$
124. 4 KJ of heat is absorbed when 1 mol of ice melts at 0°C and at constant pressure 1 atm. If molar volume of ice and water are 19 ml and 18 ml respectively, then [Take 1 atm = 10^5 Pa]
(A) $\Delta H > \Delta U$ (B) $\Delta U > \Delta H$ (C) $\Delta U = 4.1$ KJ (D) $\Delta U = 4.0001$ KJ
125. 5 moles of a liquid L are converted into its vapour at its boiling point (273°C) and at a pressure of 1 atm. If the value of latent heat of vapourisation of liquid L is 273 L atm/mole, then which of the following statements is/are correct: Assume volume of liquid to be negligible and vapour of the liquid to behave ideally.
(A) Work done by the system in the above process is 224 L atm.
(B) The enthalpy change (ΔH) for the above process is 1365 L atm (with respect to magnitude only)
(C) The internal energy of the system increases in the above process.
(D) The value of ΔU for the above process is 1589 L atm.
126. Three moles of an ideal diatomic gas undergoes a change in state from A to B reversibly as follows :
Which of the following statements is/are correct regarding the above process :
(A) Work done on the gas is 2700 R
(B) In the above process, pressure varies inversely with square of volume.
(C) Heat supplied to the gas is 4050 R
(D) Data insufficient for the calculation of work and heat supplied.
127. Following graph is constructed for the fixed amount of ideal gas.
(A) From 1 - 2 pressure will increase
(B) From 2 - 3 pressure remains constant
(C) Gas pressure at (3) is greater at state (1)
(D) From 1 - 2 pressure will decrease





128. Which of the following option show more entropy of first compound than second :
 (A) (i) 1-hexene at 273 K and 1 atm (ii) Cyclohexane at 273 K and 1 atm
 (B) (i) at NTP (ii) at NTP
 (C) (i) SO₂ (g) at 300 K and 0.1 atm (ii) SO₂ (g) at 300°C and 0.1 atm
 (D) (i) C₂H₅OH (ℓ) at 46°C and 1 atm (ii) CH₃OH (ℓ) at 46°C and 1 atm
129. Choose the correct statement (s) :
 (A) Temperature, enthalpy and entropy are state functions
 (B) For reversible and irreversible both isothermal expansion of an ideal gas, change in internal energy and enthalpy is zero
 (C) for a reaction in which $\Delta n_{\text{gas}} = 0$, entropy change is not always zero
 (D) The entropy change associated with reversible isothermal expansion of an ideal gas is equal to $2.303 R \log_{10} \frac{P_1}{P_2}$
130. One mole of an ideal diatomic gas ($C_v = 5 \text{ cal}$) was transformed from initial 25°C and 1 L to the state when temperature is 100°C and volume 10 L. Then for this process ($R = 2 \text{ calories/mol/K}$) (take calories as unit of energy and kelvin for temp)
 (A) $\Delta H = 525$
 (B) $\Delta S = 5 \ln \frac{373}{298} + 2 \ln 10$
 (C) $\Delta E = 525$
 (D) ΔG of the process can not be calculated using given information.
131. Select the correct statements :
 (A) All combustion reactions are exothermic (B) Heat of combustion are always exothermic
 (C) $\text{N}_2 + \text{O}_2 \longrightarrow 2\text{NO}$; $\Delta H = +ve$ (D) $\text{F}_2 + \frac{1}{2} \text{O}_2 \longrightarrow \text{F}_2\text{O}$; $\Delta H = +ve$
132. Consider the reactions
 (i) $\text{S (rhombic)} + 3/2 \text{O}_2(\text{g}) \longrightarrow \text{SO}_3(\text{g})$, ΔH_1 (ii) $\text{S (monoclinic)} + 3/2 \text{O}_2(\text{g}) \longrightarrow \text{SO}_3(\text{g})$, ΔH_2
 (iii) $\text{S (rhombic)} + \text{O}_3(\text{g}) \longrightarrow \text{SO}_3(\text{g})$, ΔH_3 (iv) $\text{S (monoclinic)} + \text{O}_3(\text{g}) \longrightarrow \text{SO}_3(\text{g})$, ΔH_4
 (A) $\Delta H_1 < \Delta H_2 < \Delta H_4$ (magnitude only) (B) $\Delta H_1 < \Delta H_3 < \Delta H_4$ (magnitude only)
 (C) $\Delta H_1 < \Delta H_2 = \Delta H_3 < \Delta H_4$ (magnitude only) (D) $\Delta H_1 + \Delta H_4 = \Delta H_2 + \Delta H_3$
133. In an insulated rigid chamber some CO₂, CO and carbon is placed and O₂ is passed in the chamber. The mixture is ignited and combustion takes place. Which of the following is/are correct :
 (A) ΔT for system > 0 (B) ΔT for system < 0 (C) $q_{\text{system}} = \text{zero}$ (D) $\Delta U = \text{zero}$
134. Which of the following is/are correct. (Consider all gases are ideal & reaction occurs at const. temperature & pressure.
 (A) $\text{CaCO}_3(\text{s}) \longrightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$ (–ve P-V work)
 (B) $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \longrightarrow 2\text{HCl}(\text{g})$ (–ve P-V work)
 (C) $\text{Zn}(\text{s}) + \text{Cu}^{2+}(\text{aq}) \longrightarrow \text{Zn}^{2+}(\text{aq}) + \text{Cu}(\text{s})$ (Non zero non P-V work)
 (D) $\text{NaCl}(\text{s}) \longrightarrow \text{Na}^+(\text{aq}) + \text{Cl}^-(\text{aq})$ (Non zero non P-V work)
135. If $\Delta H_f^\circ(\text{C}_2\text{H}_6)(\text{g}) = -85 \text{ kJ mol}^{-1}$, $\Delta H_f^\circ(\text{C}_3\text{H}_8)(\text{g}) = -104 \text{ kJ mol}^{-1}$, ΔH° for $\text{C}(\text{s}) \rightarrow \text{C}(\text{g})$ is 718 kJ mol^{-1} and heat of formation of H-atom is 218 kJ mol^{-1} then :
 (A) $\Delta H_{\text{C-C}} = 345 \text{ kJ}$ (B) $\Delta H_{\text{C-H}} = 414 \text{ kJ}$ (C) $\Delta H_{\text{H-H}} = 436 \text{ kJ}$ (D) $\Delta H_{\text{H-H}} = 218 \text{ kJ}$

COMPREHENSION

Comprehension # 1

A gaseous sample is generally allowed to do only expansion/compression type of work against its surroundings. The work done in case of an irreversible expansion (in the intermediate stages of expansion/compression the states of gases are not defined). The work done can be calculated using

$$dw = -P_{\text{ext}} dV$$



while in case of reversible process the work done can be calculated using

$dw = -PdV$ where P is pressure of gas at some intermediate stages. Like for an isothermal reversible process. Since $P = \frac{nRT}{V}$, so $w = \int dw = - \int_{V_i}^{V_f} \frac{nRT}{V} \cdot dV = -nRT \ln \left(\frac{V_f}{V_i} \right)$.

Since $dw = -PdV$ so magnitude of work done can also be calculated by calculating the area under the PV curve of the reversible process in PV diagram.

136. An ideal gaseous sample at initial state i (P_0, V_0, T_0) is allowed to expand to volume $2V_0$ using two different process; in the first process the equation of process is $PV^2 = K_1$ and in second process the equation of the process is $PV = K_2$. Then,

(A) work done in first process will be greater than work in second process (magnitude wise)
 (B) The order of values of work done can not be compared unless we know the value of K_1 and K_2 .
 (C) value of work done (magnitude) in second process is greater in above expansion irrespective of the value of K_1 and K_2 .
 (D) Ist process is not possible

137. There are two samples of same gas initially under similar initial state. Gases of both the samples are expanded. Ist sample using reversible isothermal process and IInd sample using reversible adiabatic process till final pressures of both the samples becomes half of initial pressure, then

(A) Final volume of Ist sample < final volume of IInd sample
 (B) Final volume of IInd sample < final volume of Ist sample
 (C) final volumes will be equal
 (D) Information is insufficient

138. In the above problem

(A) work done by gas in Ist sample > work done by gas in IInd sample
 (B) work done by gas in IInd sample > work done by gas in Ist sample
 (C) work done by gas in Ist sample = work done by gas in IInd sample
 (D) none of these

139. If four identical samples of an ideal gas initially at similar state (P_0, V_0, T_0) are allowed to expand to double their volumes by four different process.

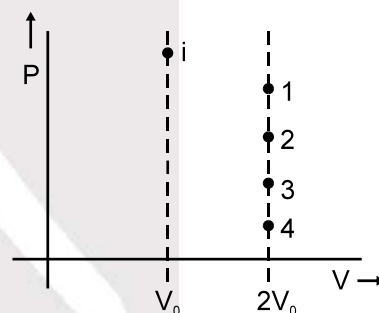
I : by isothermal irreversible process

II : by reversible process having equation $P^2V = \text{constant}$

III : by reversible adiabatic process

IV : by irreversible adiabatic expansion against constant external pressure.

Then, in the graph shown the final state is represented by four different points then, the correct match can be



(A) 1 – I, 2 – II, 3 – III, 4 – IV

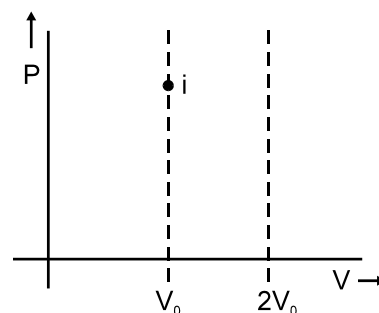
(B) 1 – II, 2 – I, 3 – IV, 4 – III

(C) 2 – III, 3 – II, 4 – I, 1 – IV

(D) 3 – II, 1 – I, 3 – IV, 4 – III

140. Two samples (initially under same states) of an ideal gas are first allowed to expand to double their volume using irreversible isothermal expansion against constant external pressure, then samples are returned back to their original volume first by reversible adiabatic process and second by reversible process having equation $PV^2 = \text{constant}$ then

(A) final temperature of both samples will be equal
 (B) final temperature of first sample will be greater than of second sample
 (C) final temperature of second sample will be greater than of first sample
 (D) none of these.



**Comprehension # 2**

Phase transitions are ubiquitous in nature. We are all familiar with the different phase of water (vapour, liquid and ice) and with the change from one to another, the change of phase are called phase transitions. There are six ways a substance can change between these three phase ; melting, freezing, evaporating, condensing ; sublimation and its reverse.

At 1 atm pressure vaporisation of 1 mole of water from liquid (75°C) to vapour (120°C).

$C_v(\text{H}_2\text{O}, \ell) = 75 \text{ J mole}^{-1} \text{ K}^{-1}$; $C_p(\text{H}_2\text{O}, \text{g}) = 33.3 \text{ J mole}^{-1} \text{ K}^{-1}$

ΔH_{vap} at 100°C = 40.7 kJ/mole.

Calculate change in internal energy when

141. Liquid water is heated from 75°C to 100°C ?
 (A) 1875 J (B) 13125 J (C) – 1875 J (D) – 13125 J
142. Liquid water at 100°C is converted to vapour at 100°C ?
 (A) 40700 J (B) 37598.878 J (C) 43801.1 J (D) 3101.2 J
143. Water vapour at heated from 100°C to 120°C ?
 (A) 666 J (B) 2998 J (C) 499.72 J (D) 166.28 J

Comprehension # 3

A gaseous mixture of propane, acetylene and CO_2 is burnt in excess of air. Total 4800 kJ heat is evolved. The total volume of $\text{CO}_2(\text{g})$ after combustion is 224 liters at NTP.

The total evolved heat is used to perform two separate process :

- (i) Vapourising 87.5% of water (liquid) obtained in the process of burning the original mixture.
 (ii) Forming 3808 liters ethylene measured at STP from its elements.

$\Delta H_{\text{H-H}} = 435 \text{ kJ/mol}$ $\Delta H_{\text{C-H}} = 416 \text{ kJ/mol}$ $\Delta H_{\text{C-C}} = 347 \text{ kJ/mol}$
 $\Delta H_{\text{C=C}} = 615 \text{ kJ/mol}$, $\Delta H_{\text{C}\equiv\text{C}} = 812 \text{ kJ/mol}$ $\Delta H_{\text{sublimation of (C,s)}} = 718 \text{ kJ/mol}$
 $\Delta H_f^\circ(\text{CO}_2, \text{g}) = -394 \text{ kJ/mol}$ $\Delta H_f^\circ(\text{H}_2\text{O}, \ell) = -286 \text{ kJ/mol}$. $\Delta H_f^\circ(\text{H}_2\text{O}, \text{g}) = -246 \text{ kJ/mol}$.

144. $\Delta H_{\text{rxn}}^\circ$ for $\text{C}_2\text{H}_2(\text{g}) + \text{H}_2(\text{g}) \longrightarrow \text{C}_2\text{H}_4(\text{g})$
 (A) –220 kJ/mol (B) –180 kJ/mol (C) –250 kJ/mol (D) –200 kJ/mol
145. Sum of enthalpies of combustion of $\text{C}_3\text{H}_8(\text{g})$ and $\text{C}_2\text{H}_2(\text{g})$ is :
 (A) –2198 kJ/mol (B) –3499 kJ/mol (C) –2798 kJ/mol (D) –3099 kJ/mol
146. Total moles of hydrocarbon gases taken in the initial mixture
 (A) 3 (B) 4 (C) 2 (D) 5

PART - IV : PRACTICE TEST-2 (IIT-JEE (ADVANCED Pattern))

Max. Time : 1 Hr.

Max. Marks : 81

Important Instructions**A. General :**

- The test is of 1 hour duration.
- The Test Booklet consists of 27 questions. The maximum marks are 81.

B. Question Paper Format

- Each part consists of five sections.



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ADVTDS - 105



4. Section-1 contains 8 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which ONE is correct.
5. Section-2 contains 6 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which ONE OR MORE THAN ONE are correct.
6. Section 3 contains 6 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9 (both inclusive).
7. Section 4 contains 2 paragraphs each describing theory, experiment and data etc. 6 questions relate to paragraph. Each question pertaining to a particular passage should have only one correct answer among the four given choices (A), (B), (C) and (D).
8. Section 5 contains 1 multiple choice questions. Question has two lists (list-1 : P, Q, R and S; List-2 : 1, 2, 3 and 4). The options for the correct match are provided as (A), (B), (C) and (D) out of which ONLY ONE is correct.

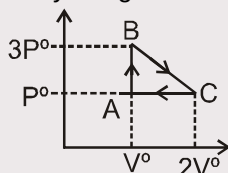
C. Marking Scheme

9. For each question in Section 1, 4 and 5 you will be awarded 3 marks if you darken the bubble corresponding to the correct answer and zero mark if no bubble is darkened. In all other cases, minus one (– 1) mark will be awarded.
10. For each question in Section 2, you will be awarded 3 marks. If you darken all the bubble(s) corresponding to the correct answer(s) and zero mark. If no bubbles are darkened. No negative marks will be answered for incorrect answer in this section.
11. For each question in Section 3, you will be awarded 3 marks if you darken only the bubble corresponding to the correct answer and zero mark if no bubble is darkened. No negative marks will be awarded for incorrect answer in this section.

SECTION-1 : (Only One option correct Type)

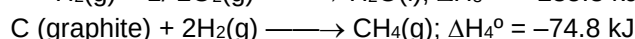
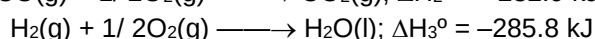
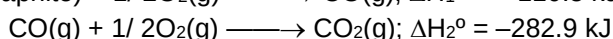
This section contains 8 multiple choice questions. Each questions has four choices (A), (B), (C) and (D) out of which Only ONE option is correct.

1. One mole of ideal monoatomic gas is carried through the reversible cyclic process as shown in figure. Calculate the max. temperature attained by the gas during the cycle.



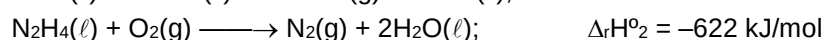
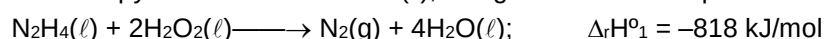
- (A) $\frac{25}{8} \left(\frac{P^0 V^0}{R} \right)$ (B) $-\frac{25}{8} \left(\frac{P^0 V^0}{R} \right)$ (C) $\frac{35}{8} \left(\frac{P^0 V^0}{R} \right)$ (D) $-\frac{35}{8} \left(\frac{P^0 V^0}{R} \right)$

2. Determine ΔU^0 at 300 K for the following reaction using the listed enthalpies of reaction :



- (A) –653.5 kJ (B) –686.2 kJ (C) –747.4 kJ (D) None of these

3. Determine enthalpy of formation for $\text{H}_2\text{O}_2(\ell)$, using the listed enthalpies of reactions :



- (A) –383 kJ/mol (B) –187 kJ/mol (C) –498 kJ/mol (D) None of these



4. The enthalpy of neutralization of a weak monoprotic acid (HA) in 1M solution with a strong base is -55.95 kJ/mol . If acid HA requires 1.4 kJ/mol heat for its complete ionization and enthalpy of neutralization of the strong monobasic acid with a strong monoacidic base is -57.3 kJ/mol . What is the % ionization of the weak acid in molar solution ?
 (A) 1% (B) 3.57% (C) 35.7% (D) 10%
5. Calculate $\Delta_f G^\circ$ for $(\text{NH}_4\text{Cl}, \text{s})$ at 310 K.
 Given : $\Delta_f H^\circ (\text{NH}_4\text{Cl}, \text{s}) = -314.5 \text{ kJ/mol}$; $\Delta_r C_p = 0$
 $S^\circ_{\text{N}_2(\text{g})} = 192 \text{ JK}^{-1} \text{ mol}^{-1}$; $S^\circ_{\text{H}_2(\text{g})} = 130.5 \text{ JK}^{-1} \text{ mol}^{-1}$;
 $S^\circ_{\text{Cl}_2(\text{g})} = 233 \text{ JK}^{-1} \text{ mol}^{-1}$; $S^\circ_{\text{NH}_4\text{Cl}(\text{s})} = 99.5 \text{ JK}^{-1} \text{ mol}^{-1}$
 All given data are at 300 K.
 (A) -198.56 kJ/mol (B) -426.7 kJ/mol (C) -202.3 kJ/mol (D) None of these
6. For a perfectly crystalline solid $C_{p,m} = aT^3$, where a is constant. If $C_{p,m}$ is 0.42 J/K-mol at 10 K, molar entropy at 10 K is
 (A) 0.42 J/K-mol (B) 0.14 J/K-mol (C) 4.2 J/K-mol (D) zero
7. The molar entropy content of 1 mole of oxygen (O_2) gas at 300 K and 1 atm is $250 \text{ J mole}^{-1} \text{ K}^{-1}$. Calculate ΔG when 1 mole of oxygen is expanded reversibly and isothermally from 300 K, 1 atm to double its volume (Take $R = 8.314 \text{ J mole}^{-1} \text{ K}^{-1}$, $\ln 10 = 2.303$)
 (A) $1.728 \text{ KJ mole}^{-1} \text{ K}^{-1}$ (B) 0
 (C) $-1.728 \text{ KJ mole}^{-1} \text{ K}^{-1}$ (D) $0.75 \text{ KJ mole}^{-1} \text{ K}^{-1}$
8. Fixed amount of an ideal mono atomic gas contained in a sealed rigid vessel ($V = 24.6 \text{ litre}$) at 1.0 bar is heated reversibly from 27°C to 127°C . Determine change in Gibb's energy (in Joule) if entropy of gas $S = 10 + 10^{-2} T \text{ (J/K)}$
 (A) 530 J (B) -530 J (C) 1778.5 J (D) 3778.5 J

Section-2 : (One or More than one options correct Type)

This section contains 6 multipole choice questions. Each questions has four choices (A), (B), (C) and (D) out of which ONE or MORE THAN ONE are correct.

9. Which of the following statement(s) is/are true?
 (A) $\Delta E = 0$ for combustion of $\text{C}_2\text{H}_6(\text{g})$ in a sealed rigid adiabatic container
 (B) $\Delta_f H^\circ (\text{S, monoclinic}) \neq 0$
 (C) If dissociation energy of $\text{CH}_4 (\text{g})$ is 1656 kJ/mol and $\text{C}_2\text{H}_6(\text{g})$ is 2812 kJ/mol , then value of C—C bond energy will be 328 kJ/mol
 (D) If $\Delta H_f(\text{H}_2\text{O}, \text{g}) = -242 \text{ kJ/mol}$; $\Delta H_{\text{vap}} (\text{H}_2\text{O}, \text{l}) = 44 \text{ kJ/mol}$ then $\Delta_f H^\circ (\text{OH}^-, \text{aq.})$ will be -142 kJ/mol
10. From the following data, mark the option(s) where ΔH is correctly written for the given reaction.
 Given : $\text{H}^+ (\text{aq}) + \text{OH}^- (\text{aq}) \longrightarrow \text{H}_2\text{O}(\text{l})$; $\Delta H = -57.3 \text{ kJ}$
 $\Delta H_{\text{solution}}$ of $\text{HA}(\text{g}) = -70.7 \text{ kJ/mol}$
 $\Delta H_{\text{solution}}$ of $\text{BOH}(\text{g}) = 20 \text{ kJ/mol}$
 $\Delta H_{\text{ionization}}$ of HA = 15 kJ/mol and BOH is a strong base.
- | Reaction | $\Delta H_r (\text{kJ/mol})$ |
|--|------------------------------|
| (A) $\text{HA}(\text{aq}) + \text{BOH}(\text{aq}) \longrightarrow \text{BA}(\text{aq}) + \text{H}_2\text{O}$ | -42.3 |
| (B) $\text{HA}(\text{g}) + \text{BOH}(\text{g}) \longrightarrow \text{BA}(\text{aq}) + \text{H}_2\text{O}$ | -93 |
| (C) $\text{HA}(\text{g}) \longrightarrow \text{H}^+(\text{aq}) + \text{A}^-(\text{aq})$ | -55.7 |
| (D) $\text{B}^+(\text{aq}) + \text{OH}^-(\text{aq}) \longrightarrow \text{BOH}(\text{aq})$ | -20 |



11. The value of $\Delta H_{\text{transition}}$ of C (graphite) $\longrightarrow$ C (diamond) is 1.9 kJ/mol at 25°C entropy of graphite is higher than entropy of diamond. This implies that :
 (A) C (diamond) is more thermodynamically stable than C (graphite) at 25°C
 (B) C (graphite) is more thermodynamically stable than C (diamond) at 25°C
 (C) diamond will provide more heat on complete combustion at 25°C
 (D) $\Delta G_{\text{transition}}$ of C (diamond) $\longrightarrow$ C (graphite) is -ve
12. Which of the following statement(s) is/are false ?
 (A) All adiabatic processes are isentropic (or isentropic) processes
 (B) When $(\Delta G_{\text{system}})_{T,P} < 0$; the reaction must be exothermic
 (C) $dG = VdP - SdT$ is applicable for closed system, both PV and non-PV work
 (D) the heat of vaporisation of water at 100°C is 40.6 kJ/mol. When 9 g of water vapour condenses to liquid at 100°C of 1 atm, then $\Delta S_{\text{system}} = 54.42 \text{ J/K}$
13. For the reaction $2\text{Ag}_2\text{O (s)} \longrightarrow 4\text{Ag(s)} + \text{O}_2 \text{ (g)}$, ΔH is 61.17 kJ mol⁻¹ and ΔS is 132 J K⁻¹ mol⁻¹. Compute the temperature above which the given reaction will be spontaneous.
 (A) $T > 463.4 \text{ K}$ (B) $T > 190.25^\circ \text{C}$ (C) $T < 190.25^\circ \text{C}$ (D) $T < 463.4 \text{ K}$
14. Select the correct enthalpy at corresponding temperature using following datas
 (i) Heat capacity of solid from 0 K to normal melting point 200 K
 $C_{P,m}(\text{s}) = 0.035 T \text{ JK}^{-1} \text{ mol}^{-1}$.
 (ii) Enthalpy of fusion = 7.5 KJ mol⁻¹,
 (iii) Enthalpy of vaporisation = 30 KJ mol⁻¹.
 (iv) Heat capacity of liquid form 200 K to normal boiling point 300 K
 $C_{P,m}(\ell) = 60 + 0.016 T \text{ JK}^{-1} \text{ mol}^{-1}$.
 (v) Heat capacity of gas from 300 K to 600 K at 1 atm
 $C_{P,m}(\text{g}) = 50.0 \text{ JK}^{-1} \text{ mol}^{-1}$.
 (A) $S_{200}(\text{s}) = 7$ (B) $S_{300}(\ell) = 70.43$ (C) $S_{300}(\text{g}) = 170.43$ (D) $S_{600}(\text{g}) = 205.09$

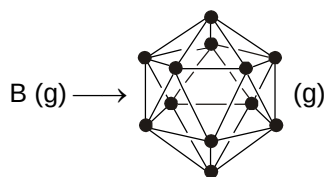
Section-3 : (One Integer Value Correct Type.)

This section contains 6 questions. Each question, when worked out will result in one integer from 0 to 9 (both inclusive)

15. Two moles of a perfect gas undergo the following processes :
 (a) a reversible isobaric expansion from (1.0 atm, 20.0 L) to (1.0 atm, 40.0 L)
 (b) a reversible isochoric change of state from (1.0 atm, 40.0 L) to (0.5 atm, 40.0 L)
 (c) a reversible isothermal compression from (0.5 atm, 40.0 L) to (1.0 atm, 20.0 L)
 Calculate the magnitude of work (W) done in L.atm [JEE 2002, 5/60]
16. The enthalpy of combustion of mol. wt. 180 glucose is - 2808 KJmol⁻¹ at 25°C. X and Y grams of glucose do you need to consume respectively cases [Assume wt = 62.5 Kg].
 (a) to climb a flight of stairs rising through 3M. (b) to climb a mountain of altitude 6000 M ?
 Assume that 25% of enthalpy can be converted to useful work.
 X and Y are related as $Y = mX$; then find m/1000.
17. A sample of certain mass of an ideal polyatomic gas is expanded against constant pressure of 1 atm adiabatically from volume 2 L, pressure 6 atm and temperature 300 K to state where its final volume is 8L. Then calculate entropy change (in J / K) in the process. (Neglect vibrational degrees of freedom) [1L atm = 100 J, log 2 = 0.3, log 3 = 0.48, ln 10 = 2.3] (approximate integer)
18. The heat of combustion of acetylene is 312 kcal. If heat of formation of CO₂ & H₂O are -94 & -68 kcal respectively. Given that heat of atomisation of C(s) & H₂(g) are 150 & 100 kcal respectively and C-H bond energy is 95 kcal. Calculate $\frac{\Delta H_{\text{C}\equiv\text{C}}}{22}$.



19. Boron exist in different allotropic forms. All allotropic form contains icosahedral units (icosahedral is a regular shape with 12 corners & 20 faces) with boron atoms at all 12 corners and all bonds are equivalent.



Calculate heat evolved at constant pressure (in kJ) per mole of boron atoms undergoing above change if $\Delta H_{BE}(B-B) = 200 \text{ kJ/mol}$. Report you Answer after dividing by 100

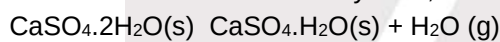
20. 1 mole of an ideal gas is allowed to expand isothermally at 27°C till its volume is tripled. If the expansion is carried out reversibly then the $\Delta S_{\text{universe}}$ will be ?

SECTION-4 : Comprehension Type (Only One options correct)

This section contains 1 paragraphs, each describing theory, experiments, data etc. 3 questions relate to the paragraph. Each question has only one correct answer among the four given options (A), (B), (C) and (D)

Paragraph for Questions 21 to 23

Concrete is produced from a mixture of cement, water, sand and small stones. It consists primarily of calcium silicates and calcium aluminates formed by heating and grinding of clay and limestone. In later steps of cement production a small amount of gypsum, $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ is added to improve subsequent hardening of concrete. The use of elevated temperatures during the final production may lead to formation of unwanted hemihydrate, $\text{CaSO}_4 \cdot 1/2\text{H}_2\text{O}$. Consider the following reaction :



The following thermodynamic data apply at 27°C , standard pressure : 1 bar

Compound	$\Delta H_f^\circ (\text{kJ/mol})$	$S^\circ (\text{JK}^{-1}\text{mol}^{-1})$
$\text{CaSO}_4 \cdot 2\text{H}_2\text{O} (s)$	-2021.0	194.0
$\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}(s)$	-1575.0	119.5
$\text{H}_2\text{O}(g)$	-242.8	188

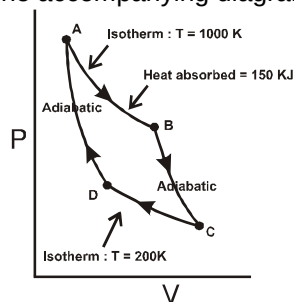
$$R = 8.314 \text{ JK}^{-1} \text{ mol}^{-1}$$

21. ΔH° for the formation of 1.45 kg of $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}(s)$ from $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}(s)$ is
 (A) +446 kJ (B) -830 kJ (C) -446 kJ (D) +818 kJ
22. Equilibrium pressure (in bar) of water vapour in closed vessel containing $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}(s)$, $\text{CaSO}_4 \cdot \text{H}_2\text{O}(s)$, $\text{H}_2\text{O}(g)$ at 27°C : ($2.303R \times 300 \times 1.5 = 8645.83$, $\log 5.5 = 0.74$)
 (A) 15×10^{-4} bar (B) 2×10^{-4} bar (C) 5.5×10^{-3} bar (D) 7.00×10^{-4} bar
23. Approximate temperature at which the equilibrium water vapour pressure is 1.00 bar
 (A) 121°C (B) 400°C (C) 200°C (D) 240°C



Paragraph for Questions 24 to 26

The accompanying diagram represents a reversible cannot cycle for an ideal gas :



24. How much heat is rejected at the lower temperature, 200 K, during the isothermal compression ?
 (A) 150 kJ (B) 30 kJ (C) 120 kJ
 (D) data are not sufficient to calculate exact value
25. What is the entropy increase during isothermal expansion at 1000 K ?
 (A) 0.15 JK^{-1} (B) 150 kJ K^{-1} (C) 150 JK^{-1} (D) 750 JK^{-1}
26. What is the Gibbs free energy change during the process $A \rightarrow B$?
 (A) 150 kJ (B) -150 kJ (C) 30 kJ (D) Data are not sufficient

SECTION-5 : Matching List Type (Only One options correct)

This section contains 1 questions, each having two matching lists. Choices for the correct combination of elements from List-I and List-II are given as options (A), (B), (C) and (D) out of which one is correct

27. Match column-I to column-II standard entropy in kJ/K-molar at 25°C

	Column-I		Column-II
1.	ΔH_{C-C}	(p)	733.48
2.	ΔH_{C-H}	(q)	97.81
3.	$\Delta H_{C=C}$	(r)	434.3
4.	$\Delta H_{C\equiv C}$	(s)	454.64
5.	$\Delta H_{C=O}$	(t)	804.22

Using the data (all values are in kJ/mol at 25°C) given below :

$\Delta H^\circ_{\text{combustion}}$ (ethane) = -1559.8	;	$\Delta H^\circ_{\text{combustion}}$ (ethene) = -1410.9
$\Delta H^\circ_{\text{combustion}}$ (acetylene) = -1299.7	;	$\Delta H^\circ_{\text{combustion}}$ (acetaldehyde) = -1192.3
ΔH°_f $\text{CO}_2(\text{g})$ = -393.5	;	ΔH°_f of $\text{H}_2\text{O}(\ell)$ = -285.8
ΔH° for $\text{C}_{(\text{s})}$ (graphite) $\rightarrow \text{C}_{(\text{g})}$ = 716.68	;	Bond energy of H-H = 435.94
Bond energy of O=O = 498.94		

	1	2	3	4	5		1	2	3	4	5
(A)	q	s	r	p	t	(B)	r	p	t	q	s
(C)	q	p	s	r	t	(D)	p	s	q	r	t

Practice Test-2 (IIT-JEE (ADVANCED Pattern))

OBJECTIVE RESPONSE SHEET (ORS)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.										
Que.	11	12	13	14	15	16	17	18	19	20
Ans.										
Que.	21	22	23	24	25	26	27			
Ans.										



APSP Answers

PART – I

- | | | | | |
|---------|---------|---------|---------|---------|
| 1. (3) | 2. (3) | 3. (4) | 4. (2) | 5. (2) |
| 6. (1) | 7. (3) | 8. (1) | 9. (3) | 10. (3) |
| 11. (3) | 12. (3) | 13. (2) | 14. (1) | 15. (2) |
| 16. (1) | 17. (1) | 18. (1) | 19. (2) | 20. (2) |
| 21. (1) | 22. (2) | 23. (4) | 24. (2) | 25. (2) |
| 26. (4) | 27. (1) | 28. (1) | 29. (2) | 30. (1) |

PART – II

- | | | | | |
|---------------------------------|---------|---------|---------|---------|
| 1. (A) | 2. (B) | 3. (B) | 4. (D) | 5. (B) |
| 6. (A) | 7. (B) | 8. (C) | 9. (C) | 10. (B) |
| 11. (B) | 12. (B) | 13. (A) | 14. (A) | 15. (B) |
| 16. (A) | 17. (C) | 18. (C) | 19. (C) | 20. (C) |
| 21. (B) | 22. (D) | 23. (D) | 24. (D) | 25. (D) |
| 26. (C) | 27. (B) | 28. (C) | 29. (C) | 30. (D) |
| 31. (D) | 32. (A) | 33. (D) | 34. (C) | 35. (B) |
| 36. (B) | 37. (C) | 38. (D) | 39. (D) | 40. (C) |
| 41. (B) | 42. (A) | 43. (D) | 44. (C) | 45. (A) |
| 46. (A) | 47. (B) | 48. (A) | 49. (B) | 50. (A) |
| 51. (B) | 52. (A) | 53. (D) | 54. (B) | 55. (A) |
| 56. (B) | 57. (B) | 58. (D) | 59. (A) | 60. (D) |
| 61. (C) | 62. (D) | 63. (C) | 64. (A) | |
| 65. (All options are incorrect) | | 66. (B) | 67. (A) | 68. (B) |

PART – III

- | | |
|---|--|
| 1. -34.465 kJ | 2. -0.7478 kJ |
| 3. (i) Isobaric | (ii) $w = 830.865 \text{ J}$, $q = -1500 \text{ J}$ |
| 4. $T_2 = 100 \text{ K}$, $w = -5.016 \text{ KJ}$ | 5. (a) -816 J (b) +2026 J (c) 1210 J |
| 6. Hydrogen is diatomic so at high temperature rotational and vibrational motion also counts. | |
| 7. $\Delta H = 990 \text{ J}$, $\Delta E = 10 \text{ J}$ | 8. Zero |
| 9. (i) Rev. Process $\Delta S_{\text{system}} = \frac{3}{2} R \ln 10$; $\Delta S_{\text{surr}} = -\frac{3}{2} R \ln 10$; $\Delta S_{\text{Total}} = 0$ | |
| (ii) Irr. Process $\Delta S_{\text{system}} = \frac{3}{2} R \ln 10$; $\Delta S_{\text{surr}} = -\frac{3}{2} R (0.9)$; $\Delta S_{\text{Total}} = \frac{3}{2} R (1.403)$ | |
| 10. (a) When $\Delta H = -ve$ and $T\Delta S = -ve$, also $T\Delta S > \Delta H$ then $\Delta G = +ve$ in such case, the reaction will not be spontaneous. | |
| (b) $-\Delta G^\circ = 2.303 RT \log K_c$. | |
| If ΔG° is negative (not to say that $\Delta G < 0$) then K_c will be +ve and will have value greater than one. | |

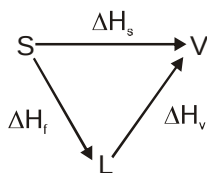


11. -43.92 KJ

12. (i) $\Delta G = 5.7 \text{ kJ/mol}$ (ii) backward shifting

13. -7.432 kJ

14. Exothermic A, C, D; endothermic : B, E



15. Since, sublimation involves

According to Hess's law, $\Delta H_{(f)} + \Delta H_{(v)} = \Delta H_{(s)}$

- | | | | | |
|--|---------------------------|---|----------------------------|-------------|
| 16. $q_1 > q_2$ | 17. 22.8 cal/g. | 18. 1312 kJ | 19. 43.73 kJ/mol | |
| 20. -827 kJ mol^{-1} | 21. (C) | 22. (C) | 23. (B) | 24. (D) |
| 25. (A) | 26. (A) | 27. (C) | 28. (D) | 29. (A) |
| 30. (B) | 31. (A) | 32. (B) | 33. (A) | 34. (D) |
| 35. (A) | 36. (B) | 37. (A) | 38. (A) | 39. (B) |
| 40. (C) | 41. (B) | 42. (A) | 43. (A) | 44. (A) |
| 45. (D) | 46. (A) | 47. (B) | 48. (A) | 49. (B) |
| 50. (B) | 51. (A) | 52. (C) | 53. (D) | 54. (D) |
| 55. (B) | 56. (D) | 57. (C) | 58. (B) | 59. (C) |
| 60. (A) | 61. (D) | 62. (D) | 63. (B) | 64. (C) |
| 65. (B) | 66. (C) | 67. (A) | 68. (C) | 69. (B) |
| 70. (C) | 71. (B) | 72. (D) | 73. (C) | 74. (A) |
| 75. (A) | 76. (D) | 77. (D) | 78. (C) | 79. (B) |
| 80. (D) | 81. (B) | 82. (C) | 83. (B) | 84. (C) |
| 85. (D) | 86. (B) | 87. (A) | 88. (A) | 89. (A) |
| 90. (C) | 91. (B) | 92. (A) | 93. (A) | 94. (B) |
| 95. (D) | 96. (B) | 97. (D) | 98. (D) | 99. (B) |
| 100. $(A - q), (B - p), (C - s), (D - r)$ | | 101. 5 | 102. 10 | 103. 9 |
| 104. 5 | 105. 96 | 106. 64 L | 107. 10 | 108. 24 |
| 109. $\Delta S_{\text{निकाय}} = 8 \text{ Cal}$ | | 110. 18 | 111. 18 | 112. 25%. |
| 113. -16 Kcal. | 114. 5 | 115. 36 kJ | 116. 20 % | 117. 9 |
| 118. 9 | 119. 9 | 120. $\Delta S_{\text{total}} = 3 \text{ J K}^{-1}$ | 121. (ACD) | 122. (ABCD) |
| 123. (BC) | 124. (BD) | 125. (ABC) | 126. (ABC) | 127. (ABC) |
| 128. (AD) | 129. (ABCD) | 130. (ABD) | 131. (ABCD) | 132. (ABD) |
| 133. (ACD) | 134. (ACD) | 135. (ABC) | 136. (C) | 137. (B) |
| 138. (A) | 139. (B) | 140. (C) | 141. (A) | 142. (B) |
| 143. (C) | 144. (A) | 145. (B) | 146. (A) | |



PART – IV

- | | | | | |
|---|------------|----------|------------|-----------|
| 1. (A) | 2. (D) | 3. (B) | 4. (B) | 5. (A) |
| 6. (B) | 7. (C) | 8. (B) | 9. (ABC) | 10. (ABC) |
| 11. (BCD) | 12. (ABCD) | 13. (AB) | 14. (ABCD) | |
| 15. $(-20 + 0 + 14) = -6 \text{ L atm}$ | 16. 2 | 17. 3 | 18. 7 | |
| 19. 5 | 20. 0 | 21. (D) | 22. (C) | 23. (A) |
| 24. (B) | 25. (C) | 26. (B) | 27. (A) | |

APSP Solutions

PART – I

6. $\Delta H_{\text{solution}} = \frac{(125 + 8) \times 4.2 \times 6}{8} \times 80 = 33516 \text{ J/mol or } 33.51 \text{ kJ/mol.}$
7. $\Delta_r C_p = 0, \therefore \Delta H_{300} = \Delta H_{310}$
8. $\text{C}_2\text{H}_5\text{OH}(\ell) + 3\text{O}_2(\text{g}) \longrightarrow 2\text{CO}_2(\text{g}) + 3\text{H}_2\text{O}(\ell)$
 $\Delta n_g = 2 - 3 = -1$
 so $\Delta U = \Delta H - \Delta n_g RT$
 $= -q + RT$
9. $\text{H}_2\text{C}_2\text{O}_4(\ell) + \frac{1}{2}\text{O}_2(\text{g}) \longrightarrow \text{H}_2\text{O}(\ell) + 2\text{CO}_2(\text{g});$
 $\Delta n_g = 3/2$
 $\Delta U_c = -\frac{0.312 \times 8.75}{1} \times 90 = -245.7 \text{ kJ/mol}$
 $\Delta H = \Delta U + \Delta n_g RT$
 $= -245.7 + \frac{3}{2} \times \frac{8.314 \times 300}{1000} = -241.947 \text{ kJ/mol.}$
10. From given reaction $\Delta n_g = 12 - 15 = -3$
 so $\Delta E^\circ = \Delta H^\circ - \Delta n_g RT = -6542 + 3RT$
 for 1.5 mole, $\Delta E^\circ = \frac{1.5}{2} \{-6542 + 3RT\} = 4900.9 \text{ kJ}$
11. $\Delta n_g = 1 - 2 = -1$
 $\Delta E = \Delta H - \Delta n_g RT = \Delta H - RT = -72.3 + 8.314 \times 300 \times 10^{-3} = -69.806 \text{ kJ/mole}$
 so for 3 mole we will get $\Delta E = -69.806 \times 3 \text{ kJ/mole} = 209.42 \text{ kJ/mole}$
12. For same amount of gas at constant temperature, lesser is the volume, lower will be the entropy.
13. $W = -P_{\text{ext}}(\Delta V) = -(1 \text{ atm}) \times 500 \times 50 \times 10^{-3} \text{ L}$
 $= -25 \text{ L atm} = -25 \times 101.3 \times 10^{-3} \text{ kJ} = -2.53 \text{ kJ}$
14. $\text{C}_2\text{H}_4 + \text{H}_2 \longrightarrow \text{C}_2\text{H}_6$
 50 ml 50 ml 0
 X X 50 ml
 $\Delta H = \Delta U + P(\Delta V)$
 $-0.31 = \Delta U + 1.5 \times 1.01 \times 10^5 (-50 \times 10^{-6}).$
 $\Delta U = -0.3024 \text{ kJ.}$



15. $\text{H}_2\text{O (s)} \longrightarrow \text{H}_2\text{O (l)}, \quad \Delta H = 1440 \text{ cal.}$
16. $\Delta E = q + w$
 $W_{BC} = \frac{1}{2} (2V^0 - V^0) (P^0 - 3P^0) + (2V^0 - V^0) (0 - P^0) = -2P^0V^0$
 $\Delta E = nC_V\Delta T = 1 \times \frac{3}{2} R \left(\frac{P_0 2V_0}{R} - \frac{3P_0 V_0}{R} \right) = -\frac{3}{2} P^0V^0$
 $q_{BC} = \Delta E - W = -\frac{3}{2} P^0V^0 + 2P^0V^0 = \frac{1}{2} P^0V^0$
17. Beaker is open and in open beaker external pressure remains constant.
 $\text{Zn} + \text{H}_2\text{SO}_4 \longrightarrow \text{ZnSO}_4 + \text{H}_2 (\text{g})$
 $\frac{130}{65} = 2 \text{ mole}$
 $PV = nRT$
 $P\Delta V = n_gRT$
 $W = -n_gRT = 2 \times 2 \times 300 = -1200 \text{ cal.}$
18. $\text{C}_3\text{H}_8(\text{g}) + 5\text{O}_2 \rightarrow 3\text{CO}_2(\text{g}) + 4\text{H}_2\text{O}(\ell)$

$$\Delta_c H = \left[\begin{array}{l} 8 \times \text{B.E. (C—H)} \\ +2 \times \text{B.E. (C—C)} \\ +5 \times \text{B.E. (O=O)} \end{array} \right] - \left[\begin{array}{l} 6 \times \text{B.E. (C=O)} \\ +8 \times \text{B.E. (O—H)} \\ +3 \times |\text{R.E.}| \text{ of } \text{CO}_2 \\ +4 \times \Delta_{\text{vap}} \text{H(H}_2\text{O)} \end{array} \right]$$
19. $\text{H}_2 (\text{g}) + \frac{1}{2} \text{O}_2 (\text{g}) \longrightarrow \text{H}_2\text{O (l)} ; \quad \Delta H_{\text{comb H}}$
 $\Delta H_{\text{comb}} = \Delta H_{\text{H—H}} + \frac{1}{2} \Delta H_{\text{O=O}} - 2\Delta H_{\text{O—H}} - \Delta H_{\text{vap}} = x_1 + \frac{x_2}{2} - 2x_3 - x_4$
20. $\frac{1}{2} \text{N}_2(\text{g}) + \frac{3}{2} \text{Cl}_2(\text{g}) \longrightarrow \text{NCl}_3(\text{g}) ; \quad \Delta H = \Delta H_1 + \frac{\Delta H_2}{2} - \frac{3}{2} \Delta H_3$
21. $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \longrightarrow \text{H}_2\text{O}(\ell)$
 so $\Delta H^\circ = -57.3 = -285.5 - \Delta H_f^\circ (\text{H}^+, \text{aq}) - \Delta H_f^\circ (\text{OH}^-, \text{aq})$
 so $\Delta H_f^\circ (\text{OH}^-, \text{aq}) = -228.5 \text{ kJ/mole}$ (as $\Delta H_f^\circ (\text{H}^+, \text{aq}) = 0$)
22. $\Delta_r H = \frac{8}{9} \times 45.54 + \frac{1}{9} \times 68.91 = 48.137 \text{ kJ.}$
23. $\text{C}_2\text{H}_5\text{—S—C}_2\text{H}_5 + \text{S} \longrightarrow \text{C}_2\text{H}_5\text{—S—S—C}_2\text{H}_5$
 $\Delta H_f^\circ (\text{C}_2\text{H}_5\text{—S—C}_2\text{H}_5) = \Delta H_f^\circ (\text{C}_2\text{H}_5\text{—S—S—C}_2\text{H}_5) - \Delta H_{\text{S—S}} + \Delta H_{\text{sub S}}$
 $-201.9 = -147.2 - \Delta H_{\text{S—S}} + 222.8$
 $\Delta H_{\text{S—S}} = 277.5 \text{ kJ/mol}$
24. $\Delta H^\circ = 2 \times \Delta H_1^\circ + 2 \times \Delta H_2^\circ - \Delta H_3^\circ$
25. $\Delta G = \Delta H - T\Delta S = -2808 - 310 \times 182.4 \times 10^{-3} = -2864.5 \text{ kJ}$
26. $\Delta S = 10.13 = 31.2 + 51.1 - 47.3 - S_{\text{H}_2\text{O}}$
 $S_{\text{H}_2\text{O}} = 45.13 \text{ cal/k mole.}$
27. $\Delta G = 29.3 \times 10^3 - (2 \times 239.7 - 152.3 - 223) \times 298 = -1721.8 \text{ Joule.}$



28. Since, expansion occurred at constant temperature,

$$\Delta S = nR \ln \frac{V_2}{V_1} = \frac{1}{32} \times 8.314 \ln \frac{3.0}{0.75} = 0.36 \text{ JK}^{-1}$$

Since, this is case of free expansion, $P_{\text{ext}} = 0$. $\Rightarrow -W = P_{\text{ext}} \Delta V = 0$, $q = 0$
Also, since, $\Delta T = 0 \Rightarrow \Delta H = \Delta E = 0$.

29. (2) $\text{Cu}_2\text{O(s)} + \frac{1}{2} \text{O}_2\text{(g)} \rightleftharpoons 2\text{CuO(s)}$

$$\Delta G^\circ_{\text{reaction}} = [2 \times (-30.4)] - [-34.98] = -25.82 \text{ kcal}$$

and $-25.82 \times 10^3 = 2.303 \times 2 \times 298 \log K$

$\therefore K 10^{19}$, a very high value, hence reaction will be almost complete with a trace of Cu_2O .

30. $\Delta_f S^\circ(\text{NH}_4\text{Cl, s})$ at 300 K

$$= S^\circ_{\text{NH}_4\text{Cl(s)}} - \left[\frac{1}{2} S^\circ_{\text{N}_2} + 2S^\circ_{\text{H}_2} + \frac{1}{2} S^\circ_{\text{Cl}_2} \right] = -374 \text{ JK}^{-1} \text{ mol}^{-1}$$

$$\Delta_f C_p = 0$$

$$\therefore \Delta_f S^\circ_{310} = \Delta_f S^\circ_{300} = -374 \text{ JK}^{-1} \text{ mol}^{-1}$$

$$\Delta_f H^\circ_{310} = \Delta_f H^\circ_{300} = -314.5$$

$$\Delta_f G^\circ_{310} = \Delta_f H^\circ - 310$$

$$\Delta S^\circ = -314.5 - \frac{310(-374)}{1000} = -198.56 \text{ kJ/mol.}$$

PART - III

1. $W = -2.303 nRT \log \frac{V_2}{V_1} = -2.303 \times 6 \times 8.314 \times 300 \log \frac{10}{1} = -34464.8 \text{ Joule} = 34.465 \text{ kJ}$

2. $\int dW = - \int P_{\text{ext}} dv$

$$W_{\text{irr}} = -P_{\text{ext}} [V_2 - V_1] = -P_{\text{ext}} \left(\frac{nRT}{P_2} - \frac{nRT}{P_1} \right) = -P_{\text{ext}} \times nRT \left(\frac{1}{P_2} - \frac{1}{P_1} \right)$$

$$= -1 \times 1 \times .082 \times 300 \left(\frac{1}{2} - \frac{1}{5} \right) = -1 \times .082 \times 300 \times \frac{3}{10} = -7.38 \text{ L.atm} = -747.8 \text{ J}$$

$$W_{\text{irr}} = -0.7478 \text{ KJ}$$

3. (i) $\left(\frac{Pv}{nRT} \right)_A = \left(\frac{Pv}{nRT} \right)_B$

$$\frac{P_A \times 4}{nR \times 400} = \frac{P_B \times 3}{nR \times 300} = \frac{P_A}{100} = \frac{P_B}{100} \Rightarrow P_A = P_B \text{ Hence process is isobaric.}$$

$$(ii) W = -P_{\text{ext}} (V_2 - V_1) = - \left(\frac{nRT_A}{V_A} \right) (V_B - V_A) = - \left(\frac{1 \times .082 \times 300}{3} \right) (3 - 4) \times 101.325 \text{ J} = 830.865 \text{ J}$$

$$q_{AB} = nC_p (T_2 - T_1) = 1 \times 15 \times (300 - 400) = -150 \text{ J}$$

4. (a) $T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$

$$300 \times V^{1/3} = T_2 \times (27 V)^{1/3} \Rightarrow T_2 = 100 \text{ K}$$

- (b) $\Delta E = q + W$

For reversible adiabatic process,

$$q = 0$$

$$\Delta U = W$$

$$\Rightarrow W = n C_v \Delta T$$

$$\Rightarrow W = 1 \times 25.08 \times (100 - 300) = -5.016 \text{ kJ}$$



$$5. \quad PV = nRT \quad \Rightarrow \quad 5 \times 1 = n \times 0.082 \times 300 \quad \Rightarrow \quad n = 0.203.$$

	Initial	Final
(a)	$V = 1 \text{ L.}$ $P = 5 \text{ atm}$ $T = 300 \text{ K}$	$V_2 = 5 \text{ L.}$ $T = 300 \text{ K}$

$$w = -2.303 nRT \log \frac{V_2}{V_1} = -2.303 \times 0.203 \times 8.314 \times 300 \log 5 = -816 \text{ J}$$

$$w = -816 \text{ J and } q = -w = 816 \text{ J}$$

So, work done by the gas = 816 J.

$$(b) \quad w = -P_{\text{ext}} (V_2 - V_1) = -5 [1-5] = +20 \times 101.325$$

$$w = 2026 \text{ J.}$$

$$(c) \quad q_1 = -w_1 = +816 \text{ J}$$

$$q_2 = -w_2 = -2026 \text{ J}$$

$$q_1 + q_2 = -1210 \text{ J}$$

So, Net heat released by the system = -1210

Net Heat gained by the surroundings = 1210

6. Helium (He) gas is monoatomic and it has three translational degree of freedom. Hence, the contribution of each translational degree of freedom towards C_v being $R/2$, so the **total contribution towards $C_v = 3 \times R / 2$** . Hydrogen molecule is diatomic. At low temperature, rotational and vibrational contribution for H_2 are zero. So, C_v for H_2 at low temperature continues to be $3R/2$. At moderate temperature, rotational contribution ($= 2 \times R / 2$) also becomes dominant and hence total contribution towards $C_v = \frac{3R}{2} + R = \frac{5R}{2}$. At even high temperature, vibrational contribution ($= 1 \times R$) also becomes

$$\text{significant. Hence total contribution towards } C_v = \frac{3R}{2} + R + R = \frac{7R}{2}$$

$$7. \quad \Delta U = q + W$$

for adiabatic process $q = 0$, hence $\Delta U = W$ and $W = -p(\Delta V) = -P (V_2 - V_1)$

$$\text{so, } \Delta U = -100 (99 - 100) = -100 (-1) = 100 \text{ bar mL} = 10 \text{ J}$$

$$\text{Now } \Delta H = \Delta U + \Delta (PV)$$

Here ΔU already calculated above and

$$\Delta PV = (P_2 V_2 - P_1 V_1)$$

$$\text{So, } \Delta H = 100 + (100 \times 99 - 1 \times 100) = 9900 \text{ bar mL} = 990 \text{ J}$$

$$8. \quad dS = \frac{dq_{\text{rev.}}}{T} \quad \Rightarrow \quad \sum \frac{dq_{\text{rev.}}}{T} = \Delta S$$

As 'S' being a state function, $\Delta S = 0$ in a cyclic process.

$$\therefore \sum \frac{dq_{\text{rev.}}}{T} = 0.$$

$$9. \quad (i) \quad \Delta S_{\text{system}} = nC_v \ln \frac{T_2}{T_1} = \frac{3}{2} R \ln 10 \text{ and}$$

$$\Delta S_{\text{surr}} = \int -\frac{dq_{\text{sys}}}{T} = \int_{T_1}^{T_2} -\frac{nC_v dT}{T} = -nC_v \ln \frac{T_2}{T_1} = -\frac{3}{2} R \ln 10$$

(ii) Entropy is a state function.

$$\Delta S_{\text{system}} = \frac{3}{2} R \ln 10.$$

$$\Delta S_{\text{surr}} = \frac{q}{T} = -\frac{nC_v (T_2 - T_1)}{T_2} = -\frac{3}{2} R (0.9).$$

$$\Delta S_{\text{Total}} = \frac{3}{2} R \ln 10 - \frac{3}{2} R (0.9) = \frac{3}{2} R (1.403)$$



10. (a) When $\Delta H = -ve$ and $T\Delta S = -ve$, also $T\Delta S > \Delta H$ then $\Delta G = +ve$ in such case, the reaction will not be spontaneous.
 (b) $-\Delta G^\circ = 2.303 RT \log K_c$.
 If ΔG° is negative (not to say that $\Delta G < 0$) then K_c will be +ve and will have value greater than one.
11. $2NO (9, 1 \times 10^{-5} \text{ atm}) + Cl_2 (9, 1 \times 10^{-2}) \rightleftharpoons 2NOCl (9, 1 \times 10^{-2})$
 $\Delta G^\circ = -2.303 RT \left(\log \frac{(1 \times 10^{-2})^2}{(1 \times 10^{-5})^2 (1 \times 10^{-2})} \right) = -2.303 RT \left[\log \frac{1 \times 10^{-2}}{1 \times 10^{-10}} \right]$
 $= -2.303 RT [\log 1 \times 10^8] = -2.303 \times 8 \times 298 \times 8 = -43.92 \text{ kJ}.$
12. (a) (i) ΔG° for the reaction
 $\Delta G^\circ_{\text{reac.}} = 2\Delta G^\circ_f (NO_2) - \Delta G^\circ_f (N_2O_4)$
 $100 - 100 = 0$
 Now, $\Delta G = 2.303 RT \log Q_P + \Delta G^\circ$
 Here $Q_P = \frac{P_{NO_2}^2}{P_{N_2O_4}} = \frac{100}{10} = 10 \text{ atm}$
 So, $\Delta G = 2.303 RT \log Q_P + 0 = 2.303 RT \log Q_P = 2.303 RT \log_{10} 10 = 2.303 RT = 5.7 \text{ kJ/mole}$
 (ii) Since Q_P is more than K_P
 (Calculate K_P by putting the value of ΔG° in the equation $\Delta G^\circ = 2.303 RT \log K_P$ as $\Delta G^\circ = 0$ that's why K_P comes as 1.)
 Hence, the reaction will proceed in backward direction.
13. $\Delta H = \Delta U + \Delta nRT$
 $\Delta H - \Delta U = -3RT$
 $= -3 \times 8.314 \times 298 = -7432 \text{ J} = -7.432 \text{ kJ}$
16. $\Delta C_p = -ve$; $C_p (\text{products}) < C_p (\text{Reactants})$ so $q_2 < q_1$
17. $\Delta H_2 - 24 = -0.024 \times (523 - 473) \text{ cal/g.} \therefore \Delta H_2 = 22.8 \text{ cal/g.}$
18. Equal volume of $H_2(g)$ & $CO(g)$
 Total volume = 112 L
 So, volume of CO = volume of H_2 = 56 L
 Mole of CO = Mole of H_2 = 2.5 mole
- | | | |
|------------|---|---|
| | $H_2(g) + \frac{1}{2} O_2(g) \longrightarrow H_2O(g)$ | $\Delta H = -241.8 \text{ kJ}$ |
| For | 1 mole | $\Delta H = -241.8 \text{ kJ}$ |
| For | 2.5 mole | $\Delta H = -241.8 \times 2.5$ |
| Similarly, | $CO(g) + \frac{1}{2} O_2(g) \longrightarrow CO_2(g)$ | $\Delta H = -283 \text{ kJ}$ |
| | | $\Delta H = -283 \times 2.5 \text{ kJ}$ |
- Total Heat evolved = $[-241.8 + (-283)] 2.5 = -1312 \text{ kJ}$
19. $\Delta_r H = \frac{5}{8} \times 40 + \frac{3}{8} \times 50 = 43.75 \text{ kJ/mol}$
20. Equation (vi) can be generated by the following manipulations.
 Eq. (i) – Eq. (ii) – Eq. (iii) – Eq. (iv) – $\frac{1}{2}$ Eq. (v)
 Carrying out the corresponding manipulations on ΔH°_L , we get
 $\Delta H^\circ_L = (-563 - 419 - 88 + 322 - 79) \text{ kJ mol}^{-1} = -827 \text{ kJ/mole}$
21. In a reversible process, the driving and the opposite forces are nearly equal, hence the system and the surroundings always remain in equilibrium with each other.
22. (C) An ideal gas under going expansion in vacuum shows, $\Delta E = 0$, $W = 0$ and $q = 0$.
23. $W_2 - P(V_2 - V_1) = nR\Delta T = 1 \times R \times 2 = 2R.$



24. Mechanical work is important only in gases as they undergo appreciable change in volume.

26. Let final common temperature is T_f .
Heat gained by Zinc piece = Heat lost by water
 $0.4 (T_f - 20) \times 65.38 = 4.2 (100 - T_f) \times 180 \Rightarrow T_f = 97.3^\circ\text{C}$

27. Internal energy of a gas = $\frac{f}{2} nRT$

Internal energy of a gas $\propto f$

28. Mechanical work is done on the system and internal energy increases.

29. $\therefore$ The process is of free expansion then $W = 0$.

30. For polytropic process ($n = 2$)

$$PV^2 = C$$

$$TV = C$$

$$T_1 V_1 = T_2 V_2$$

$$300 \times 1 = T_2 \times 3 \Rightarrow T_2 = 100 \text{ K}$$

$$\Delta U = n C_{V,m} \Delta T$$

$$= 1 \times \frac{fR}{2} (T_2 - T_1) = 1 \times \frac{5 \times R}{2} \times (100 - 300) = \frac{1 \times 5 \times R}{2} \times 200 = -500 R = -4.2 \text{ kJ}$$

32. Final temperature of gas must be double of the initial temperature

$$\Delta H = \Delta U + \Delta (PV) = n C_V \Delta T + \Delta (PV) \quad (\Delta T = T_f - T_i = 2T_i)$$

$$= \frac{nR}{(\gamma - 1)} T_i + \Delta (PV) = \frac{3}{2} \times 100 + 100 = 250 \text{ J.}$$

33. Work is not a state function because it depends upon the path followed.

34. $q + w = \Delta U$ is a state function.

35. $W_{\text{Total}} = \text{area enclosed}$

$$V = \frac{nRT}{P}$$

$$V_A = \frac{2R \times 300}{1} = \frac{600}{1} R$$

$$V_B = \frac{2R \times 300}{2} = \frac{300}{1} R; V_C = \frac{2}{2} R \times 400 = \frac{400}{1} R$$

$$V_D = \frac{800}{1} R$$

$$W_{AB} = -nRT_A \ln \frac{V_B}{V_A} = -2R (300) \ln \frac{1}{2} = 600 R \ln 2$$

$$W_{BC} = -2(400 - 300) R = -200 R$$

$$W_{CD} = -2 R (400) \ln \frac{V_D}{V_C} = -800 R \ln 2$$

$$W_{AD} = -1(600 R - 800 R) = 200 R$$

$$W_{\text{Total}} = W_{AB} + W_{BC} + W_{CD} + W_{AD} = -200 R \ln 2 = -100 R \ln 4$$

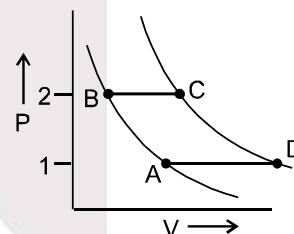
36. Since ice-water ratio is maintained as 1 : 1. So, $\Delta T = 0$.

37. (A) $w = -nR \Delta T = -2 \times 8.314 \times 100 = -1662.8 \text{ J}$

$$\Delta U = n \int_{300}^{400} C_{V,m} dT = 2 \times \int_{300}^{400} (20 + 10^{-2} T) dT = 2 \left[20 \times 100 + \frac{10^{-2}}{2} (400^2 - 300^2) \right] = 4700 \text{ J}$$

$$4700 = q - 1662.8$$

$$\therefore q = 6362.8 \text{ J.}$$





38. latent heat of vaporisation of water = 2.25×10^6 J/kg
 $\Delta H = 2.25 \times 10^6$ J/kg.
 work done = $-P_{\text{ext}} (V_2 - V_1)$
 $\Delta H = 2.25 \times 10^6$ J/kg
 $\Delta H = \Delta U + P\Delta V$
- (a) Now, volume of water $V = \left(\frac{m}{d}\right) = \frac{1}{1000} \text{ M}^3 = 1 \text{ L}$
- (b) volume of steam = $\frac{1000}{0.6} = 1666.67 \text{ L}$
 $2.25 \times 10^6 = \Delta U + 1 [1666.67 - 1] 101.325$
 $\Delta U = 22.5 \times 10^5 - 1.68 \times 10^5 = 20.8 \times 10^5 = 2.08 \times 10^6 \text{ J}$
39. Volume = $5 \times 10 \times 3 = 150 \text{ m}^3 = 150 \times 10^3 \text{ L}$; $T = 27^\circ\text{C} = 300 \text{ K}$; $P = 1 \text{ atm}$
 $PV = nRT \Rightarrow 1 \times 150 \times 10^3 = n \times 0.082 \times 300 \Rightarrow n = 6097.6$
 Total energy released per second = $150 \times 50 = 7500 \text{ J/sec.}$
 Amount of energy released by persons = energy gained by air $\Rightarrow [7500] t_{\text{sec}} = n C [dT]$
 $\Rightarrow 7500 t_{\text{sec}} = 6097.6 \times \frac{7}{2} \times 8.312 \times 15 \Rightarrow t_{\text{sec}} = 354.87 \text{ sec.} = \frac{354.87}{60} = 5.91 \text{ minutes.}$
40. At constant volume $\Delta U = q_v = nC_{v,m} \Delta T = \frac{3}{2} R \times 300 = 450 R$
 At constant pressure $\Delta H = q_p = nC_{p,m} \Delta T = \frac{5}{2} R (-150) = -375 R$
41. FACTUAL
42. FACTUAL
43. Molar mass = 102 gram/mole
 $P = 650 \text{ torr}$; $T = 77 + 273 = 350 \text{ K}$
 $Q = i \times t = 0.25 \times 600 = 150$
 $E = Q \times V = 150 \times 12 = 1800 \text{ J}$
 This heat is supplied to the system at constant pressure that's why this is used for change in enthalpy
 $\therefore$ For vaporisation of 1.8 gram, amount of heat required $q = 1800 \text{ J}$
 $\therefore$ For vaporisation of 102 gram, amount of heat required $q = \frac{1800}{1.8} \times 102 \text{ J}$
 $= 102 \times 10^3 \text{ J} = 102 \text{ KJ/mole}$
 $\Delta H = \Delta U + P\Delta V$
 $\Delta H = \Delta U + \Delta n_g RT$
 For determination of ΔU per mol ($\Delta n_g = 1$)
 $102 \text{ (KJ/mol)} = \Delta U + (1 \times 8.3 \times 350) \times 10^{-3} \Delta U = 102 - 2.9 = 99.1 \text{ KJ/mole}$
44. In a perfect gas is no energy of interaction between the molecules of the gas, so change the distance between the molecular by changing the volume of the gas cannot affect the internal energy of the gas. U is independent of V . Hence internal pressure is zero.
45. $W = +ve$ because work done on system.
 $W = 0$ because thermally insulated container.
47. State 1 $\longrightarrow$ State 2
 (1 L, 10 atm, 300 K) (4L, 5 atm, 600 K)
 Heat given, $q = 50 \times (600 - 300) = 15000 \text{ J}$
 $W = -P_{\text{ext}} (V_2 - V_1) = -1 (4-1) = -3 \text{ L atm} = -300 \text{ J}$
 $\Delta E = q + W = 15000 - 300 = 14700 \text{ J}$
 $\Delta H = \Delta E + P_2 V_2 - P_1 V_1 = 14700 \text{ J} + (20 - 10) \times 100 \text{ J} = 15.7 \text{ KJ}$



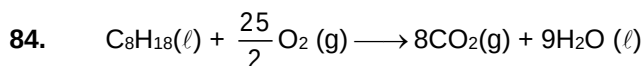
48. $\therefore H = E + PV$ and $\Delta H = \Delta E + P\Delta V$
 $P\Delta V = nRT$.
 $\therefore \Delta H = \Delta E + nR\Delta T$
 For isothermal and reversible process
 $\Delta T = 0$.
 $\therefore \Delta H = \Delta E + 0$.
 $\therefore \Delta E = 0$. $\therefore \Delta H$ is also equal to zero.
49. $\Delta U = W$
 $nC_v(T_2 - T_1) = -P \times (V_2 - V_1)$
 $\frac{3}{2} R (T_2 - T_1) = -1 \Rightarrow \therefore T_2 = T - \frac{2}{3 \times 0.0821}$
50. From first law of Thermodynamics, $\Delta E = q + w \Rightarrow nC_v dT = nC_d T - PdV$ (1)
 Now according to process, $P = V$ and according to ideal gas equation, $PV = nRT$
 We have, $V^2 = nRT$
 On differentiating, $2VdV = nRdT$ and $PdV = VdV = \frac{nR}{2} dT$
 So, from first equation we have, $nC_v dT = nC_d T - \frac{nR}{2} dT$
 So, $C_v = C - \frac{R}{2}$ Hence, $C = \frac{4R}{2}$
51. $W = -P\Delta V = -1 \times 10^5 (1 \times 10^{-2} - 1 \times 10^{-3}) = -1 \times 10^5 \times 9 \times 10^{-3} = -900 \text{ J}$.
52. $N_2 + 3H_2 \rightarrow 2NH_3$ $\Delta n = 2 - 4 = -2$
 $\Delta H = \Delta U + \Delta nRT = \Delta U - 2RT$. $\therefore \Delta H < \Delta U$.
53. In isolated system, the expansion of gas is carried out adiabatically. Since heat exchange between system and surrounding is not possible i.e. $q = 0$ and secondary w_{rev} is always greater than w_{irr} therefore for reversible process there must be comparatively higher decreases in internal energy i.e. ΔU for reversible process will be more negative. Hence, final temperature in reversible process will be smaller than irreversible process.
 $\therefore (T_f)_{irr} > (T_f)_{rev}$
54. $\Delta S = nC_{v,m} \ln \frac{T_2}{T_1} + nR \ln \frac{V_2}{V_1} = C_{v,m} \ln 2 + R \ln \left(\frac{1}{2}\right) = (C_{v,m} - R) \ln 2$
55. $\Delta S_{system} = nR \ln \frac{V_2}{V_1} = 2 \times R \times \ln 2 = 11.52 \text{ J/K}$
 $\Delta S_{surrounding} = -\frac{3.41 \times 1000}{310} = -11 \text{ J/K}$
 $\Delta S_{total} = +11.52 - 11 = +0.52 \text{ J/K}$
56. $\Delta S = nC_v \ln \frac{T_2}{T_1} + nR \ln \frac{V_2}{V_1}$
 For adiabatic process ($Q = 0$)
 $\Delta E = W \Rightarrow nC_v \ln \frac{T_2}{T_1} = -nR \ln \frac{V_2}{V_1} \therefore \Delta S = 0$.
57. $\Delta S = nC_{p,m} \ln \frac{T_2}{T_1} = 2.5 \times 18 \times 4.2 \ln \left(\frac{360}{300}\right) = 34.02 \text{ J/K}$



58. $\text{H}_2\text{O (l)} \longrightarrow \text{H}_2\text{O (g)}$ $\Delta H_{\text{vap}} = \Delta S_{\text{vap}} T_{\text{B.P.}} = x_1 T_1$
 $\Delta H_f(\text{H}_2\text{O, g}) = \Delta H_f(\text{H}_2\text{O, l}) + \Delta H_{\text{vap}} = x_2 + x_1 T_1$
 $\Delta H_{\text{reaction}} = 2\Delta H_f(\text{CO}_2, \text{g}) + 3\Delta H_f(\text{H}_2\text{O, g}) - \Delta H_f(\text{C}_2\text{H}_6, \text{g})$
 $= 2x_3 + 3(x_2 + x_1 T_1) - x_4 = 2x_3 + 3x_2 + 3x_1 T_1 - x_4.$
59. Lesser the ΔG° , lesser its stability & easily it can decompose.
60. The compressor has to run for longer time releasing more heat to the surroundings.
61. All these are statements of second law of thermodynamics.
62. As dew formation is spontaneous process, therefore, entropy or randomness of the universe will increase. As randomness of the system has decreased but randomness of the surrounding will increase larger so that change is positive.
65. $\Delta S(\text{gas}) = nR \ln \frac{P_1}{P_2} = 3 \times 8.314 \times \ln 2 = 17.29 \text{ J/K}$
 Since $\Delta T = 0$, therefore $\Delta U = 0$,
 So, $q = -W$
 $= nRT \left[1 - \frac{P_2}{P_1} \right]$
 $q = T \times 12.47 \text{ J/K}$
 $\Delta S_{\text{surr}} = -\frac{q}{T} = -12.47 \text{ J/K}$
 Change in $\Delta S = +4.82 \text{ J/K}$
66. $0.40 = aT_1^3 + bT_1 \Rightarrow 0.40 = a \times (1000) + b \times 10 \Rightarrow 0.4 = 1000a + 10b \dots(1)$
 $0.92 = aT_2^3 + bT_2 \Rightarrow 0.92 = a \times 8000 + 20b \dots(2)$
 from Eqs. (1) and (2)
 $a = 2 \times 10^{-5}, b = 0.038$
 $S_m = \int \frac{aT^3 + bT}{T} \cdot dT = \frac{a[T_2^3 - T_1^3]}{3} + b[T_2 - T_1] = 0.813 \text{ J/K mol}$
67. (A) $\Delta G_{200}^\circ = \Delta H_{200}^\circ - T \Delta S_{200}^\circ$
 $\Delta H_{200}^\circ = 20 - 4 = 16 \text{ kJ/mol}$
 $\Delta H_{T_2}^\circ = \Delta H + \Delta C_p [T_2 - T_1]$
 $\Delta H_{400}^\circ = \Delta H_{200}^\circ + \frac{20 \times 200}{1000} \text{ kJ/mol} = 16 + 4 = 20 \text{ kJ/mol}.$
68. $\Delta S_{\text{reaction}} = 6(87) - 4X - 205 \Rightarrow x = 145.75 \text{ J/mole K}$
69. It is because of the fact that for spontaneity, the value of $\Delta G = (\Delta H - T\Delta S)$ should be < 0 . If ΔS is $-ve$, the value of $T\Delta S$ shall have to be less than ΔH or the value of ΔS has to be less than $\frac{\Delta H}{T}$ i.e., $\frac{x}{298}$.
71. $\Delta S = nC_v \ln \left(\frac{T_f}{T_i} \right) + nR \ln \left(\frac{V_f}{V_i} \right) = \frac{5}{2} \times 2 \ln \frac{373}{298} + 2 \ln 10$
72. (D) No. of moles of sucrose = $\frac{34.2}{342} = 0.1$
 $-(\Delta G)_{T,P} = \text{useful work done by the system}$
 $-\Delta G = -\Delta H + T \cdot \Delta S = + (6000 \times 0.1) + \frac{180 \times 0.1 \times 300}{1000} = 605.4 \text{ kJ}.$



73. $\Delta G = \Delta H - T\Delta S$
 $\Delta H = \sum \Delta H_p - \sum \Delta H_r$
 $\Delta H = -110.5 + 266.3 = 155.8 \text{ KJ}$
 $\Delta S = \sum S_p - \sum S_r$
 $\Delta S = 197.6 + 27.28 - (57.5 + 5.74) = 161.64 \text{ J/mole K}$
 For reaction to be spontaneous, $\Delta G < 0$
 $\Delta H - T\Delta S < 0 \Rightarrow T > \frac{\Delta H}{\Delta S} = \frac{155800}{161.64} = 964 \text{ K}$
74. Given that, $\Delta H_{\text{vaps}} = 30 \text{ kJ/mol} = 30 \times 10^3 \text{ J/mol}$.
 $\Delta S_{\text{vaps}} = 75 \text{ J/mol}$.
 We know that, $\Delta S = \frac{\Delta H_{\text{vap}}}{T_{\text{B,P}}}$
 $\therefore \Delta H = T\Delta S \Rightarrow 30 \times 10^3 = T \times 75$.
 $T = 400 \text{ K}$.
75. For spontaneous process, $\Delta G = -\text{ve}$, $K > 1$ and $E^0_{\text{cell}} = +\text{ve}$.
76. If $\Delta n = -\text{ve}$ then $\Delta H < \Delta U$.
77. $\text{CS}_2 (\ell) + 3\text{O}_2 (\text{g}) \longrightarrow \text{CO}_2 (\text{g}) + 2\text{SO}_2 (\text{g}) \quad \Delta H = -256 \text{ Kcal}$
 Let $\Delta H_f (\text{CO}_2, \text{g}) = -4x$ and $\Delta H_f (\text{SO}_2, \text{g}) = -3x$
 $\Delta H_{\text{reaction}} = \Delta H_f (\text{CO}_2, \text{g}) + 2 \Delta H_f (\text{SO}_2, \text{g}) - \Delta H_f (\text{CS}_2, \ell)$
 $-265 = -4x - 6x - 26$
 $x = +23.9$
 $\therefore \Delta H_f (\text{SO}_2, \text{g}) = 3x = -71.7 \text{ Kcal / mol}$.
78. $\text{CH}_2 = \text{CH}_2 + \text{H}_2 \longrightarrow \text{CH}_3-\text{CH}_3 \quad \Delta H = X_1$
 So, Hydrogenation energy of benzene should be $3x_1$
 $\Delta H_{\text{calculated}} = 3x_1$
 So, Resonance energy = $[3x_1 - x_2]$
79. Required reaction $\Rightarrow 6\text{C}_{(\text{graphite})} + 3\text{H}_{2(\text{g})} \longrightarrow \text{C}_6\text{H}_{6(\text{g})}$
 1. $6\text{C}_{(\text{g})} + 6\text{H}_{(\text{g})} \longrightarrow \text{C}_6\text{H}_{6(\text{g})}$
 $\Delta H_1 = -(6 \times \Delta H_{\text{C-H}} + 3\Delta H_{\text{C=C}} + 3\Delta H_{\text{C-C}}) = -(6 \times 416 + 3 \times 591 + 3 \times 331) = -5262 \text{ kJ mole}^{-1}$
 2. $6\text{C}_{(\text{graphite})} \longrightarrow 6\text{C}_{(\text{g})} \quad \Delta H_2 = 6 \times 718 = 4308$
 3. $3\text{H}_{2(\text{g})} \longrightarrow 6\text{H}_{(\text{g})} \quad \Delta H_3 = 3 \times 436 = 1308$
 Adding (1) + (2) + (3)
 $6\text{C}_{(\text{graphite})} + 3\text{H}_{2(\text{g})} \longrightarrow \text{C}_6\text{H}_{6(\text{g})}$
 $\Delta H = \Delta H_f (\text{C}_6\text{H}_6, \text{g}) = \Delta H_1 + \Delta H_2 + \Delta H_3 = 354 \text{ kJ mole}^{-1}$
 Hence resonance energy = $\Delta H_f (\text{cal.}) - \Delta H_f (\text{theo}) = 354 - 83 = 271 \text{ kJ mole}^{-1}$
81. In neutralization of strong acid and base only H^+ and OH^- ions react in every case. So, heat of neutralization is remain constant and equal to -13.7 kcal (approximate).
82. Eqn. (ii) is for heat of neutralisation of strong acid with strong base. Eqn. (i) for weak acid with strong base. The difference $q_2 - q_1$ is heat of dissociation of weak acid.
83. $\text{H}_3\text{PO}_3 \longrightarrow 2\text{H}^+ + \text{HPO}_3^{2-}; \Delta_r H = ?$
 $2\text{H}^+ + 2\text{OH}^- \longrightarrow 2\text{H}_2\text{O};$
 $\Delta_r H = -55.84 \times 2 = -111.68$
 $-106.68 = \Delta_{\text{ion}} H - 55.84 \times 2$
 $\Delta_{\text{ion}} H = 5 \text{ kJ/mol}$.



$$\Delta n_g = 8 - \frac{25}{2} = \frac{16 - 25}{2} = -\frac{9}{2} = -4.5$$

$$\Delta H - \Delta E = \Delta n_g RT = -\frac{4.5 \times 8.314 \times 298}{1000} = -11.15 \text{ kJ}$$

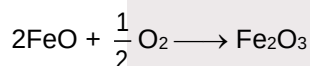
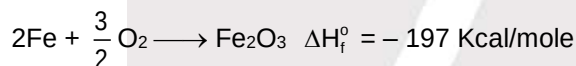
85. $\Delta C_P = C_P(\text{HCl}) - \frac{1}{2} C_P(\text{H}_2) - \frac{1}{2} C_P(\text{Cl}_2) = 6.81 - \frac{1}{2} \times 6.82 - \frac{1}{2} \times 7.71$
 $= 6.81 - 3.41 - 3.855 = -0.45$

$$\Delta H_{348} = \Delta H_{298} + (\Delta C_P)(\Delta T) = -22060 + (-0.45) \times 50 = -22082.5 \text{ cal}$$

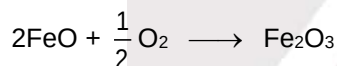
86. Let mixture contain x mole of steam & y mole oxygen then,
 Here, temperature is constant. So, amount of heat released = amount of heat gain

$$x \times 132 = y(220) \Rightarrow \frac{x}{y} = \frac{220}{132} = \frac{1.66}{1} \Rightarrow y : x = 1 : 0.6$$

87. 12g C = 1 mole, suppose CO produced = x moles.
 Then CO_2 produced = (1 - x) mole
 $\therefore x(-21.41) + (1 - x)(-94.05) = -57.5$
 This on solving gives x = 0.54 mole.

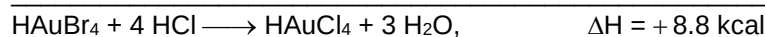
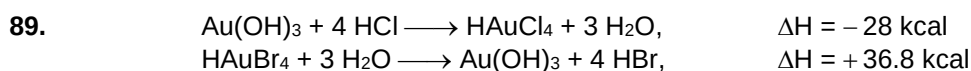


$$\Delta H = -197 + 65 \times 2 \Rightarrow \Delta H = -67 \text{ Kcal/mole}$$



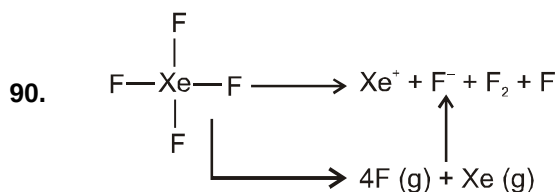
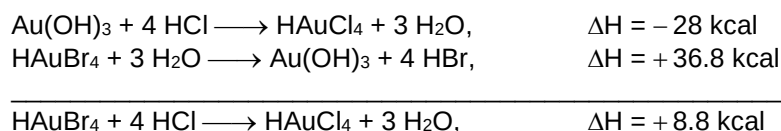
$$\frac{\frac{1}{3}}{\frac{2}{3} - 2x} = \frac{\frac{2}{3}}{\frac{1}{3} + x} \Rightarrow x = \frac{1}{5}$$

$$\text{So, energy released} = \frac{1}{5} \times 67 = 13.4 \text{ kcal/mole}$$



When 100% conversion, heat absorbed = 8.8 Kcal

$$\% \text{ conversion} = \frac{0.44}{8.8} \times 100 = 5\%$$



$$\Delta H = 4E_{\text{Xe-F}} + \Delta H_{\text{Ioni}} [\text{Xe} \longrightarrow \text{Xe}^+] - \Delta E_{\text{F-F}} - \Delta H_{\text{eg}} [\text{F} \longrightarrow \text{F}^-]$$

$$= 4 \times 34 + 279 - 85 - 38 = 136 + 279 - 123 = 415 - 213 = 292 \text{ kcal/mole.}$$

91. 1 M $\text{H}_2\text{SO}_4 = 2\text{g eq. of } \text{H}_2\text{SO}_4$

hence, $y = 2x$ or $x = \frac{1}{2}y$.

92. $Q = m S \Delta T_1 \Rightarrow \frac{Q_R}{2} = \frac{m}{2} S \Delta T_1 \Rightarrow \text{So, } \Delta T_1 = \Delta T_2$

93. Assuming density of solution is 1g/cc
and specific heat is 4.2 J/g-K

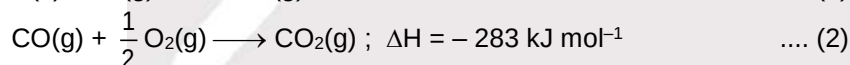
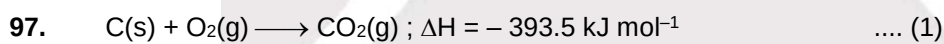
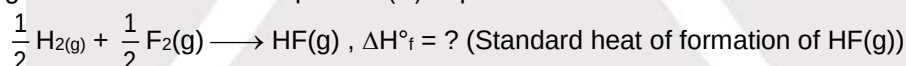
$$q = ms\Delta T = 100 \times 4.2 \times 3$$

millimoles of acid neutralized = 5

$$\Delta H = -100 \times 4.2 \times 3 \times \frac{1000}{5} = -2.52 \times 10^2 \text{ kJ/mole.}$$

95. HCOOH is a weak acid so heat of neutralization is less than x .

96. The amount of heat either evolved or absorbed when one gram mole of a substance is formed from its **constituent elements**, is known as the standard heat of formation (ΔH°_f).
Standard state temperature is 25°C or 298K and pressure of gaseous substance is one atmosphere.
Thus in given thermochemical equation '(B)' represents the standard heat of formation of HF .



On subtraction equation (2) from equation (1), we get $\text{C(s)} + \text{O}_{2(\text{g})} \rightarrow \text{CO(g)}; \Delta H = -110.5 \text{ kJ mol}^{-1}$.
The enthalpy of formation of carbon monoxide per mole = $-110.5 \text{ kJ mol}^{-1}$.

98. Let the bond dissociation energy of XY , X_2 and Y_2 be x, x and $x, \frac{x}{2} \text{ KJ/mol}$ respectively,



$\Delta H_{\text{reaction}} = [(\text{sum of bond dissociation energy of all reactants}) - (\text{sum of bond dissociation energy of all product})]$

$$= \left[\frac{1}{2} \Delta H_{\text{X}_2} + \frac{1}{2} \Delta H_{\text{Y}_2} - \Delta H_{\text{XY}} \right] = \frac{x}{2} + \frac{0.5x}{2} - x = -200$$

$$\therefore x = \frac{200}{0.25} = 800 \text{ KJ mol}^{-1}.$$



99. $\frac{1}{2} \text{I}_2(\text{s}) + \frac{1}{2} \text{Cl}_2(\text{g}) \rightarrow \text{ICl}(\text{g})$ $\Delta H_f, \text{ICl}(\text{g}) = \left[\frac{1}{2} \Delta H_{\text{I}_2(\text{s}) \rightarrow \text{I}_2(\text{g})} + \frac{1}{2} \Delta H_{\text{I}-\text{I}} + \frac{1}{2} \Delta H_{\text{Cl}-\text{Cl}} \right] - [\Delta H_{-\text{Cl}}]$

$$= \left[\frac{1}{2} \times 62.76 + \frac{1}{2} \times 151.0 + \frac{1}{2} \times 242.3 \right] - [211.3] = 16.73 \text{ kJ/mol.}$$

101. $\Delta W = \int_1^{10} P dv = -20 \int_1^{10} \frac{dv}{V} = -20 [\ln(V)]_1^{10} = -20 \ln(10) = -4605 \text{ J}$

$$\Delta q = \Delta U - \Delta W$$

$$= 400 - (-4605) = 5005 \text{ J} \approx 5 \text{ kJ.}$$

102. $n = 4$
 $V_1 = 20 \text{ lit.}$ $V_2 = 60 \text{ lit}$
 $\Delta W = -P_{\text{ext}} (V_2 - V_1)$

$$= -\frac{125}{1000} \times (60 - 20) \times 100 = -500 \text{ J.}$$

In case of adiabatic process $\Delta q = 0$

$$\Delta U = 0 + \Delta W$$

$$nC_p \Delta T = \Delta W$$

$$4 \times \frac{3}{2} \times R \Delta T = -500$$

$$\Delta T = -6 \text{ K}$$

103. $\text{mole} = \frac{7.8}{78} = 0.1$

$$\Delta H = n \times \Delta H_{\text{vap}}$$

$$= 0.1 \times 29.7 \text{ KJ} = 2.97 \text{ KJ} = 2970 \text{ J}$$

$\Delta H = \text{vaporization}$

$$t = \frac{\Delta H}{v.i} = \frac{2970}{11.4 \times 5}$$

$$t = 540 \text{ second}$$

$$t = 9 \text{ minute}$$

104. $\text{C}_{(\text{graphite})} \longrightarrow \text{C}_{(\text{diamond})}$
 1 mole 1 mole
 $\rho = 2 \text{ g/cm}^3$ $\rho = 3 \text{ g/cm}^3$
 $V = \frac{12}{2} \text{ ml}$ $V = \frac{12}{3} \text{ ml}$
 $\Delta H = \Delta U + P \Delta V$
 $\Delta U - \Delta H = -P \Delta V$

$$= -5 \times 10^5 \left(\frac{12}{3} - \frac{12}{2} \right) \times 10^{-3} \text{ L.atm} = -5 \times 10^5 \times \frac{12}{6} \times 10^{-1} \text{ J} = 100 \text{ KJ/mole} = 10 \times 10^4 \text{ J} = 100 \text{ KJ/mole}$$

$$\frac{\Delta U - \Delta H}{20} = \frac{100}{20} = 5 \text{ Ans.}$$

105. Avg. $C_{V, m} = \frac{n_1 C_{V, m_1} + n_2 C_{V, m_2}}{n_1 + n_2} = \frac{1 \times 3R + 2 \times \frac{3}{2}R}{3} = 2R$

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1} \Rightarrow 320 \times 1^{\frac{3}{2}-1} = T_2 \times (4)^{\frac{3}{2}-1} \Rightarrow T_2 = 160 \text{ K}$$

$$\Delta U = (n_1 + n_2) C_{V, \text{avg}} \Delta T = 3 \times 2R \times (160 - 320) = -960 R$$

$$|(\Delta U/10R)| = |-960 R/10R| = 96$$



106. This is adiabatic irreversible process so, for this process

$PV^\gamma = \text{Constant}$, is not applicable

$$W = -P_{\text{ext}}(V_2 - V_1)$$

But for adiabatic process

$$W = dU = \left(\frac{P_2 V_2 - P_1 V_1}{\gamma - 1} \right)$$

$$PV = nRT \Rightarrow 10 \times 10 = n \times 0.082 \times 273 \quad n = 4.47 \text{ moles}$$

$$-P_{\text{ext}}(V_2 - V_1) = \left(\frac{P_2 V_2 - P_1 V_1}{\gamma - 1} \right) \Rightarrow -1 \times (V_2 - 10) = \frac{1 \times V_2 - 10 \times 10}{1.67 - 1}$$

$$\Rightarrow (10 - V_2) = \frac{V_2 - 100}{0.67} \Rightarrow 6.7 - 0.67 V_2 = V_2 - 100 \Rightarrow 106.7 = 1.67 V_2 \Rightarrow V_2 = 64$$

107. $\Delta E = q + w \quad 60 = 50 + w \quad w = 10 \text{ J}$

108. $\Delta E = 0 \Rightarrow 20 \times 4.2 - 60 + W = 0$

$$W = -24 \text{ J.}$$

109. $\Delta S_{\text{sys}} = nR \ln \frac{V_2}{V_1} + nC_v \ln \frac{T_2}{T_1}$

$$\Delta S_{\text{system}} = nR \ln \frac{V_2}{V_1} = 2 \times 8.314 \ln 7.5 = \mathbf{33.6 \text{ J}} = 8 \text{ Cal.}$$

110. $\text{Hg}(\ell) \rightarrow \text{Hg}(\text{g})$ at 433 K and 4.19 mm of Hg is reversible and $\Delta G = 0$ but ΔG for $\text{Hg}(\text{g})$ (4.19 mm of Hg) $\rightarrow \text{Hg}(\text{g})$ (760 mm of Hg)

$$\text{with } \Delta G = nRT \ln \frac{P_2}{P_1} \quad \Delta G = 18 \text{ J/mol.}$$

111. (i) In case of adiabatic reversible expansion, $dq_{\text{rev}} = 0 \Rightarrow \Delta S = 0$.

(ii) In case of free expansion (Adiabatically)

$$W = 0, \quad q = 0, \quad \Delta U = 0 \Rightarrow nC_v(T_2 - T_1) = 0 \Rightarrow T_2 = T_1$$

$$\Delta S = nR \ln \left(\frac{V_2}{V_1} \right) = 2R \ln(3) = \mathbf{18 \text{ J/K.}}$$

112. Heat generated = $C_T \Delta T = 1260 \times 0.667 \text{ cal.}$

$$\therefore n_{\text{CH}_4} = \frac{2.108 \times 1}{210.8} = 0.01$$

$$n_{\text{total}} = \frac{PV}{RT} = 0.04 \quad \therefore \text{mol\%} = \frac{0.01}{0.04} \times 100 = 25\% \text{ Ans.}$$

113. Total mass of solution = $(150 + 150) = 300 \text{ g}$ $\{\rho_{\text{H}_2\text{O}} = 1 \text{ g/mL}\}$

$$Q_{\text{total}} = m\Delta T = 300 \times 1 \times (29 - 25) = 1200 \text{ Cal}$$

Since heat is liberated.

$$\text{Heat of neutralization} = \frac{Q}{150} \times 1000 \times \frac{1}{0.5} = \frac{1200}{150} \times 1000 \times \frac{1}{0.5} = 16 \text{ kcal}$$

Hence Heat of neutralization = -16 kcal

114. $2\text{C}_{(\text{s})} \rightarrow 5\text{C}_{(\text{g})} \quad \Delta H_1 = 5 \times 718 = 3590$
 $4\text{H}_{2(\text{g})} \rightarrow 8\text{H}_{(\text{g})} \quad \Delta H_2 = 4 \times 435 = 1740$
 $5\text{C}_{(\text{g})} + 8\text{H}_{(\text{g})} \rightarrow \text{C}_5\text{H}_{8(\text{g})} \quad \Delta H_3 = -(8 \times 913 + 2 \times 348 + 2 \times 615) = -5230$
 (1) + (2) + (3)
 $5\text{C}_{(\text{s})} + 4\text{H}_{2(\text{g})} \rightarrow \text{C}_5\text{H}_{8(\text{g})} \quad \Delta H_{\text{rxn}} = \Delta H_f(\text{C}_5\text{H}_8) = \Delta H_1 + \Delta H_2 + \Delta H_3 = 100$
 so resonance energy = $79 - 100 = -21 \text{ kJ mole}^{-1}$
 Hence resonance energy in $\text{Kcalmole}^{-1} = 5$



115. (i) $C(s) + O_2(g) \longrightarrow CO_2(g)$; $\Delta H_f^\circ = -390 \text{ KJ/mol}$; $\Delta H_f^\circ = \frac{585}{18} \times 12$
- (ii) $H_2(g) + \frac{1}{2} O_2(g) \longrightarrow H_2O(l)$; $\Delta H_f^\circ = -280 \text{ KJ/mol}$; $\Delta H_f^\circ = \frac{15540}{55.5}$
- (iii) $C_6H_6(l) + \frac{15}{2} O_2(g) \longrightarrow 6CO_2(g) + 3H_2O(l)$; $\Delta H_f^\circ = -48$
- $\therefore \Delta H^\circ = [6(-390) + 3(-280)] - 48 = -3228 \text{ KJ/mol}$
- Mass of benzene is $= 0.87 \times 100 = 87 \text{ g}$
- $\therefore$ Heat evolved from 87 g benzene $= 3600 \text{ KJ}$.
- Hence, $\frac{\Delta H}{100} = 36 \text{ kJ}$.

116. $\Delta H_{\text{ionization}} = \Delta H_{\text{neutralization of WA + SB}} - \Delta H_{\text{neutralization of SA + SB}}$
 $= -56.1 - (-57.3) = 1.2 \text{ KJ eq}^{-1}$
 Enthalpy of ionization for making 100% ionization when there is no ionization at all $= 1.5 \text{ KJ eq}^{-1}$
 $\therefore$ % of ionization $= \frac{1.5 - 1.2}{1.5} \times 100 = 20\%$

117. $Fe_2O_3(s) + 3H_2(g) \longrightarrow 2Fe(s) + 3H_2O(l)$
 $\Delta H_{300K}^\circ = -35 \text{ kJ}$
 $\Delta H^\circ = -26 \text{ KJ}$
 $\Delta C_P = 2C_P(Fe, s) + 3C_P(H_2O, l) - C_P(Fe_2O_3, s) - 3C_P(H_2, g)$
 $\Delta C_P = 80$
 $-26 \times 10^3 = -80 \times 26720 + 80(\Delta T)$
 $\Delta T = 9$

118. $Na(s) + \frac{1}{2}Br_2(g) \longrightarrow NaBr(s)$
- $\downarrow \Delta H_{\text{sub}} = 137$ $\downarrow \Delta H_{\text{diss}} = 72$
 $Na(g)$ $Br(g)$
 $\downarrow \Delta H_{1st} = 496$ $\downarrow \Delta H_{1st} = -325$
 $Na^+(g) + Br^-(g)$
- $\Delta H_{\text{Lattice energy}} = -742$
- so, $\Delta H_{rxn} = \Delta H_f(NaBr) = \Delta H_{\text{sub}} + \Delta H_{1st \text{ ionisation}}(Na) + \Delta H_{\text{diss.}}(Br_2(g)) + \Delta H_{1st \text{ electron gain enthalpy}} - \Delta H_{L.E.}$
 $= 137 + 496 + 72 - 325 - 742 = -362$
- for 0.1 mole $\Rightarrow \Delta H_{rxn} = -36.2 \text{ kJ mole}^{-1}$
- so, in kcal $\Rightarrow \Delta H_{rxn} = \frac{-36.2}{4.2} = 8.6 \approx 9 \text{ kcal}$

119. $n = \frac{3.5}{28}$
 $\Delta T = T_2 - T_1 = 298.45 - 298 = 0.45$
 $C_V = 2.5 \text{ kJ K}^{-1} = 2500 \text{ JK}^{-1}$; $C_P = C_V + R = 2500 + 8.314 = 2508.314 \text{ JK}^{-1}$
 $Q_P = C_P \Delta T = 1128.74 \text{ J}$
 $\Delta H = \frac{Q_P}{n} = \frac{1128.74}{3.5/28} = 9030 \text{ J mol}^{-1} = 9.030 \text{ KJ mol}^{-1} = 9 \text{ KJ mol}^{-1}$.

120. $q = -w = nRT \ln \frac{V_2}{V_1} = 2700 \text{ J}$
- $\Delta S_{\text{gas}} = nR \ln \frac{V_2}{V_1} = 8.314 \times 2.303 \times \log 3 = 9 \text{ JK}^{-1}$
- $q = 2700 - 900 = 1800 \text{ J}$
- As entropy is a state function and ΔS of system will be same as above one.
- $\Delta S_{\text{gas}} = 9 \text{ J K}^{-1}$
- $\Delta S_{\text{surr}} = -\frac{1800}{300} = -6 \text{ J K}^{-1}$; $\Delta S_{\text{total}} = 3 \text{ J K}^{-1}$



121. I – Work done by the system in case of infinite stage expansion is more than single stage expansion.
 III – Single stage process is a irreversible process.
 IV – We have 5 points on PV diagram.

$$123. \quad \Delta H = (n_1 C_{p,m1} + n_2 C_{p,m2}) \Delta T = \left(0.5 \times \frac{7}{2} R + 0.5 \times 4 R \right) (-100) = -375 R$$

$$\text{and } \Delta U = \left(0.5 \times \frac{5}{2} R + 0.5 \times 3 R \right) (-100) = -275 R.$$

$$124. \quad \Delta H = 4 \text{ KJ}$$

$$\Delta H = \Delta U + \Delta(PV)$$

$$\Delta U = 4 - P(V_2 - V_1)$$

$$= 4 - \frac{10^5(18-19) \times 10^{-6}}{10^3} = 4 + 1 \times 10^{-4} = 4.0001 \text{ KJ}$$

$$125. \quad 1 \times V_f = 5 \times R \times 546 = 224 \text{ L}$$

$$W = -P_{\text{ext}}(\Delta V) = -1 \text{ atm}(224 \text{ L}) = -224 \text{ L atm}$$

$$\therefore \text{work done by system} = 224 \text{ L atm}$$

$$\text{Enthalpy change } (\Delta H) = q = 273 \times 5 = 1365 \text{ L atm}$$

$$\Delta U = q + W = 1365 - 224 = 1141 \text{ L atm.}$$

$$126. \quad \text{From graph, slope} = -1$$

$$\therefore \text{eq} = \log_{10} V + \log_{10} T = \text{constant}$$

$$\therefore VT = \text{constant or } PV^2 = \text{constant}$$

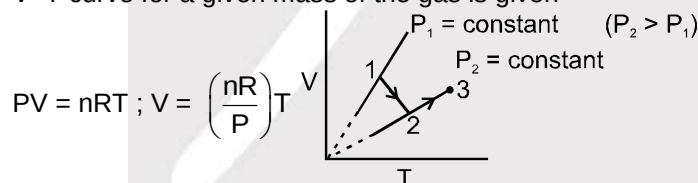
$$C = C_v + \frac{R}{1-n} \therefore C = \frac{5R}{2} + \frac{R}{1-2} = \frac{3R}{2}$$

$$\therefore q = nC\Delta T = 3 \times \frac{3R}{2} \times (1000-100) = 4050 R$$

$$\Delta U = nC_v\Delta T = 3 \times \frac{5R}{2} \times (1000-100) = 6750 R$$

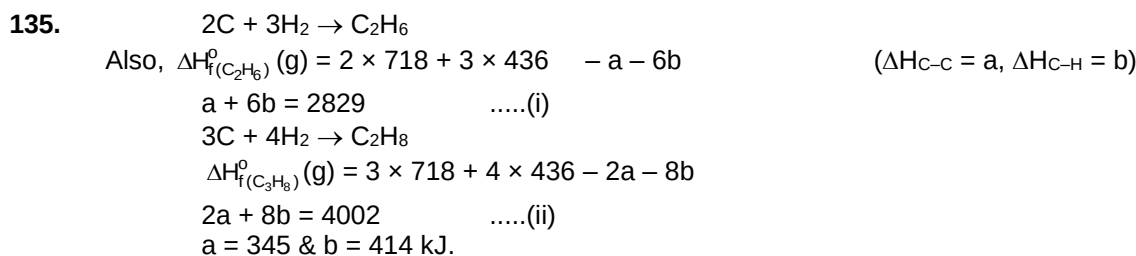
$$\therefore W = \Delta U - q = 6750 R - 4050 R = 2700 R$$

127. V-T curve for a given mass of the gas is given



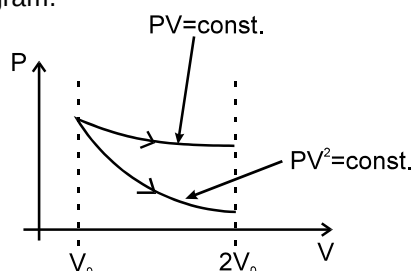
$$130. \quad \Delta S = nC_v \ln \left(\frac{T_f}{T_i} \right) + nR \ln \left(\frac{V_f}{V_i} \right) = 5 \ln \frac{373}{298} + 2 \ln 10$$

$$\Delta H = nC_p \Delta T = n(C_v + R) \Delta T = 1 \times 7 \times 75 = 525 \text{ cal}$$

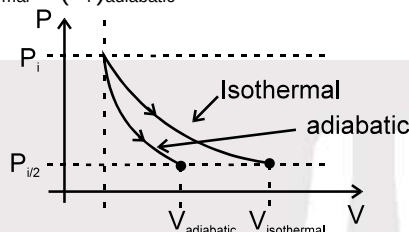




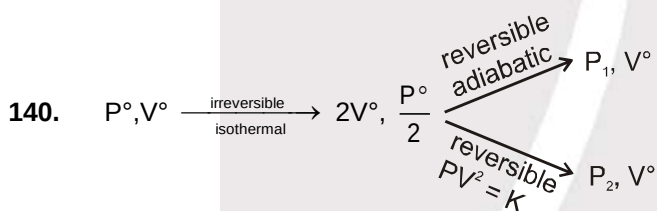
136. Work done in isothermal process will be more than $PV^2 = \text{const}$, process whatever be the value of K_1 and K_2 as is shown in the diagram.



137. Clearly $(V_f)_{\text{isothermal}} > (V_f)_{\text{adiabatic}}$



138. Work done in isothermal process is more than in adiabatic process as shown in the diagram above.



For reversible adiabatic,

$$\frac{P^\circ}{2} (2V^\circ)^\gamma = P_1 V_0^\gamma \Rightarrow P_1 = P^\circ 2^{\gamma-1}$$

For reversible $PV^2 = K$

$$\frac{P^\circ}{2} (2V^\circ)^2 = P_2 V_0^2 \Rightarrow P_2 = 2P^\circ$$

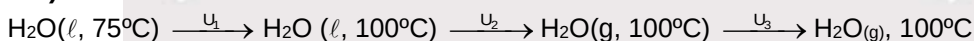
So, $P_2 > P_1$

Since final volume is same

$$P \propto T$$

So, $T_2 > T_1$

(141 to 143)



$$U_1 = nC_{V,m(\ell)} \Delta T = 1 \times 75 \times (100 - 75) = 1875 \text{ J}$$

$$U_2 = n \times \Delta H_{\text{vap}} - \Delta n_g RT = 1 \times 40700 - 1 \times 8.314 \times 373 = 37598.878 \text{ J}$$

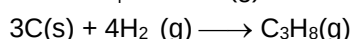
$$U_3 = nC_{V,m(\text{g})} \Delta T = n(C_{p,m(\text{g})} - R) \Delta T$$

$$= 1 \times (33.3 - 8.314) \times (120 - 100) = 24.986 \times 20 = 499.72 \text{ J}$$

(144 to 146)

Let moles of $\text{C}_3\text{H}_8 = x$, moles of $\text{C}_2\text{H}_2 = y$ & moles of $\text{CO}_2 = z$.

Calculation of ΔH_f° of $\text{C}_3\text{H}_8(\text{g})$



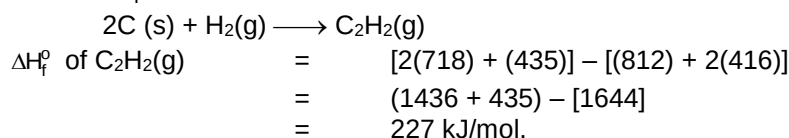
$$\Delta H_f^\circ \text{ of } \text{C}_3\text{H}_8(\text{g}) = [3(718) + 4(435)] - [2(347) + 8(416)]$$

$$= 3894 - 4022 = -128 \text{ kJ / mol.}$$

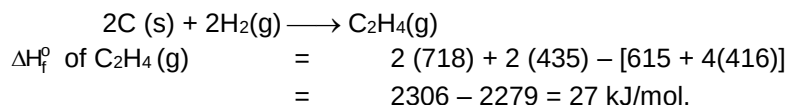




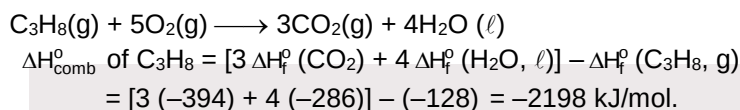
Calculations of ΔH_f° of $C_2H_2(g)$



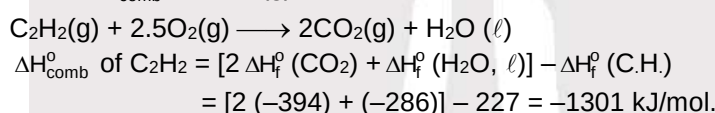
Calculations of ΔH_f° of $C_2H_4(g)$



Calculation of $\Delta H_{\text{comb}}^\circ$ of $C_3H_8(g)$



Calculation of $\Delta H_{\text{comb}}^\circ$ of $C_2H_2(g)$



Now total heat released during combustion

$$\begin{aligned}
 2198x + 1301y &= 4800 \quad (i) \\
 \text{Combustion} &= 3x + 2y + z = 10 \quad (ii)
 \end{aligned}$$

Total moles of $H_2O(l)$ formed = $4x + y$.

$$\text{moles of } C_2H_2(g) \text{ to be prepared} = \frac{3808}{22.4} = 170.$$

Total heat absorbed during evaporation of water formation of 170 moles C_2H_2 = 4800.

$$\begin{aligned}
 [(4x + y) \times 0.875 \times 40] + (170 \times 27) &= 4800 \\
 4x + y &= 6 \quad (iii)
 \end{aligned}$$

on solvent (i), (ii) and (iii)

$$\text{we get } x = 1, y = 2, \text{ and } z = 3$$

PART - IV

1. BC is a straight line & equation can be obtained by using

$$\begin{aligned}
 (y - y_1) &= \frac{y_2 - y_1}{x_2 - x_1} (x - x_1) \\
 (P - 3P^\circ) &= \frac{P^\circ - 3P^\circ}{2V^\circ - V^\circ} (V - V^\circ) \Rightarrow (P - 3P^\circ) = -\frac{2P^\circ}{V^\circ} (V - V^\circ) \quad \dots (1)
 \end{aligned}$$

Replacing P by T by using $P = \frac{RT}{V}$ in equation (1)

$$\left(\frac{RT}{V} - 3P^\circ \right) = -\frac{2P^\circ}{V^\circ} (V - V^\circ) \Rightarrow T = \frac{2P^\circ V(V - V^\circ)}{V^\circ R} + \frac{3P^\circ V}{R}$$

For T to be maximum

$$\begin{aligned}
 \frac{dT}{dV} &= 0 \\
 \frac{dT}{dV} &= -\frac{2P^\circ(2V - V^\circ)}{V^\circ R} + \frac{3P^\circ}{R} = 0 \Rightarrow V = \frac{5}{4} V^\circ
 \end{aligned}$$

$$T_{\text{max}} = -\frac{2P^\circ \times \frac{5}{4} V^\circ \left(\frac{5}{4} V^\circ - V^\circ \right)}{V^\circ R} + \frac{3P^\circ \times \frac{5}{4} V^\circ}{R} = \frac{25}{8} \left(\frac{P^\circ V^\circ}{R} \right)$$



$$2. \quad \Delta H^\circ = -3\Delta H_1^\circ + \Delta H_2^\circ + 2\Delta H_3^\circ + 3\Delta H_4^\circ$$

$$= -747.4 \text{ kJ}$$

$$\Delta H^\circ = \Delta U^\circ + \Delta n_g RT; \text{ where } \Delta n_g = -8$$

$$-747.4 = \Delta U^\circ - \frac{8 \times 8.314 \times 300}{1000}$$

$$\Delta U^\circ = -727.44 \text{ kJ.}$$

$$3. \quad \text{For } H_2(g) + O_2(g) \longrightarrow H_2O_2(l)$$

$$\Delta_f H^\circ (H_2O_2, l) = \Delta_f H^\circ + \frac{\Delta_r H_2^\circ}{2} - \frac{\Delta_r H_1^\circ}{2}$$

$$4. \quad HA \longrightarrow H^+ + A^-; \Delta_r H = 1.4 \text{ kJ/mol}$$

$$\Delta H_{\text{neutralization}} = \Delta H_{\text{ionization}} + \Delta_r H \text{ of } (H^+ + OH^- \rightarrow H_2O)$$

$$-55.95 = \Delta H_{\text{ionization}} - 57.3$$

$$\Delta H_{\text{ionization}} \text{ for 1 M HA} = 1.35 \text{ kJ/mol}$$

$$\% \text{ heat utilized by 1 M acid for ionization} = \frac{1.35}{1.4} \times 100 = 96.43\%$$

$$\text{So, acid is } 100 - 96.43 = 3.57\% \text{ ionized.}$$

$$5. \quad \Delta_f S^\circ (NH_4Cl, s) \text{ at } 300 \text{ K}$$

$$= S_{NH_4Cl(s)}^\circ - \left[\frac{1}{2} S_{N_2}^\circ + 2S_{H_2}^\circ + \frac{1}{2} S_{Cl_2}^\circ \right]$$

$$= -374 \text{ JK}^{-1} \text{ mol}^{-1}$$

$$\Delta_f C_p = 0$$

$$\therefore \Delta_f S_{310}^\circ = \Delta_f S_{300}^\circ$$

$$= -374 \text{ JK}^{-1} \text{ mol}^{-1}$$

$$\Delta H_{310}^\circ = \Delta H_{300}^\circ = -314.5$$

$$\Delta_f G_{310}^\circ = \Delta_f H^\circ - 310 \Delta S^\circ = -314.5 - \frac{310(-374)}{1000} = -198.56 \text{ kJ/mol.}$$

$$6. \quad 0.42 = a(10)^3 \Rightarrow a = 0.42 \times 10^{-3}$$

$$S_m = \int_0^{10} \frac{C_{p,m}}{T} dT = \int_0^{10} \frac{a}{T^2} dT = \frac{a}{3} [10^3 - 0] = \frac{0.42}{3} = 0.14 \text{ J/K-mol}$$

$$7. \quad \Delta G = \Delta H - \Delta(TS)$$

$$= \Delta H - T\Delta S \quad (\text{isothermal})$$

$$= 0 - T\Delta S = -T \left(\int \frac{dq_{\text{rev}}}{T} \right)$$

$$= - \int dq_{\text{rev}} = -q_{\text{rev}} = W_{\text{rev}}$$

$$\text{as process is isothermal so } \Delta E = 0 = q_{\text{rev}} + W_{\text{rev}}$$

$$\text{so } \Delta G = -nRT \ln \left(\frac{V_f}{V_i} \right)$$

$$= -RT \ln 2 = -8.314 \times 300 \times 0.693 \times 10^{-3} \text{ KJ mol}^{-1} \text{ K}^{-1}$$

$$= -1.728 \text{ KJ mol}^{-1} \text{ K}^{-1}$$

$$8. \quad \Delta G = \int_{P_1}^{P_2} V dP - \int_{T_1}^{T_2} S dT = 24.6 \left(\frac{4}{3} - 1 \right) \times 100 - \int (10 + 10^{-2} T) dT$$

$$= 820 - \left[10 (T_2 - T_1) + \frac{10^{-2} (T_2^2 - T_1^2)}{2} \right] = 820 - \left[10 (400 - 300) + 10^{-2} \frac{[(400)^2 - (300)^2]}{2} \right]$$

$$= 820 - 1000 - 350 = -530 \text{ J}$$



9. (A) ΔE will be -ve for combustion reactions
 (B) ΔH_f° (S, rhombic) = 0
 (C) $\text{CH}_4(\text{g}) \longrightarrow \text{C}(\text{g}) + 4\text{H}(\text{g}) \quad \Delta H = 1656 \text{ kJ/mole} = 4 \Delta H_{\text{C-H}}$
 $\text{C}_2\text{H}_6(\text{g}) \longrightarrow 2\text{C}(\text{g}) + 6\text{H}(\text{g}) \quad \Delta H = 2812 \text{ kJ/mole} = 6 \Delta H_{\text{C-H}} + \Delta H_{\text{C-C}}$
 so $\Delta H_{\text{C-C}} = (2812 - \frac{6}{4} \times 1656) \text{ kJ/mole} = 328 \text{ kJ/mole}$
 (D) Can not be concluded from given data.
10. (D) $\text{BOH}(\text{g}) + \text{aqueous} \longrightarrow \text{B}^+(\text{aq}) + \text{OH}^-(\text{aq}) \quad \Delta H = -20 \text{ kJ/mole}$
11. More heat is evolved due to combustion of less stable form.
12. (A) Reversible adiabatic process is isoentropic
 (B) Reaction is spontaneous, need not be exothermic.
 (C) Only when PV work is involved
- (D) $\Delta S_{\text{system}} = \frac{\Delta H_{\text{system}}}{T_b} = -\frac{40600}{373 \times 2} = -54.42 \text{ J/K}$
13. The given reaction will be spontaneous when ΔG would be negative when $\Delta H - T\Delta S$ is negative. That is,
 or $\frac{\Delta H - T\Delta S}{T} < 0$ or $T\Delta S > \Delta H$ or $T > 463.4 \text{ K}$
14. $S_{200(\text{B})} = \int_0^{200} \frac{C_p}{T} dT = 7 \text{ J mol}^{-1} \text{ K}^{-1}$
 $S_{200(\text{f})} = 7 + \frac{7500}{200} = 44.5 \text{ J mol}^{-1} \text{ K}^{-1}$
 $S_{300(\text{f})} = 44.5 + \int_{200}^{300} \frac{C_p}{T} dT = 44.5 + 60 \ln \frac{300}{200} + 1.6 = 70.43 \text{ J mol}^{-1} \text{ K}^{-1}$
 $S_{300(\text{g})} = 70.43 + \frac{30000}{300} = 170.43 \text{ J mol}^{-1} \text{ K}^{-1}$
 $S_{600(\text{g})} = 170.43 + 50 \ln \frac{600}{300} = 205.09 \text{ J K}^{-1} \text{ mol}^{-1}$
15. (a) State A to state B (Isobaric expansion)
 Work done by the gas = $W_1 = -p(V_B - V_A) = -1(40 - 20) = -20 \text{ L atm}$
 (b) State B to state C (isochoric process)
 Work done by the gas = $W_2 = 0$ ($\because \Delta V = 0$)
 (c) State C to state A (Isothermal compression)
 Work done on the gas = $W_3 = -2.303 nRT \log \frac{V_A}{V_C}$
 $= -2.303 pV \log \frac{V_A}{V_C}$ ($pV = p_A V_A = p_B V_B = nRT = 20$) $= -20 \ln(2) = 20 \times \ln(2) = 14$
 $\therefore$ total work done by the gas = $W_1 + W_2 + W_3 = -20 + 0 + 14 = -6 \text{ L atm}$
16. (a) $0.25 \times 2808 \times \frac{x}{180} \times 10^3 = 62.5 \times 9.81 \times 3$
 $x = 0.47 \text{ g}$
 (b) $0.25 \times 2808 \times 10^3 \times \frac{y}{180} = 62.5 \times 9.81 \times 6000$
 $Y = 940 \text{ g}$
 $Y = mX \quad \Rightarrow \quad 940 = m \times 0.47 \quad \Rightarrow \quad m = 2000$



17. $W = -P_{\text{ext}}(V_f - V_i) = -(1 \text{ atm})(8 - 2) \text{ L} = -6 \text{ L atm}$
as $q = 0$ so

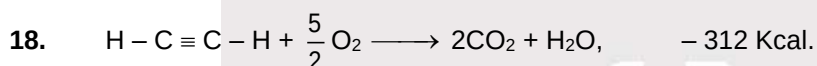
$$\Delta E = W \Rightarrow 3(8P_f - 12) = -6$$

$$8P_f = 12 - \frac{6}{3} = 10 \Rightarrow P_f = \frac{5}{4} \text{ atm}$$

so, $\frac{T_f}{T_i} = \frac{\frac{5}{4} \times 3}{6 \times 2} = \frac{10}{12}$

so $\Delta S = nC_{V,m} \ln \left(\frac{T_f}{T_i} \right) + nR \ln \left(\frac{V_f}{V_i} \right) = 3 \times \frac{12}{300} \ln \left(\frac{10}{12} \right) + \frac{12}{300} \ln 4$
 $= 3.312 \text{ J/K}$

Ans. 3 J/K



$$-312 = [2(-94) + (-68)] - \Delta H_f(\text{C}_2\text{H}_2)$$

$$\Delta H_f(\text{C}_2\text{H}_2) = 56 \text{ Kcal.}$$



$$\downarrow 2 \times 150 \quad \downarrow 100$$



$$56 = [(2 \times 150) + 100] - [(2 \times 95) + x]$$

$$x = 154 \text{ Kcal.}$$

$$\frac{\Delta H_{\text{C}=\text{C}}}{22} = 7$$

19. Heat evolved per mol of 'B' atoms = $\frac{5}{2} \times 200 = 500 \text{ kJ}$

20. In isothermal reversible process.

$$\Delta S = \frac{q_{\text{rev}}}{T}$$

$$q = -w = 2.303 RT \log \frac{V_2}{V_1}$$

$$= 2740.6 \text{ J/mole}$$

$$\Delta S_{\text{surr.}} = \frac{q_{\text{rev}}}{T} = \frac{2740.6}{300} = 9.135 \text{ J/K mole}$$

$$\Delta S_{\text{surr.}} = -\Delta S_{\text{system}} = -9.135 \text{ JK}^{-1} \text{ mole}^{-1}$$

$$\Delta S_{\text{universe}} = 0$$

21. $\Delta H^\circ = \Delta H_f^\circ \text{ of } \text{CaSO}_4 \cdot \frac{1}{2} \text{H}_2\text{O}(\text{s}) + \Delta H_f^\circ \text{ of } \frac{3}{2} \text{H}_2\text{O}(\text{g}) - \Delta H_f^\circ \text{ of } \text{CaSO}_4 \cdot 2\text{H}_2\text{O}(\text{s}) = +83 \text{ kJ mol}^{-1}$

22. $\Delta S^\circ = 119.5 + (188 \times 1.5) - 194 = 207.5 \text{ J/K mol}$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = 19550 \text{ J mol}^{-1}$$

$$\Delta G^\circ = -2.303 RT \log K_p = -2.303 R \times 300 \log (P_{\text{H}_2\text{O}})^{3/2}$$

$$\log P_{\text{H}_2\text{O}} = -\frac{19550}{8645.83} = -2.26; \quad P_{\text{H}_2\text{O}} = 4 \times 10^{-3} \text{ atm}$$



23. $P_{H_2O} = 1, K_p = 1$

$\Delta G^\circ = 0$ at eqm.

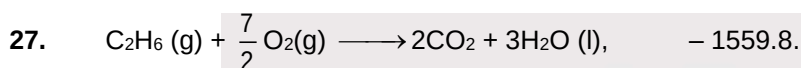
$$\Delta H^\circ - T\Delta S^\circ \Rightarrow T = \frac{\Delta H^\circ}{\Delta S^\circ} = \frac{83000}{207.5} = 400 \text{ K} = 127^\circ\text{C}$$

(24 to 26)

1. $-q_c = q_h \left(\frac{T_c}{T_h} \right) = 150 \times \frac{200}{1000}$ Heat rejected = 30 kJ

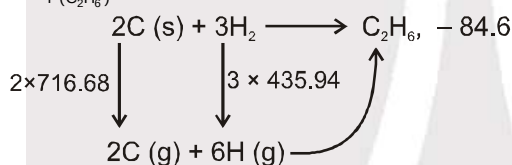
2. $\Delta S = \frac{q_{rev}}{T} = \frac{q_h}{T_h} = \frac{150 \times 10^3}{1000} = 150 \text{ JK}^{-1}$

3. $\Delta G = \Delta H - T\Delta S$; $\Delta H = 0$; $\Delta S = 150 \text{ JK}^{-1}$; $\Delta G = -150 \text{ kJ}$



$$-1559.8 = 2 \times (-393.5) + 3 \times (-285.8) - \Delta H_f(C_2H_6).$$

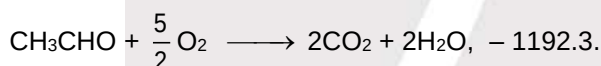
$$\Rightarrow \Delta H_f(C_2H_6) = -84.6.$$



$$\Delta H' = 0 - 6 \times BE_{C-H} - BE_{C-C}.$$

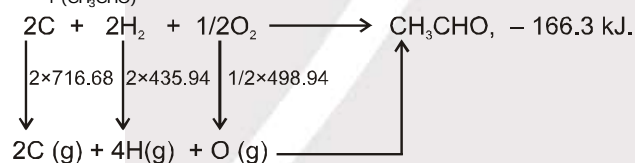
$$2 \times 716.68 + 3 \times 435.94 - 6BE_{C-H} - BE_{C-C} = -84.6.$$

$$6BE_{C-H} + BE_{C-C} = 2741.18 \text{ kJ.} \quad \dots(i)$$



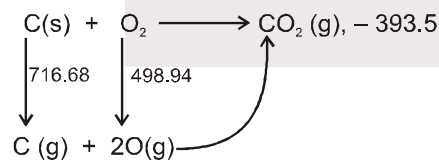
$$-1192.3 = 2 \times (-393.5) + 2 \times (-285.8) - \Delta H_f(CH_3CHO)$$

$$\Delta H_f(CH_3CHO) = -116.3 \text{ kJ.}$$



$$\Delta H' = 0 - 4 \times BE_{C-H} - BE_{C=O}.$$

$$-166.3 = 2 \times 716.68 + 2 \times 435.94 + \frac{1}{2} \times 498.94 - 4BE_{C-H} - BE_{C=O} \quad \dots(ii)$$



$$\Delta H' = 0 - 2 \times BE_{C=O}.$$

$$716.68 + 498.94 - 2 \times BE_{C=O} = -393.5.$$

$$BE_{C=O} = 804.56 \text{ kJ.} \quad \dots(iii)$$

Now we can find all bond energies.